

INTRODUCTION

The branch of Physics called optics deals with the behavior of light and other electromagnetic waves. Under many circumstances, the wavelength of light is negligible compared with the dimensions of the device as in the case of ordinary mirrors and lenses. A light beam can then be treated as a ray whose propagation is governed by simple geometric rules. The part of optics that deals with such phenomenon is known as geometric optics.

PROPAGATION OF LIGHT

Light travels along straight line in a medium or in vacuum. The path of light changes only when there is an object in its path or where the medium changes. We call this rectilinear (straight-line) propagation of light. Light that starts from a point A and passes through another point B in the same medium actually passes through all the points on the straight line AB. Such a straight line path of light is called a ray of light. Light rays start from each point of a source and travel along straight lines till they fall on an object or a surface separating two media (mediums). A bundle of light rays is called a beam of light.

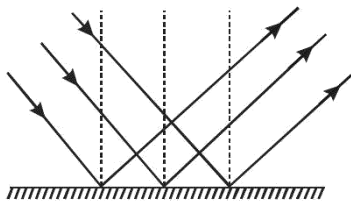
Apart from vacuum and gases, light can travel through some liquids and solids. A medium in which light can travel freely over large distances is called a transparent medium. Water, glycerine, glass and clear plastics are transparent. A medium in which light cannot travel is called opaque. Wood, metals, bricks, etc., are opaque. In materials like oil, light can travel some distance, but its intensity reduces rapidly. Such materials are called translucent.

REFLECTION OF LIGHT

When light rays strike the boundary of two media such as air and glass, a part of light is bounced back into the same medium. This is called Reflection of light.

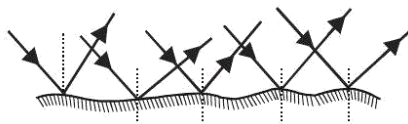
(i) Regular / Specular reflection:

When the reflection takes place from a perfect plane surface then after reflection rays remain parallel. It is called Regular reflection.



(ii) Diffused reflection

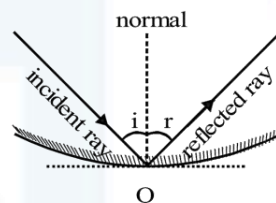
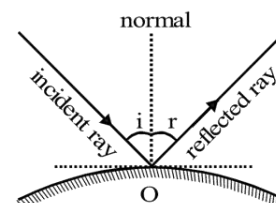
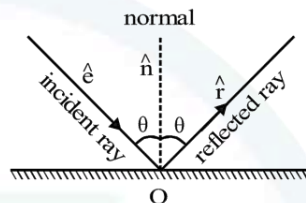
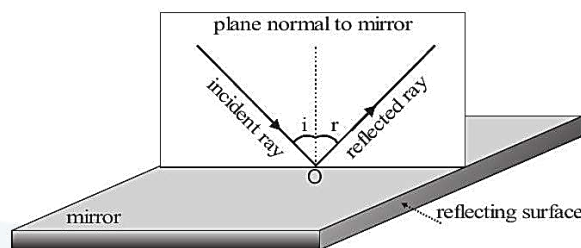
When the surface is rough, light is reflected from the surface from bits of its plane surfaces in irregular directions. This is called diffused reflection. This process enables us to see an object from any position.



LAWS OF REFLECTION

Incident ray, reflected ray and normal lies in the same plane.

The angle of reflection is equal to the angle of incident i.e. $\angle i = \angle r$.



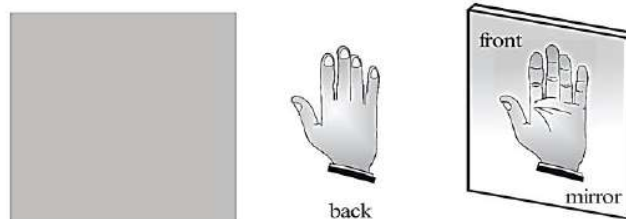
In vector form

$$\hat{r} = \hat{e} - 2(\hat{e} \cdot \hat{n})\hat{n}$$

REFLECTION FROM PLANE MIRROR

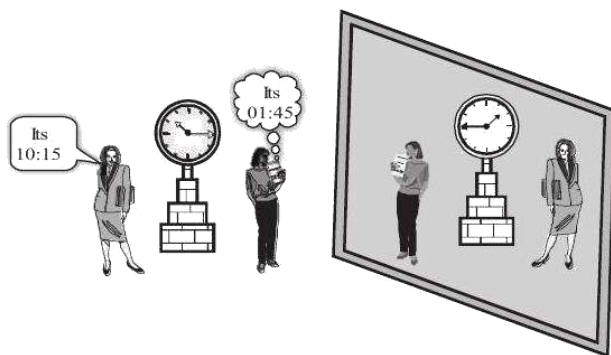
Plane mirror is the perpendicular bisector of the line joining object and image.

The image formed by a plane mirror suffers lateral-inversion, i.e., in the image formed by a plane mirror left is turned into right and vice-versa with respect to object.



When a watch placed in front of a plane mirror then watch is object and its time is object time and image of watch observed by a person standing in front of mirror then time seen by person.

- (i) Object Time = A^H
Image Time = $12 - A^H$
- (ii) Object Time = $A^H B^M$
Image Time = $11 - 60' - A^H B^M$
- (iii) Object Time = $A^H B^M C^S$
Image Time = $11 - 59' - 60'' - A^H B^M C^S$



- A plane mirror behaves like a window to virtual world.



- To see complete image in a plane mirror the minimum length of plane mirror should be half the height of a person.

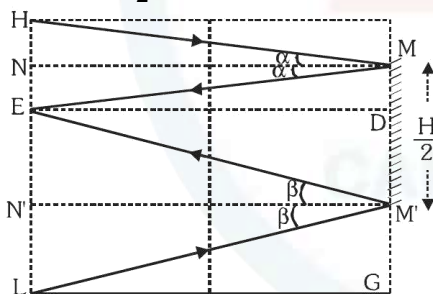
- From figure. $\triangle HNM$ and $\triangle ENM$ are congruent

$$\therefore EN = HN \quad \therefore MD = EN = \frac{1}{2}HE$$

Similarly $\triangle EN'M'$ and $\triangle LNM'$ are congruent

$$\text{Length of the mirror } MM' = MD + M'D = \frac{1}{2}HE + \frac{1}{2}EL$$

$$= \frac{1}{2}(HE + EL) = \frac{1}{2}HL$$

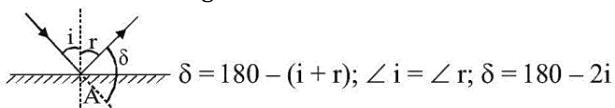


$\therefore$ Minimum of length of mirror is just half of the person.

This result does not depend on position of eye (height of the eye from ground).

This result is independent of distance of person in front of mirror.

Deviation for a single mirror



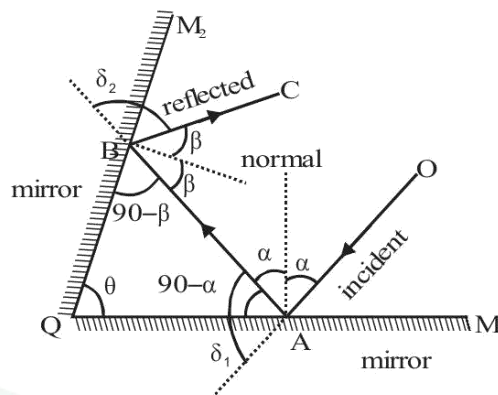
Total deviation produced by the combination of two plane mirrors which are inclined at an angle θ from each other.

$$\delta = \delta_1 + \delta_2 = 180 - 2\alpha + 180 - 2\beta = 360 - 2(\alpha + \beta)$$

$$\text{From } \triangle QAB, \theta + 90 - \alpha + 90 - \beta = 180$$

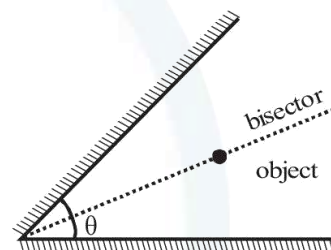
$$\Rightarrow \theta = \alpha + \beta$$

$$\text{Putting the value of } \theta \text{ in (i) from (ii), } \delta = 360 - 2\theta$$



If there are two plane mirror inclined to each other at an angle θ the number of image (n) of a point object formed are determined as follows.

- If $\frac{360^\circ}{\theta} = m$ is even then number of images $n = m - 1$
- If $\frac{360^\circ}{\theta} = m$ is odd. There will be two case.

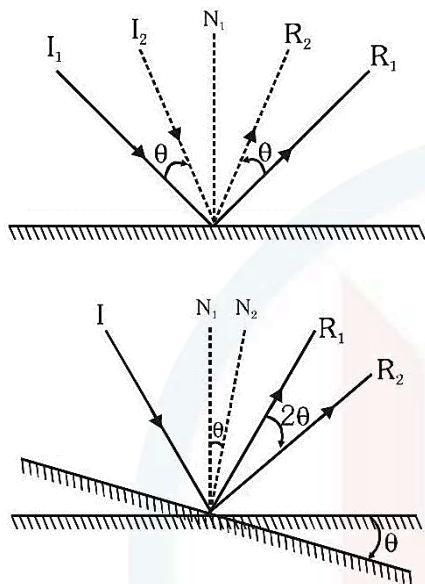


- When object is not on bisector, then number of images $n = m$
- When object is at bisector, then number of images $n = m - 1$
- If $\frac{360^\circ}{\theta} = m$ is a fraction, and the object is placed symmetrically then no. of images $n = \text{nearest even integer}$

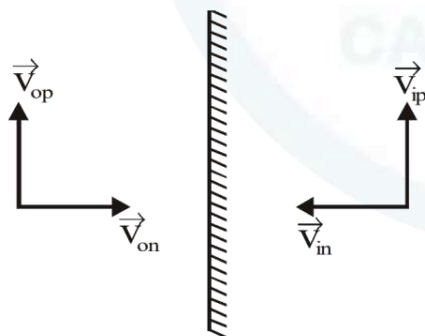
S.N	θ in degree	$m = \frac{360^\circ}{\theta}$	Number of images formed if object is placed	
			asymmetrically	symmetrically
1.	0	∞	∞	∞
2.	30	12	11	11
3.	45	8	7	7
4.	60	6	5	5
5.	72	5	5	4
6.	75	4.8	—	4
7.	90	4	3	3
8.	112.5	3.2	—	4
9.	120	3	3	2

If the object is placed between two plane mirrors then images are formed due to multiple reflections. At each reflection, a part of light energy is absorbed. Therefore, distant images get fainter.

Keeping the mirror fixed if the incident ray is rotated by some angle, the reflected ray is also rotated by the same angle but in opposite sense. (See Fig. 1)



Keeping the incident ray fixed, if the mirror is rotated by some angle, then the reflected ray rotates by double the angle in the same sense. (See Fig. 2)



Stationary Plane Mirror

$$\vec{v}_{on} = -\vec{v}_{in}, \vec{v}_{op} = \vec{v}_{ip}$$

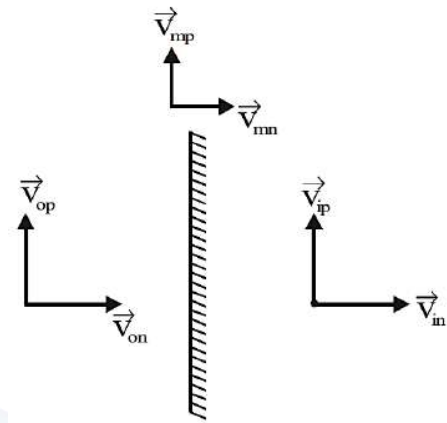
though speed of object and image are the same

v_{op} = component of velocity of object along parallel to mirror.

v_{on} = component of velocity of object along normal to mirror.

v_{ip} = component of velocity of image along parallel to mirror.

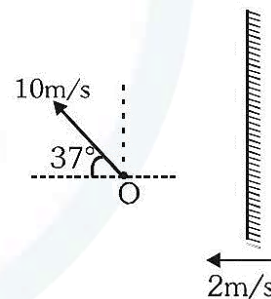
v_{in} = component of velocity of image along normal to mirror.



If mirror is moving $\vec{v}_{ip} = \vec{v}_{op}$ and $(\vec{v}_{im})_n = -(\vec{v}_{om})_n$
 $\Rightarrow \vec{v}_{in} - \vec{v}_{mn} = -(\vec{v}_{on} - \vec{v}_{mn}) \Rightarrow \vec{v}_{in} = 2\vec{v}_{mn} - \vec{v}_{on}$
 $\vec{v}_{mn}$ = component of velocity of mirror along normal.
 $\vec{v}_{op}$ = component of velocity of object along mirror.
 $\vec{v}_{on}$ = component of velocity of object along normal
 $\vec{v}_{ip}$ = component of velocity of image along mirror.
 $\vec{v}_{in}$ = component of velocity of image along normal.

Example: Find the velocity of the image.

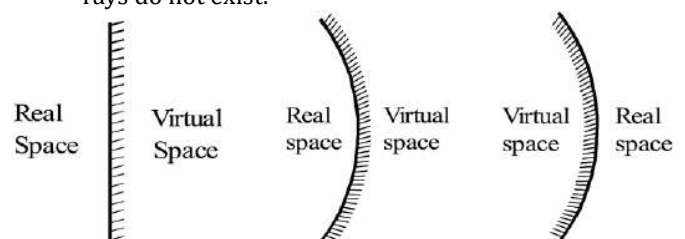
Solution: $\vec{v}_{ox} = (-10 \cos 37^\circ)\hat{i} = -8\hat{i}$ and
 $\vec{v}_{oy} = (10 \sin 37^\circ)\hat{j} = 6\hat{j}$
 $\vec{v}_{ix} = 2\vec{v}_{mx} - \vec{v}_{ox} = 2(-2\hat{i}) - (-8\hat{i})$
 $= 4\hat{i}$ and $\vec{v}_{iy} = \vec{v}_{oy} = 6\hat{j}$



REAL AND VIRTUAL SPACES

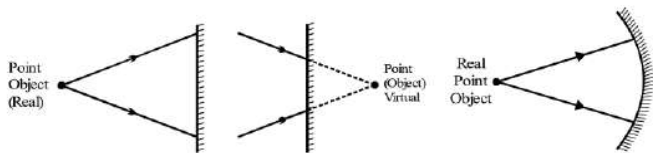
A mirror, plane or spherical divides the space into two;

- Real space, a side where the reflected rays exist.
- Virtual space is on the other side where the reflected rays do not exist.



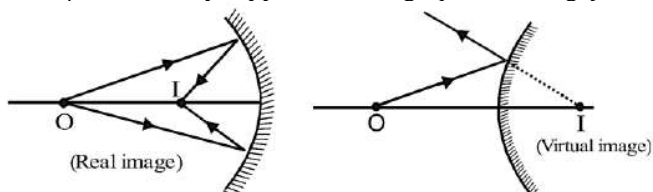
OBJECT

Object is decided by incident rays only. The point object is that point from which the incident rays actually diverge (Real object) or towards which the incident rays appear to converge (virtual object).



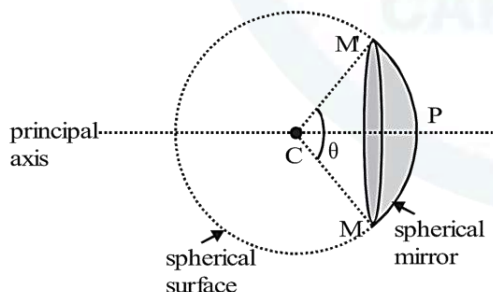
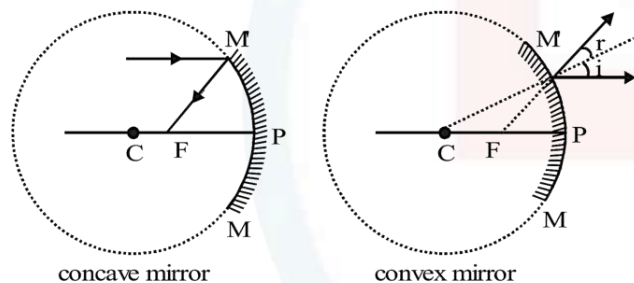
IMAGE

Image is decided by reflected or refracted rays only. The point image is that point at which the refracted / reflected rays reflected from the mirror, actually converge (real image) or from which the refracted / reflected rays appear to diverge (virtual image).



SPHERICAL (CURVED) MIRROR

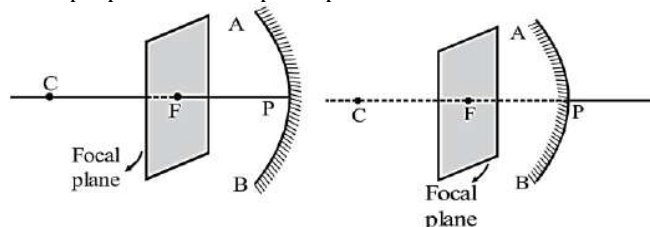
Curved mirror is part of a hollow sphere. If reflection takes place from the inner surface then the mirror is called concave and if its outer surface acts as reflector it is convex.



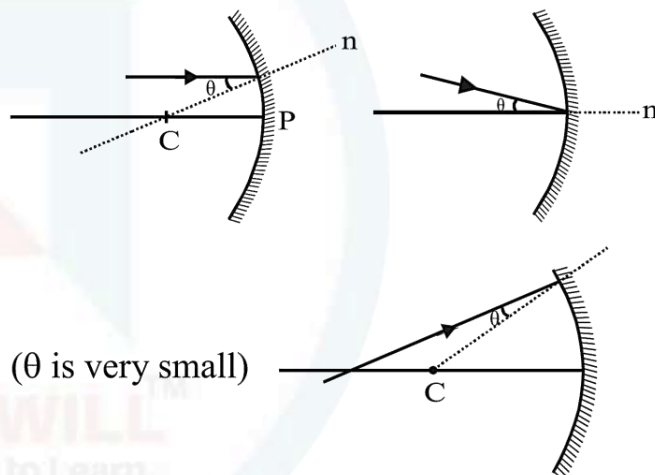
DEFINITIONS FOR THIN SPHERICAL MIRRORS

- Pole is any point on the reflecting surface of the mirror. For convenience we take it as the midpoint P of the mirror (as shown).
- Principal-section is any section of the mirror such as MM' passing through the pole is called principal-section.
- Centre of curvature is the centre C of the sphere of which the mirror is a part.
- Radius of curvature is the radius R of the sphere of which the mirror is a part.
- Principal-axis is the line CP, joining the pole and centre of curvature of the mirror.

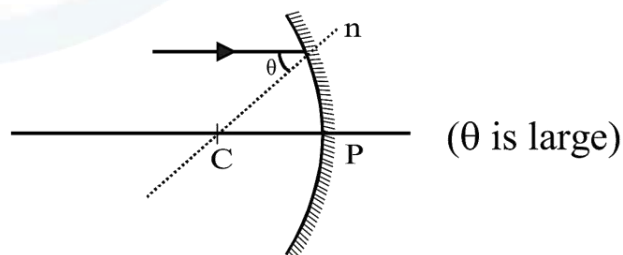
- Principal-focus is an image point F on principal axis for which object is at infinity.
- Focal-length is the distance PF between pole P and focus F along principal axis.
- Aperture, in reference to a mirror, means the effective diameter of the light reflecting area of the mirror.
- Focal Plane is the plane passing through focus and perpendicular to principal axis.



- Paraxial Rays Those rays which make small angle with normal at point of incidence and hence are close to principal axis. (θ is very small)

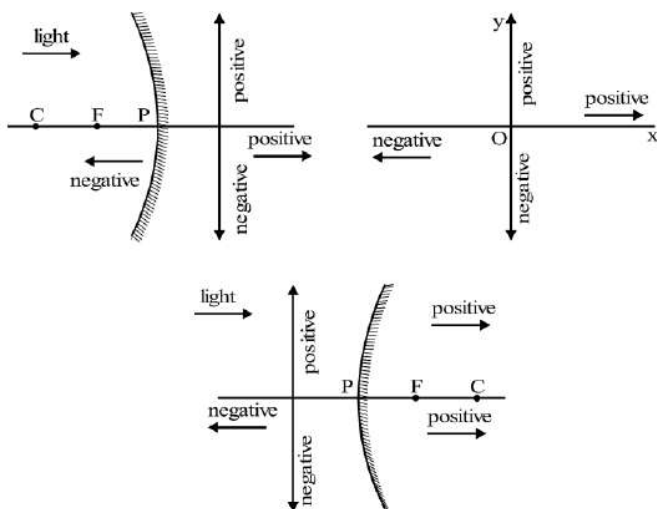


- Marginal rays: Rays having a large angle of incidence



SIGN- CONVENTION

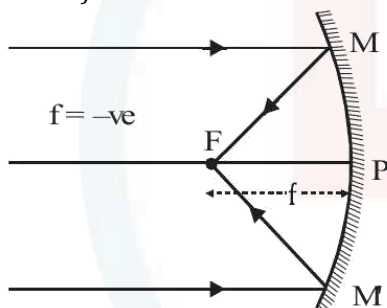
Along principal axis, distances are measured from the pole (pole is taken as the origin). Distance in the direction of light are taken to be positive while opposite to be negative. The distances above principal axis are taken to be positive while below it negative. Whenever and wherever possible the ray of light is taken to travel from left to right.



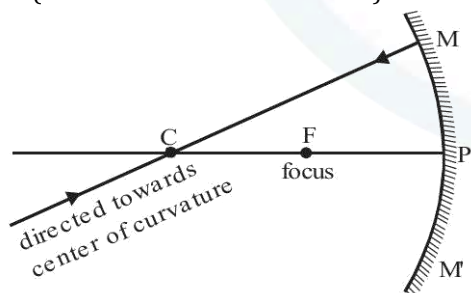
RULES FOR IMAGE FORMATION (FOR PARAXIAL RAYS ONLY)

(These rules are based on the laws of reflection $\angle i = \angle r$)

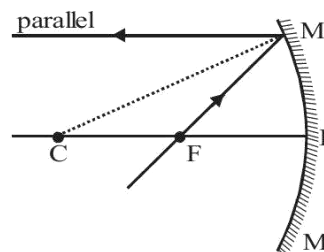
- A ray parallel to principal axis after reflection from the mirror passes or appears to pass through its focus (by definition of focus).



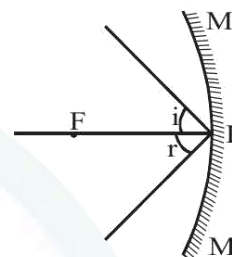
- A ray passing through or directed towards centre of curvature, after reflection from the mirror, retraces its path (as for it $\angle i = 0$ and so $\angle r = 0$).



- A ray passing through or directed towards focus, after reflection from the mirror, becomes parallel to the principal axis.



- Incident and reflected rays at the pole of a mirror are symmetrical about the principal axis $\angle i = \angle r$.

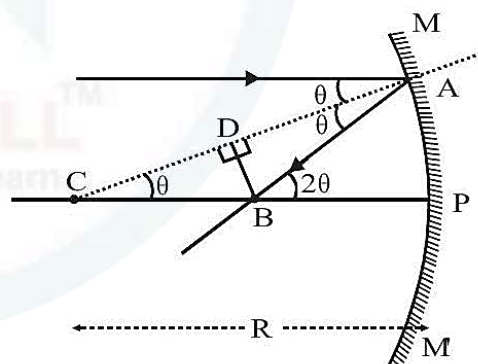


RELATIONS FOR SPHERICAL MIRRORS

Relation between f and R for the spherical mirror

For Marginal rays

$$\begin{aligned} \text{In } \triangle ABC, AB &= BC \\ AC &= CD + DA = 2 BC \cos \theta \\ \Rightarrow R &= 2 BC \cos \theta \\ \Rightarrow BC &= \frac{R}{2 \cos \theta} \text{ and} \\ BP &= PC - BC = R - \frac{R}{2 \cos \theta} \end{aligned}$$



Note: B is not the focus; it is just a point where a marginal ray after reflection meets.

For paraxial rays (parallel to principal axis)

(θ small so $\sin \theta \approx \theta$, $\cos \theta \approx 1$, $\tan \theta \approx \theta$). Hence $BC = \frac{R}{2}$ and $BP = \frac{R}{2}$

Thus, point B is the midpoint of PC (i.e. radius of curvature) and is defined as FOCUS so $BP = f = \frac{R}{2}$

(Definition: Paraxial rays parallel to the principal axis after reflection from the mirror meet the principal axis at focus)

- BP is the focal length (f) $f = \frac{R}{2}$

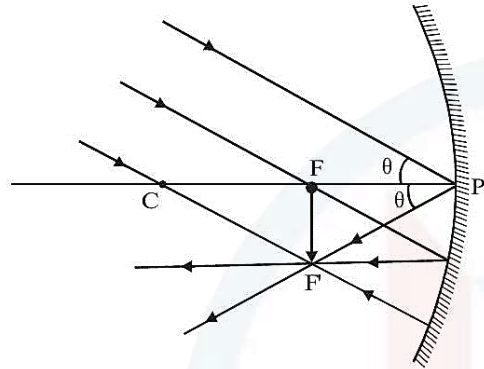
Paraxial rays (not parallel to principal axis)

Such rays after reflection meet at a point in the focal plane (F'), such that

$$\frac{FF'}{FP} = \tan \theta \approx \theta$$

$$\Rightarrow \frac{FF'}{f} = \theta$$

$$\Rightarrow FF' = f\theta$$



Relation between u, v and f for curved mirror

An object is placed at a distance u from the pole of a mirror and its image is formed at a distance v (from the pole)

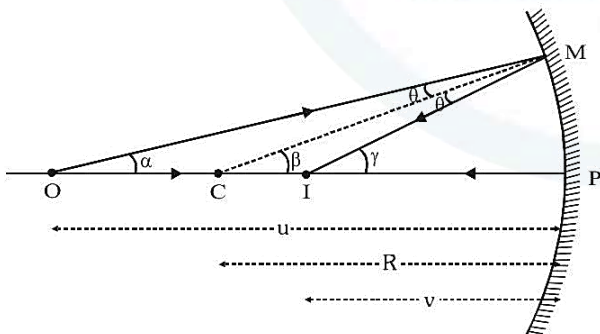
If angle is very small: $\alpha = \frac{MP}{u}$, $\beta = \frac{MP}{v}$, $\gamma = \frac{MP}{R}$

from $\triangle CMO$, $\beta = \alpha + \theta \Rightarrow \theta = \beta - \alpha$

from $\triangle CMI$, $\gamma = \beta + \theta \Rightarrow \theta = \gamma - \beta$

so we can write $\beta - \alpha = \gamma - \beta \Rightarrow 2\beta = \gamma + \alpha$

$$\therefore \frac{2}{R} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$



Sign convention for object/image for spherical mirrors

Real object u - ve Real image v - ve
Virtual object u + ve Virtual image v + ve

MAGNIFICATION

Transverse or lateral magnification

$$\text{Linear magnification } m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_i}{h_o}$$

$\triangle ABP$ and $\triangle A'B'P$ are similar so

$$\frac{-h_i}{h_o} = \frac{-v}{-u} \Rightarrow \frac{h_i}{h_o} = -\frac{v}{u}$$

$$\text{Magnification } m = -\frac{v}{u};$$

$$m = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f} = \frac{h_i}{h_o}$$

If one dimensional object is placed perpendicular to the principal axis then linear magnification is called transverse or lateral magnification.

$$M = \frac{h_i}{h_o} = -\frac{v}{u}$$

Magnification	Image	Magnification	Image
$ m > 1$	enlarged	$ m < 1$	diminished
$m < 0$	inverted	$m > 0$	erect

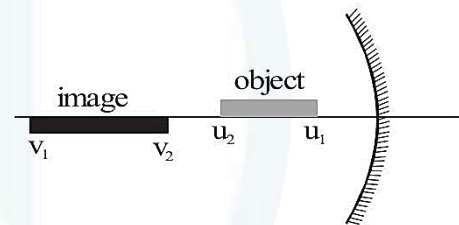
Longitudinal magnification

If one dimensional object is placed with its length along the principal axis then linear magnification is called longitudinal magnification.

Longitudinal magnification:

$$m_L = \frac{\text{length of image}}{\text{length of object}} = \frac{|v_2 - v_1|}{|u_2 - u_1|}$$

For small objects only: $m_L = -\frac{dv}{du}$



differentiation of $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ gives us $-\frac{dv}{v^2} - \frac{du}{u^2} = 0$

$$\Rightarrow -\frac{dv}{du} = \left[\frac{v}{u}\right]^2 \text{ so } m_L = -\frac{dv}{du} = \left[\frac{v}{u}\right]^2 = m^2$$

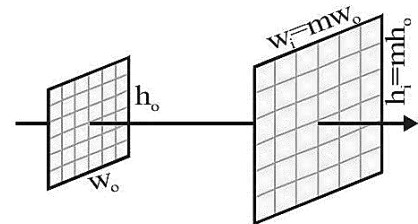
Superficial magnification

If two dimensional object placed with its plane perpendicular to principal axis its magnification is known as superficial magnification

$$\text{Linear magnification } m = \frac{h_i}{h_o} = \frac{w_i}{w_o}$$

$$h_i = mh_o, w_i = mw_o \text{ and } A_{\text{obj}} = h_o \times w_o$$

$$\text{Area of image: } A_{\text{image}} = h_i \times w_i = mh_o \times mw_o = m^2 A_{\text{obj}}$$



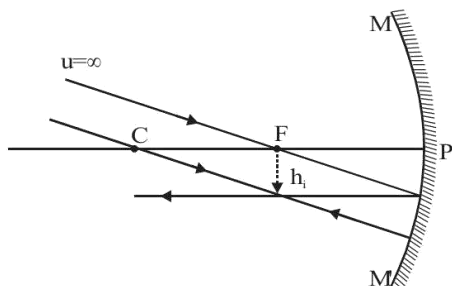
$$\text{Superficial magnification } m_s = \frac{\text{area of image}}{\text{area of object}}$$

$$= \frac{(ma) \times (mb)}{(a \times b)} = m^2$$

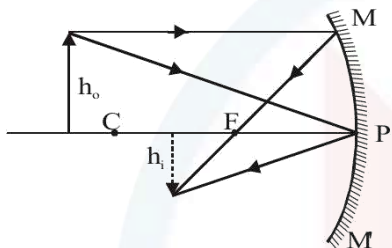
IMAGE FORMATION BY SPHERICAL MIRRORS

Concave mirror

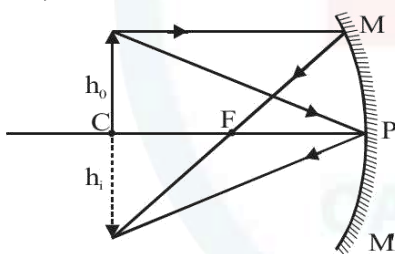
- (i) **Object:** Placed at infinity
Image: real, inverted, diminished at F
 $|m| \ll 1$ & $m < 0$



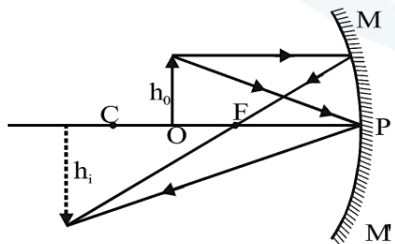
- (ii) **Object:** Placed in between infinity and C
Image: real, inverted, diminished in between C and F
 $|m| < 1$ & $m < 0$



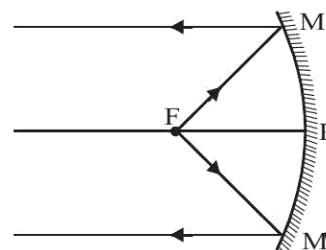
- (iii) **Object:** Placed at C
Image: real, inverted, equal at C
 $(m = -1)$



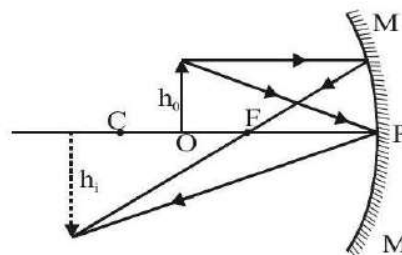
- (iv) **Object:** Placed in between F and C
Image: real, inverted, enlarged beyond C
 $|m| > 1$ & $m < 0$



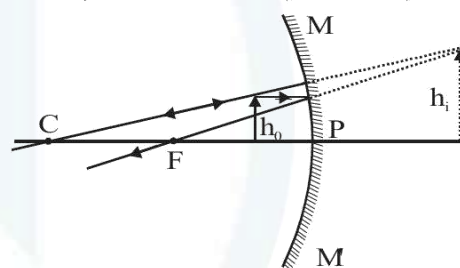
- (v) **Object:** Placed at F
Image: real, inverted, very large (assumed) at infinity
 $(m \ll -1)$



- (iv) **Object:** Placed in between F and C
Image: real, inverted, enlarged beyond C
 $|m| > 1$ & $m < 0$



- (vi) **Object:** Placed between F and P
Image: virtual, erect, enlarged and behind the mirror
 $(m > +1)$

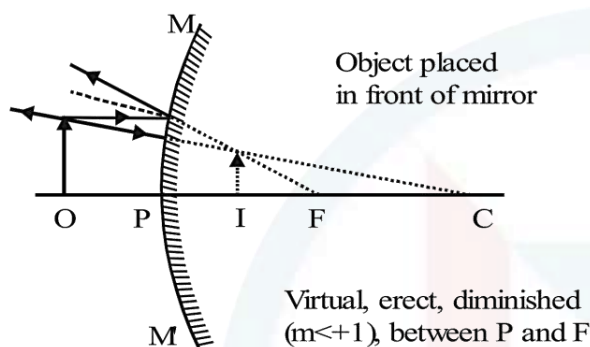
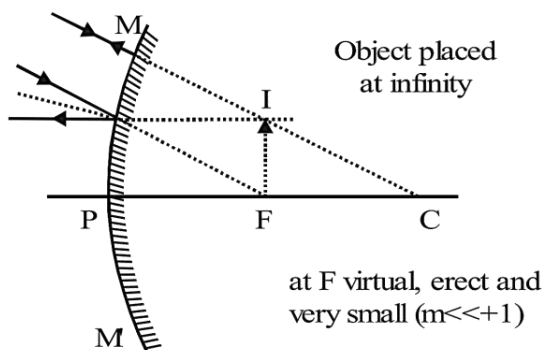


For concave mirror

Object	Image	Magnification
$-\infty$	F	$ m \ll 1$ & $m < 0$
$-\infty - C$	C - F	$ m < 1$ & $m < 0$
C	C	$m = -1$
C - F	$-\infty - C$	$ m > 1$ & $m < 0$
Just before F towards C	$-\infty$	$m \ll -1$
Just after F towards P	$+\infty$	$m \gg 1$

Convex mirror

Image is always virtual and erect, whatever be the position of the object and m is always positive.



Example: The focal length of a concave mirror is 30 cm. Find the position of the object in front of the mirror, so that the image is three times the size of the object.

Solution: As the object is in front of the mirror it is real and for real object the magnified image formed by concave mirror can be inverted (i.e., real) or erect (i.e., virtual), so there are two possibilities.

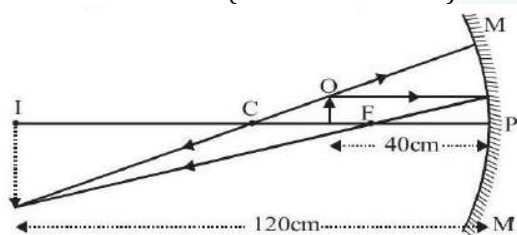
(a) If the image is inverted (i.e., real)

$$m = \frac{f}{f-u}$$

$$\Rightarrow -3 = \frac{-30}{-30-u}$$

$$\Rightarrow u = -40 \text{ cm}$$

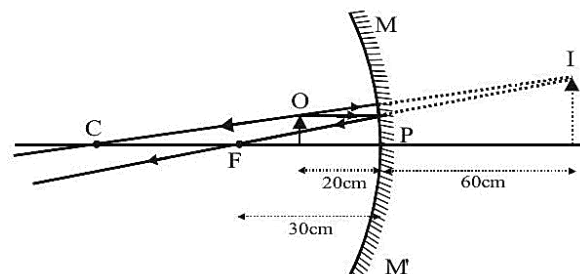
Object must be at a distance of 40 cm in front of the mirror (in between C and F).



(b) If the image is erect (i.e., virtual)

$$m = \frac{f}{f-u} \Rightarrow 3 = \frac{-30}{-30-u} \Rightarrow u = -20 \text{ cm}$$

Object must be at a distance of 20 cm in front of the mirror (in between F and P).



Example: A thin rod of length $\frac{f}{3}$ is placed along the principal axis of a concave mirror of focal length f such that its image which is real and elongated, just touches the rod. What is magnification?

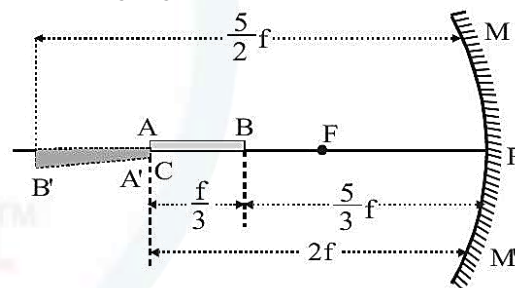
Solution: Image is real and enlarged, the object must be between C and F.

One end A' of the image coincides with the end A of rod itself.

So $v_A = u_A$, $\frac{1}{v_A} + \frac{1}{u_A} = \frac{1}{-f}$ i.e., $v_A = u_A = -2f$ so it clear that the end A is at C. $\therefore$ the length of rod is $\frac{f}{3}$

$\therefore$ Distance of the other end B from P is

$$u_B = 2f - \frac{f}{3} = \frac{5f}{3}$$



if the distance of image of end B from P is v_B then $\frac{1}{v_B} + \frac{1}{-\frac{5f}{3}} = \frac{1}{-f} \Rightarrow v_B = -\frac{5}{2}f$

$\therefore$ the length of the image

$$|v_B| - |v_A| = \frac{5}{2}f - 2f = \frac{1}{2}f \text{ and magnification}$$

$$m = \frac{|v_B| - |v_A|}{|u_B| - |u_A|} = \frac{\frac{1}{2}f}{\frac{1}{3}f} = -\frac{3}{2}$$

Negative sign implies that image is inverted with respect to object and so it is real.

Example: A concave mirror of focal length 10 cm and convex mirror of focal length 15 cm are placed facing each other 40 cm apart. A point object is placed between the mirror on their common axis and 15 cm from the concave mirror. Find the position of image produced by the reflection first at concave mirror and then at convex mirror.

Solution: For M_1 mirror O act as an object, let its image be I_1 then,

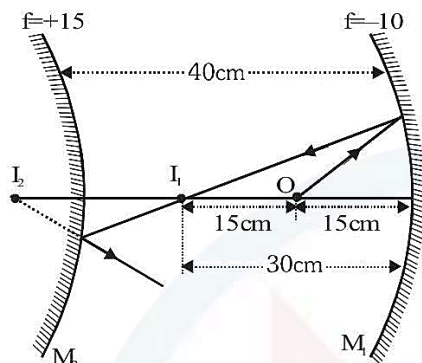
$$u = -15 \text{ cm}, f = -10 \text{ cm} \Rightarrow \frac{1}{v} + \frac{1}{-15} = \frac{1}{-10}$$

$$\Rightarrow v = -30 \text{ cm}$$

Image I_1 will act as a object for mirror M_2 its distance from mirror M_2 .

$$\text{So } u_1 = -(40 - 30) \text{ cm} = -10 \text{ cm}$$

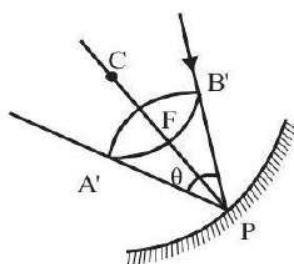
$$\frac{1}{v_1} + \frac{1}{u_1} = \frac{1}{f} \Rightarrow \frac{1}{v_1} - \frac{1}{10} = \frac{1}{15} \Rightarrow v_1 = +6 \text{ cm}$$



So final image I_2 is formed at a distance 6 cm behind the convex mirror and is virtual.

Example: The sun subtends an angle θ radians at the pole of a concave mirror of focal length f . What is the diameter of the image of the sun formed by the mirror.

Solution: Since the sun is at large distance very distant, u is very large and so $\frac{1}{u} \approx 0$

$$\therefore \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = -\frac{1}{f} \Rightarrow v = -f$$


The image of sun will be formed at the focus and will be real, inverted and diminished
 $A'B'$ = height of image and

$$\theta = \frac{\text{Arc}}{\text{Radius}} = \frac{A'B'}{FP}$$

$$\Rightarrow \theta = \frac{d}{f} \Rightarrow d = f\theta$$

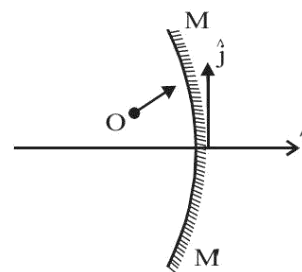
VELOCITY OF IMAGE OF MOVING OBJECT (SPHERICAL MIRROR)

(a) **Velocity component along axis (Longitudinal velocity)**

When an object is coming from infinite towards the focus of concave mirror

$$\therefore \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow -\frac{1}{v^2} \frac{dv}{dt} - \frac{1}{u^2} \frac{du}{dt} = 0 \Rightarrow \vec{v}_{ix} = -\frac{v^2}{u^2} \vec{v}_{ox}$$

$$= -m^2 \vec{v}_{ox}$$



$\left[\vec{v}_{ix} = \frac{dv}{dt} = \text{velocity of image along principal-axis}; \right.$
 $\left. \vec{v}_{ox} = \frac{du}{dt} = \text{velocity of object along principal-axis} \right]$
 $X_1, \dots, X_n$

(b) **Velocity component perpendicular to axis (Transverse velocity)**

$$m = \frac{h_i}{h_o} = -\frac{v}{u} = \frac{f}{f-u} \Rightarrow h_i = \left(\frac{f}{f-u} \right) h_o$$

$$\frac{dh_i}{dt} = \left(\frac{f}{f-u} \right) \frac{dh_o}{dt} + \frac{fh_o}{(f-u)^2} \frac{du}{dt}$$

$\left[\frac{dh_i}{dt} = \text{velocity of image } \perp^r \text{ to principal-axis} \right]$
 $\left[\frac{dh_o}{dt} = \text{velocity of object } \perp^r \text{ to principal-axis} \right]$

$$\vec{v}_{iy} = \left[m \vec{v}_{oy} + \frac{m^2 h_o}{f} \vec{v}_{ox} \right] \hat{j}$$

Note: Here principal axis has been taken to be along x-axis.

POWER OF A MIRROR

The power of a mirror is defined as $P = -\frac{1}{f(m)}$

$$= -\frac{100}{f(cm)}$$

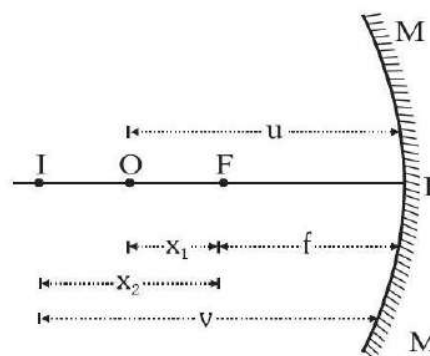
NEWTON'S FORMULA

In case if spherical mirrors if object distance (x_1) and image distance (x_2) are measured from focus instead of pole,

$$u = -(f + x_1) \text{ and } v = -(f + x_2), \text{ by } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow -\frac{1}{(f+x_2)} - \frac{1}{(f+x_1)} = -\frac{1}{f}$$

on solving $x_1 x_2 = f^2$ This is Newton's formula.

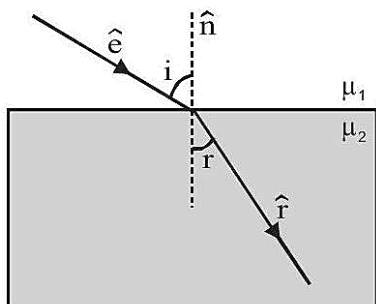


REFRACTION

Refraction is the phenomenon in which direction of propagation of light changes at the boundary when it passes from one medium to the other. In case of refraction frequency does not change.

Laws of Refraction

- (i) Incident ray, refracted ray and normal always lie in the same plane. In vector form $(\hat{e} \times \hat{n}) \cdot \hat{r} = 0$
- (ii) The product of refractive index and sine of angle of incidence at a point in a medium is constant. $\mu_1 \sin i = \mu_2 \sin r$ (Snell's law) In vector form $\mu_1 |\hat{e} \times \hat{n}| = \mu_2 |\hat{r} \times \hat{n}|$



Absolute refractive index

It is defined as the ratio of speed of light in free space 'c' to that in a given medium v. μ or $n = \frac{c}{v}$

Denser is the medium, lesser will be the speed of light and so greater will be the refractive index,

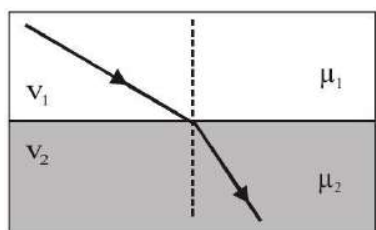
$$\therefore v_{\text{glass}} < v_{\text{water}},$$

$$\therefore \mu_{\text{G}} > \mu_{\text{W}}$$

Relative refractive index

When light passes from one medium to the other, the refractive index of medium 2 relative to 1 is written as ${}_1\mu_2$ and is defined as

$${}_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{(c/v_2)}{(c/v_1)} = \frac{v_1}{v_2}$$



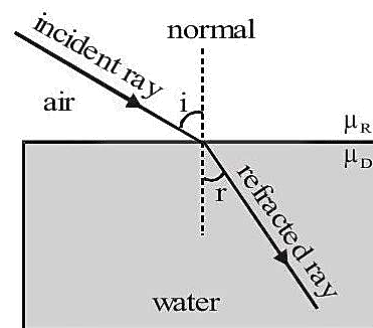
Bending of light ray

According to Snell's law, $\mu_1 \sin i = \mu_2 \sin r$

- (i) If light passes from rarer to denser medium $\mu_1 = \mu_R$ and $\mu_2 = \mu_D$

$$\text{so that } \frac{\sin i}{\sin r} = \frac{\mu_D}{\mu_R} > 1 \Rightarrow \angle i > \angle r$$

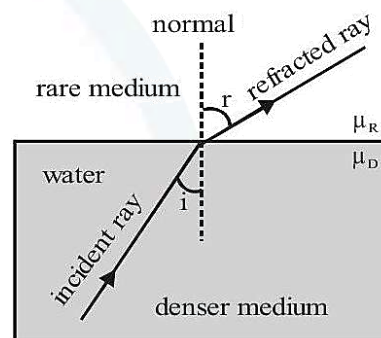
In passing from rarer to denser medium, the ray bends towards the normal.



- (ii) If light passes from denser to rarer medium $\mu_1 = \mu_D$ and $\mu_2 = \mu_R$

$$\frac{\sin i}{\sin r} = \frac{\mu_R}{\mu_D} < 1 \Rightarrow \angle i < \angle r$$

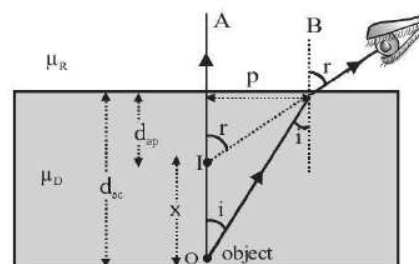
In passing from denser to rarer medium, the ray bends away from the normal.



APPARENT DEPTH AND NORMAL SHIFT

If a point object in denser medium is observed from rarer medium and boundary is plane, then from Snell's law we have $\mu_D \sin i = \mu_R \sin r$... (i) If the rays OA and OB are close enough to reach the eye. $\sin i \approx \tan i = \frac{p}{d_{ac}}$ and $\sin r \approx \tan r = \frac{p}{d_{ap}}$ here d_{ac} = actual depth, d_{ap} = apparent depth So that equation

$$(i) \text{ becomes } \mu_D = \frac{p}{d_{ac}} = \mu_R \frac{p}{d_{ap}} \Rightarrow \frac{d_{ac}}{d_{ap}} = \frac{\mu_D}{\mu_R} = \frac{\mu_1}{\mu_2}$$

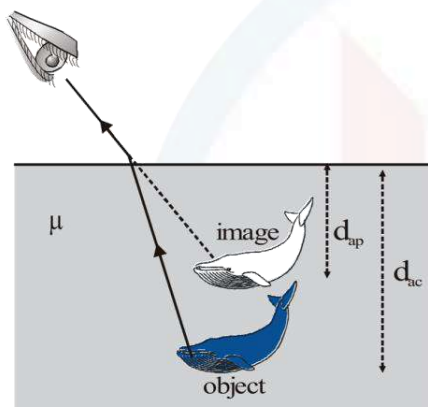
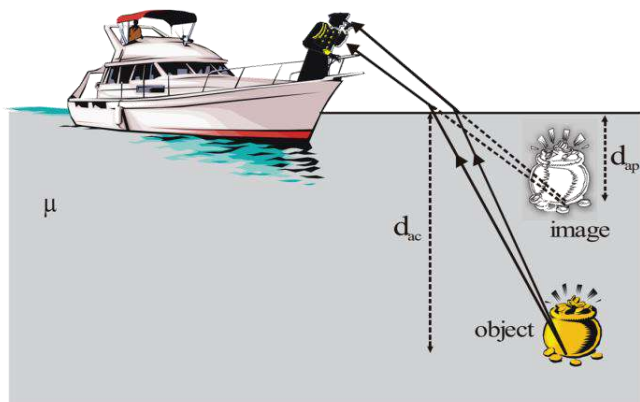


$$(\text{If } \mu_R = 1, \mu_D = \mu) \text{ then } d_{ap} = \frac{d_{ac}}{\mu} \text{ so } d_{ap} < d_{ac}$$

The distance between object and its image, called normal shift (x)

$$x = d_{ac} - d_{ap} \left[\because d_{ap} = \frac{d_{ac}}{\mu} \right]; x = d_{ac} - \frac{d_{ac}}{\mu}$$

$$= d_{ac} \left[1 - \frac{1}{\mu} \right] \dots \text{(iii) If } d_{ac} = d \text{ then } x = d \left[1 - \frac{1}{\mu} \right]$$



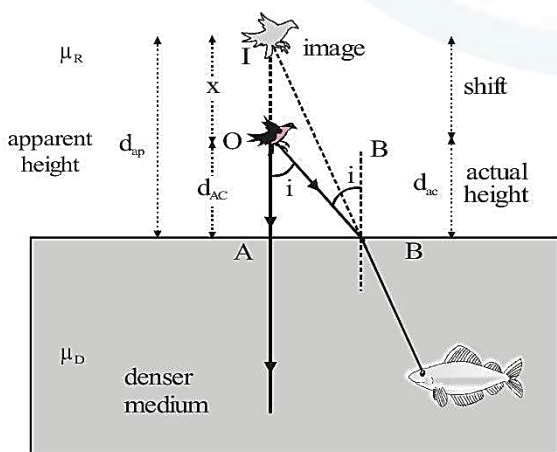
Object in a rarer medium is seen from a denser medium

$$\frac{d_{ac}}{d_{ap}} = \frac{\mu_1}{\mu_2} = \frac{\mu_R}{\mu_D} = \frac{1}{\mu} (< 1)$$

$$d_{ap} = \mu d_{ac} \text{ i.e., } d_{ap} > d_{ac}$$

A high flying object appears to be higher than in reality.

$$x = d_{ap} - d_{ac} \Rightarrow x = [\mu - 1]d_{ac}$$



LATERAL SHIFT

The perpendicular distance between incident and emergent ray is known as lateral shift.

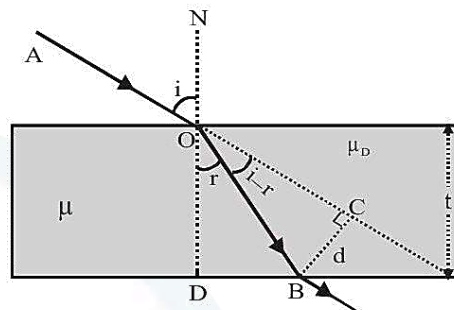
Lateral shift $d = BC$ and t = thickness of slab

$$\text{In } \triangle BOC: \sin(i - r) = \frac{BC}{OB} = \frac{d}{OB}$$

$$\Rightarrow d = OB \sin(i - r) \dots$$

$$\text{In } \triangle OBD: \cos r = \frac{OD}{OB} = \frac{t}{OB} \Rightarrow OB = \frac{t}{\cos r}$$

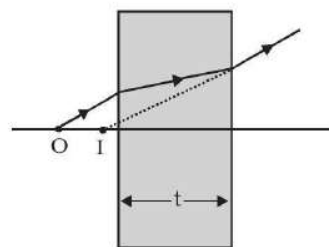
$$\text{From (i) and (ii) } d = \frac{t}{\cos r} \sin(i - r)$$



TRANSPARENT GLASS SLAB (Normal shift)

When an object is placed in front of a glass slab, it shifts the object in the direction of incident light and forms an image at a distance x .

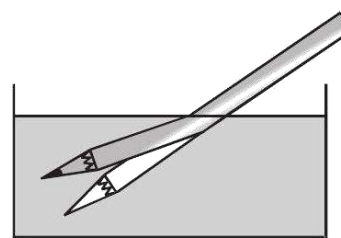
$$x = t \left[1 - \frac{1}{\mu} \right]$$



SOME ILLUSTRATIONS OF REFRACTION

Bending of an object

When a point object in a denser medium is seen from a rarer medium it appears to bend by $\frac{d}{\mu}$



Twinkling of stars

Due to fluctuations in refractive index of atmosphere the refraction becomes irregular and the light sometimes reaches the eye and sometimes it does not. This gives rise to twinkling of stars.

Example: A 20 cm thick glass slab of refractive index 1.5 is kept in front of a plane mirror. An object is kept in air at a distance 40 cm from the mirror.

Find the position of image w.r.t an observer near the object. What is effect of separation between glass slab and the mirror on image.

Solution: Shifting in object due to glass slab

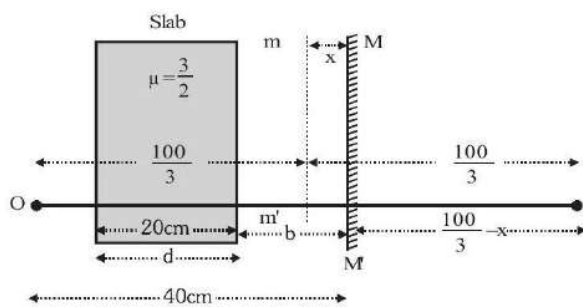
$$x = d \left(1 - \frac{1}{\mu} \right) = 20 \left[1 - \frac{1}{1.5} \right] = \frac{20}{3} \text{ cm}$$

Distance of object from mirror (as seen by mirror) = $40 - \frac{20}{3} = \frac{100}{3}$ cm

Image will be formed at a distance $\frac{100}{3}$ cm from mirror M.

Shifting in image due to glass slab = $\frac{20}{3}$ cm

So distance of image from mirror



$$= \frac{100}{3} - \frac{20}{3} = \frac{80}{3} \text{ cm}$$

Distance of image from the actual plane mirror is independent of separation b between glass slab and the mirror. If the distance is more then brightness of image will be less.

Example: If one face of a prism angle 30° and $\mu = \sqrt{2}$ is silvered, the incident ray retraces its initial path. What is the angle of incidence?

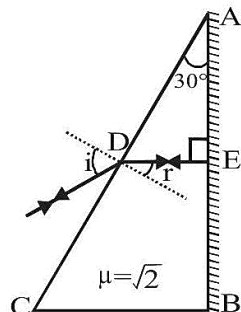
Solution: As incident ray retraces its path the ray is incident normally on the silver face of the prism as shown in figure.

Further, as in $\triangle AED$ $30^\circ + 90^\circ + \angle D = 180^\circ \Rightarrow \angle D = 60^\circ$

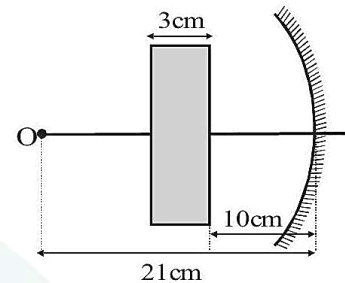
Now as by construction, $\angle D + \angle r = 90^\circ$
 $\Rightarrow \angle r = 90^\circ - 60^\circ = 30^\circ$

$\therefore$ from Snell's law at surface AC, $1 \sin i = \sqrt{2} \sin 30^\circ = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$

$\therefore \sin i = \frac{1}{\sqrt{2}} \Rightarrow i = 45^\circ$



Example: An object is placed 21 cm in front of a concave mirror of radius of curvature 20 cm. A glass slab of thickness 3 cm and refractive index 1.5 is placed closed to the mirror in space between the object and the mirror. Find the position of final image formed if distance of nearer surface of the slab from the mirror is 10 cm.



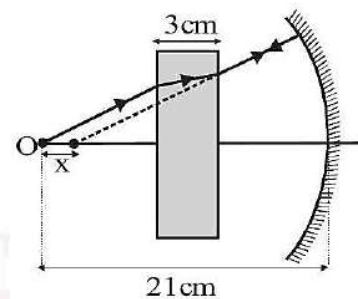
Solution: Shift by slab $x = d \left(1 - \frac{1}{\mu} \right) = 3 \left(1 - \frac{1}{1.5} \right) = 1 \text{ cm}$

for image formed by mirror

$$u = -(21 - 1) \text{ cm} = -20 \text{ cm}$$

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f} \Rightarrow \frac{1}{-20} + \frac{1}{v} = \frac{1}{-10} \Rightarrow v = -20 \text{ cm}$$

shift in the direction of light $v = -(20 + 1) = -21 \text{ cm}$.

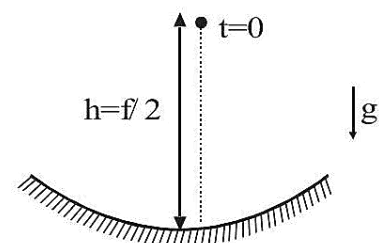


Example: A particle is dropped along the axis from a height $\frac{f}{2}$ on a concave mirror of focal length f as shown in figure. Find the maximum speed of image.

Solution: $v_{IM} = -m^2 v_{OM} = -m^2 (gt)$ where

$$M = \frac{f}{f-u} = \frac{-f}{-f + \left(\frac{f}{2} - \frac{gt^2}{2} \right)} = \frac{2f}{f+gt^2}$$

$$\Rightarrow v_1 = - \left(\frac{2f}{f+gt^2} \right)^2 (gt) = \frac{-4f^2 gt}{(f+gt^2)^2}$$

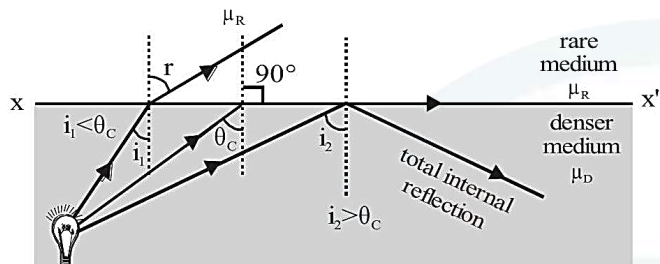


For maximum speed $\frac{dv_1}{dt} = 0 \Rightarrow t = \sqrt{\frac{f}{3g}}$

$$\Rightarrow v_{\text{Imax}} = \frac{3}{4} \sqrt{3fg}$$

TOTAL INTERNAL REFLECTION

When light ray travel from denser to rarer medium it bend away from the normal if the angle of incident is increased, angle of refraction will also increased. At a particular value of angle the refracted ray subtend 90° angle with the normal, this angle of incident is known as critical angle (θ_c). If angle of incident further increase the ray come back in the same medium this phenomenon is known as total internal reflection.



CONDITIONS

Angle of incident $>$ critical angle [$i > \theta_c$]

Light should travel from denser to rare medium $\Rightarrow$

Glass to air, water to air, Glass to water

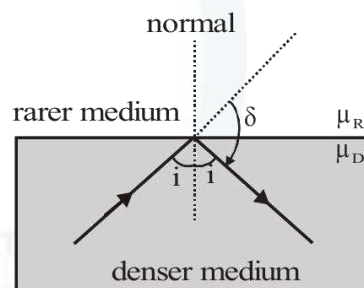
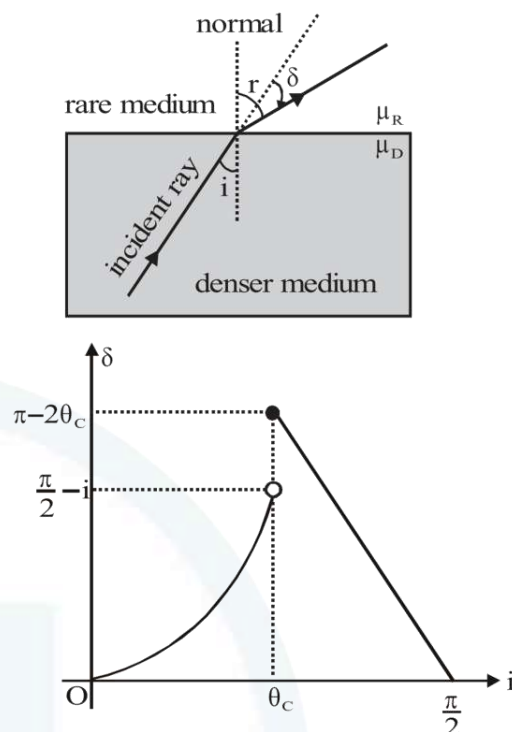
Snell's Law at boundary xx' , $\mu_D \sin \theta_c = \mu_R \sin 90^\circ \Rightarrow \sin \theta_c = \frac{\mu_R}{\mu_D}$

Graph between angle of deviation (δ) and angle of incidence (i) as rays goes from denser to rare medium

If $i < \theta_c$, $\mu_D \sin i = \mu_R \sin r$; $r = \sin^{-1} \left(\frac{\mu_D}{\mu_R} \sin i \right)$

so $\delta = r - i = \sin^{-1} \left(\frac{\mu_D}{\mu_R} \sin i \right) - i$

If $i > \theta_c$; $\delta = \pi - 2i$

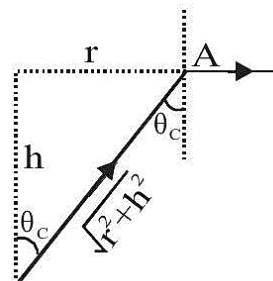


A point object is situated at the bottom of tank filled with a liquid of refractive index μ upto height h . It is found light from the source come out of liquid surface through a circular portion above the object

$$\sin \theta_c = \frac{r}{\sqrt{r^2 + h^2}} \text{ \& sin } \theta_c = \frac{1}{\mu} \Rightarrow \frac{1}{\mu} = \frac{r}{\sqrt{r^2 + h^2}} \Rightarrow \frac{1}{\mu^2} = \frac{r^2}{r^2 + h^2}$$

$$\Rightarrow \mu^2 r^2 = r^2 + h^2 \Rightarrow (\mu^2 - 1)r^2 = h^2 \Rightarrow \text{radius of circular portion}$$

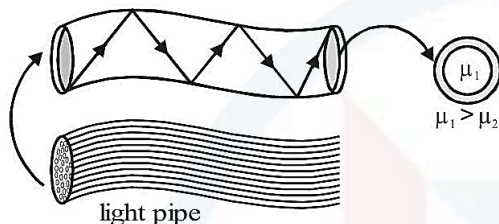
$$r = \frac{h}{\sqrt{\mu^2 - 1}} \text{ and area} = \pi r^2$$



SOME ILLUSTRATIONS OF TOTAL INTERNAL REFLECTION

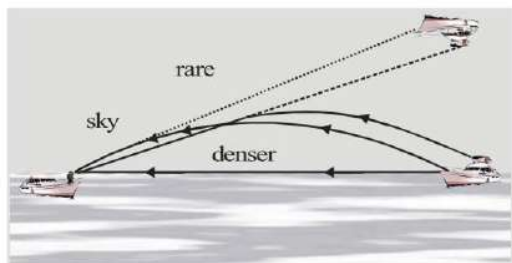
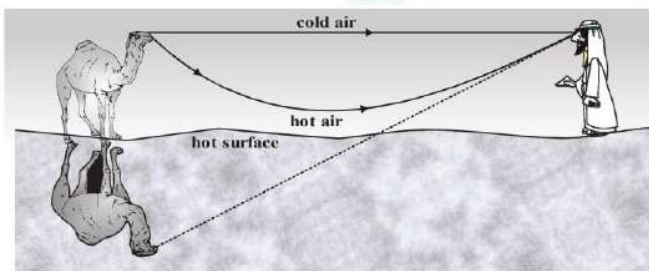
Sparkling of diamond: The sparkling of diamond is due to total internal reflection inside it. As refractive index for diamond is 2.5 so $\theta_c = 24^\circ$. Now the cutting of diamond are such that $i > \theta_c$. So TIR will take place again and again inside it. The light which beams out from a few places in some specific directions makes it sparkle.

Optical Fibre: In it light through multiple total internal reflections is propagated along the axis of a glass fibre of radius of few microns in which index of refraction of core is greater than that of surroundings.



Mirage and looming: Mirage is caused by total internal reflection in deserts where due to heating of the earth, refractive index of air near the surface of earth becomes lesser than above it. Light from distant objects reaches the surface of earth with $i > \theta_c$ so that TIR will take place and we see the image of an object along with the object as shown in figure.

Similar to 'mirage' in deserts, in polar regions 'looming' takes place due to TIR. Here μ decreases with height and so the image of an object is formed in air if ($i > \theta_c$) as shown in figure.



Example: A rectangular block of glass is placed on a printed page laying on a horizontal surface. Find the minimum value of the refractive index of glass for which the letters on the page are not visible from any of the vertical faces of the block.

Solution: The situation is depicted in figure. Light will not emerge out from the vertical face BC if at it $i > \theta_c$ or $\sin i > \sin \theta_c \Rightarrow \sin i > \frac{1}{\mu}$ [as

$$\sin \theta_c = \frac{1}{\mu} \dots (i)$$

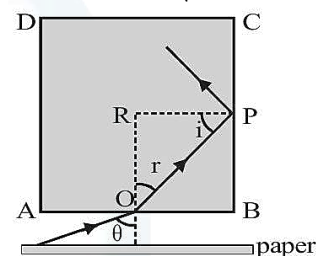
But from Snell's law at O $1 \times \sin \theta = \mu \sin r$

$$\text{And in } \triangle OPR, r + 90^\circ + i = 180^\circ \Rightarrow r + i = 90^\circ \Rightarrow r = 90^\circ - i$$

$$\text{So } \sin \theta = \mu \sin(90^\circ - i) = \mu \cos i$$

$$\Rightarrow \cos i = \frac{\sin \theta}{\mu}$$

$$\text{so } \sin i = \sqrt{1 - \cos^2 i} = \sqrt{1 - \left[\frac{\sin \theta}{\mu}\right]^2}$$



so substituting the value of $\sin i$ from equation (ii) in (i),

$$\sqrt{1 - \frac{\sin^2 \theta}{\mu^2}} > \frac{1}{\mu} \text{ i.e., } \mu^2 > 1 + \sin^2 \theta$$

$$\therefore (\sin^2 \theta)_{\max} = 1 \therefore \mu^2 > 2 \Rightarrow \mu > \sqrt{2}$$

$$\therefore \mu_{\min} = \sqrt{2}$$

REFRACTION AT TRANSPARENT CURVED SURFACE

μ_1 = refractive index of the medium in which actual incident ray lies.

μ_2 = refractive index of the medium in which actual refractive ray lies.

O = Object

P = pole

C = centre of curvature

R = PC = radius of curvature

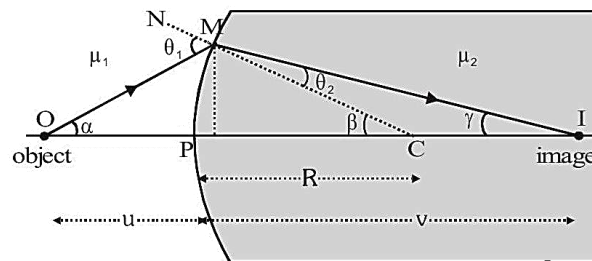
Refraction from curved surface

$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$$

if angle is very small: $\mu_1 \theta_1 = \mu_2 \theta_2$

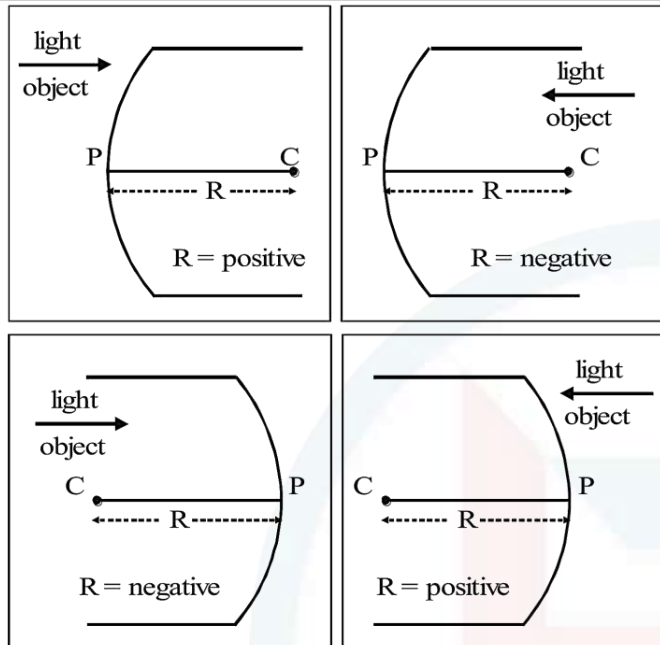
$$\text{But } \theta_1 = \alpha + \beta$$

$$\beta = \theta_2 + \gamma$$



from (i), (ii) and (iii) $\mu_1(\alpha + \beta) = \mu_2(\beta - \gamma)$
 $\Rightarrow \mu_1\alpha + \mu_1\beta = \mu_2\beta - \mu_2\gamma \Rightarrow \mu_1\alpha + \mu_2\gamma = (\mu_2 - \mu_1)\beta$
 $\Rightarrow \frac{\mu_1 PM}{-u} + \frac{\mu_2 PM}{v} = \frac{(\mu_2 - \mu_1)PM}{R} \Rightarrow \frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

SIGN CONVENTION FOR RADIUS OF CURVATURE



These are valid for all single refraction surfaces - convex, concave or plane. In case of plane refracting surface $R \rightarrow \infty$, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R} \Rightarrow \frac{\mu_2}{v} - \frac{\mu_1}{u} = 0$ i.e. $\frac{u}{v} = \frac{\mu_1}{\mu_2}$ or $\frac{d_{Ac}}{d_{Ap}} = \frac{\mu_1}{\mu_2}$

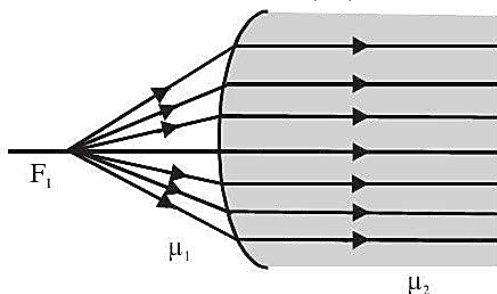
FOCAL LENGTH OF A SINGLE SPHERICAL SURFACE

A single spherical surface as two principal focus points which are as follows-

- (i) First focus: The first principal focus is the point on the axis where when an object is placed, the image is formed at infinity. That is when

$$u = f_1, v = \infty, \text{ then from } -\frac{\mu_1}{u} + \frac{\mu_2}{v} = \frac{(\mu_2 - \mu_1)}{R}$$

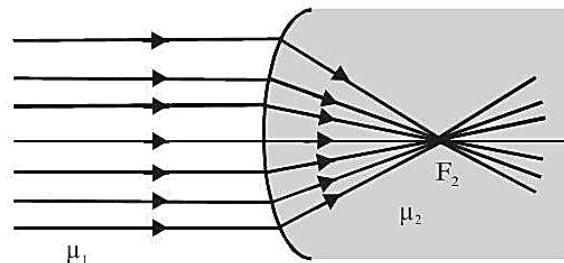
$$\text{We get } -\frac{\mu_1}{f_1} = \frac{\mu_2 - \mu_1}{R} \Rightarrow f_1 = \frac{-\mu_1 R}{(\mu_2 - \mu_1)}$$



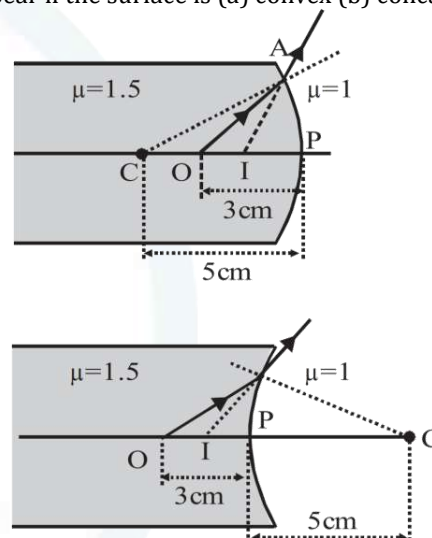
- (ii) Second focus: Similarly, the second principal focus is the point where parallel rays focus. That is $u_1 = -\infty, v_1 = f_2$, then

$$\frac{\mu_2}{f_2} = \frac{\mu_2 - \mu_1}{R}; f_2 = \frac{\mu_2 R}{(\mu_2 - \mu_1)}$$

- (iii) Ratio of Focal length: $\frac{f_1}{f_2} = -\frac{\mu_1}{\mu_2}$



Example: An air bubble in glass ($\mu = 1.5$) is situated at a distance 3 cm from a spherical surface of diameter 10 cm as shown in Figure. At what distance from the surface will the bubble appear if the surface is (a) convex (b) concave.



Solution: In case of refraction from curved surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$$

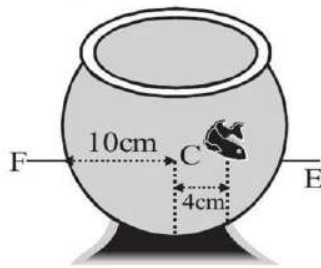
- (a) $\mu_1 = 1.5, \mu_2 = 1, R = -5$ cm and $u = -3$ cm
 $\Rightarrow \frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{1-1.5}{(-5)} \Rightarrow v = -2.5$ cm the bubble will appear at a distance 2.5 cm from the convex curved surface inside the glass.

- (b) $\mu_1 = 1.5, \mu_2 = 1, R = 5$ cm and $u = -3$ cm $\Rightarrow \frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{1-1.5}{(5)} \Rightarrow v = -1.66$ cm the bubble will appear at a distance 1.66 cm from the concave curved surface inside the glass.

Note: If the surface is plane then $R \rightarrow \infty$

$$\text{case (a) or (b) would yield } \frac{1}{v} - \frac{(1.5)}{(-3)} = \frac{(1-1.5)}{\infty} \Rightarrow v = -2 \text{ cm}$$

Example: In a thin spherical fish bowl of radius 10 cm filled with water of refractive index (4/3), there is a small fish at a distance 4 cm from the centre C as shown in Figure. Where will the fish appear to be, if seen from (a) E and (b) F (neglect the thickness of glass)?



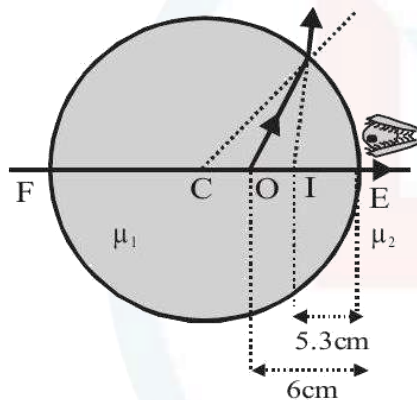
Solution: In the case of refraction from curved surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$$

- (a) Seen from E $\mu_1 = \frac{4}{3}$, $\mu_2 = 1$, $R = -10$ cm &
 $u = -(10 - 4) = -6$ cm

$$\Rightarrow \frac{1}{v} - \frac{\frac{4}{3}}{-6} = \frac{1 - \frac{4}{3}}{-10} \Rightarrow v = \frac{90}{17} = -5.3 \text{ cm}$$

i.e., fish will appear at a distance 5.3 cm from E towards F

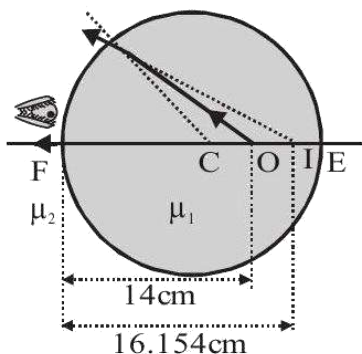


(lesser than actual distance, i.e., 6 cm)

- (b) Seen from F $\mu_1 = \frac{4}{3}$, $\mu_2 = 1$, $R = -10$ cm and
 $u = -(10 + 4) = -14$ cm

$$\Rightarrow \frac{1}{v} - \frac{\frac{4}{3}}{-14} = \frac{1 - \frac{4}{3}}{-10} \Rightarrow v = \frac{-210}{13} = -16.154 \text{ cm}$$

so fish will appear at a distance 16.154 cm from F toward E

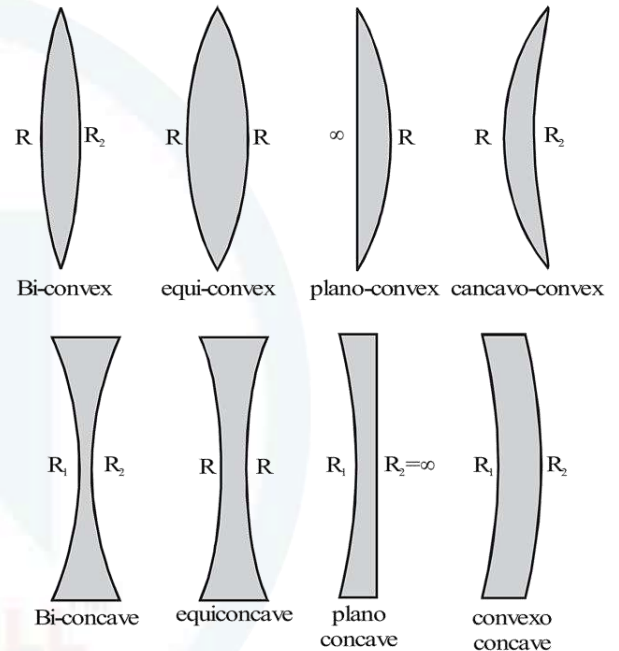


(more than actual distance, i.e., 14 cm)

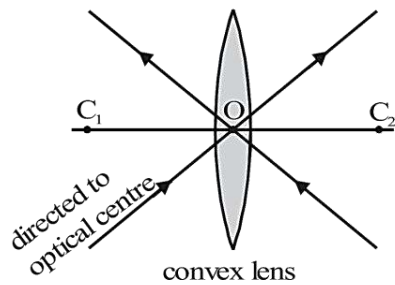
LENS

A lens is a piece of transparent material with two refracting surfaces such that at least one is curved and refractive index of its material is different from that of the surroundings.

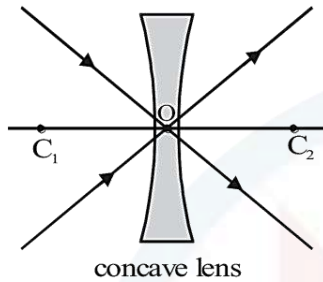
A thin spherical lens with refractive index greater than that of surroundings behaves as a convergent or convex lens, i.e., converges parallel rays if its central (i.e. paraxial) portion is thicker than marginal one. However if the central portion of a lens is thinner than marginal, it diverges parallel rays and behaves as divergent or concave lens. This is how we can identify convergent and divergent lenses.



Optical Centre: O is a point for a given lens through which any ray passes undeviated



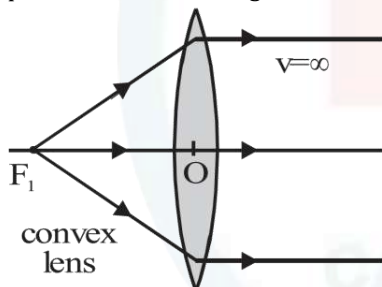
convex lens



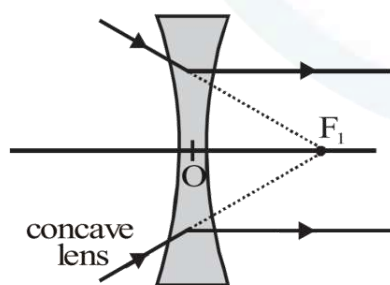
concave lens

Principal Axis: C_1C_2 is a line passing through optical centre and perpendicular to the lens.

Principal Focus: A lens has two surfaces and hence two focal points. First focal point is an object point on the principal axis for which image is formed at infinity.

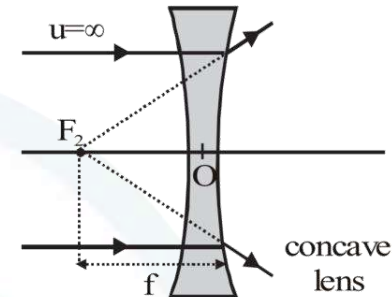
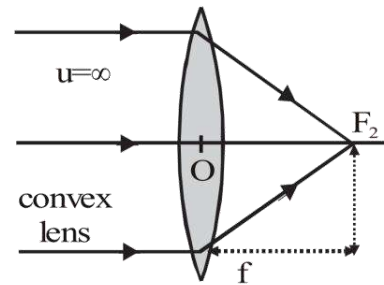


convex lens



concave lens

While second focal point is an image point on the principal axis for which object lies at infinity



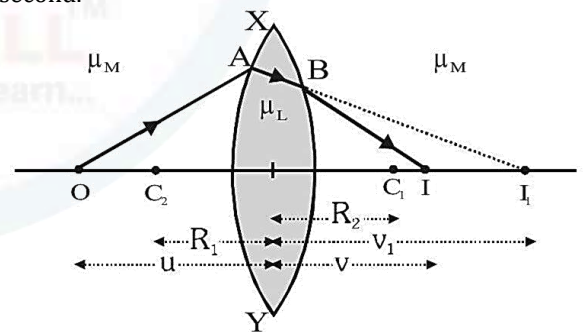
Focal Length f : is defined as the distance between optical centre of a lens and the point where the parallel beam of light converges or appears to converge.

Aperture: In reference to a lens, aperture means the effective diameter. Intensity of image formed by a lens which depends on the light passing through the lens will depend on the square of aperture, i.e., $\text{Intensity} \propto (\text{Aperture})^2$

LENS-MAKER'S FORMULA

In case of image formation by a lens

Image formed by first surface acts as object for the second.



So, from the formula of refraction at curved surface.

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\text{For first surface } A \quad \frac{\mu_L}{v_1} - \frac{\mu_M}{u} = \frac{\mu_L - \mu_M}{R_1} \dots (i)$$

$$[\because \mu_2 = \mu_L, \mu_1 = \mu_M]$$

$$\text{For second surface } B \quad \frac{\mu_M}{v} - \frac{\mu_L}{v_1} = \frac{\mu_M - \mu_L}{R_2} = -\frac{\mu_L - \mu_M}{R_2}$$

$$[\because \mu_2 = \mu_M, \mu_1 = \mu_L, \mu_1 = \mu_2, u \rightarrow v_1]$$

By adding (i) and (ii)

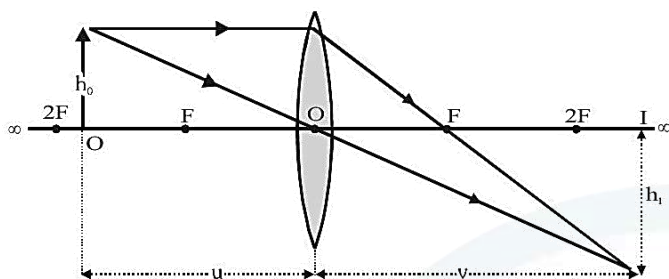
$$\mu_M \left[\frac{1}{v} - \frac{1}{u} \right] = (\mu_L - \mu_M) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\Rightarrow \frac{1}{v} - \frac{1}{u} = \frac{\mu_L - \mu_M}{\mu_M} \left[\frac{1}{R_1} - \frac{1}{R_2} \right] = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right].$$

$$\therefore \mu = \frac{\mu_L}{\mu_M}$$

Now if object is at infinity, Image will be formed at the focus, $u = -\infty, v = f$

$$\text{So } \frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$



This is known as lens makers formula by equating (iii) and (iv) $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ this is known as lens formula

$$\text{Magnification: } m = \frac{\text{height of image}}{\text{height of object}} = \frac{h_1}{h_0} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$$

RULES FOR IMAGE FORMATION

A ray passing through optical centre proceeds undeviated through the lens

A ray passing through first focus or directed towards it, after refraction from the lens, becomes parallel to the principal axis.

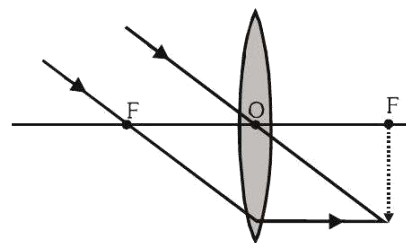
A ray passing parallel to the principal axis after refraction through the lens passes or appears to pass through F_2

For Convergent or convex lens

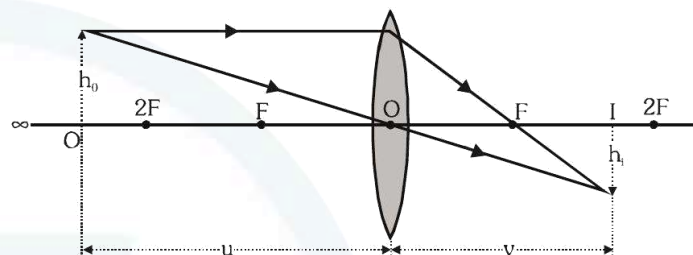
Object	Image	Magnification
$-\infty$	F	$ m \ll 1 \text{ \& } m < 0$
$-\infty - 2F$	$F - 2F$	$ m < 1 \text{ \& } m < 0$
$2F$	$2F$	$m = -1$
$F - 2F$	$\infty - 2F$	$ m > 1 \text{ \& } m < 0$
Just before F towards C	$+\infty$	$m \ll -1$
Just before F towards P	$-\infty$	$m \gg 1$
$F - 0$	In front of lens	$m > 1$

IMAGE FORMATION FOR CONVEX LENS (CONVERGENT LENS)

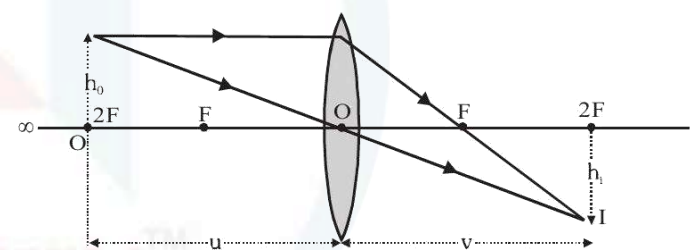
- (i) Object is placed at infinity
Image:
at F real inverted very small in size $|m| \ll 1 \text{ \& } m < 0$



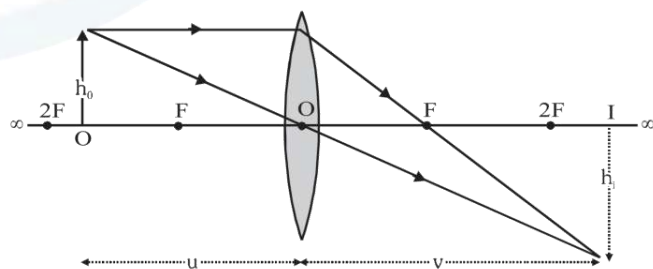
- (ii) Object is placed in between $\infty - 2F$
Image:
real ($F - 2F$) inverted small in size (diminished)
 $|m| < 1 \text{ \& } m < 0$



- (iii) Object is placed at $2F$
Image:
real (at $2F$) inverted equal (of same size)
($m = -1$)



- (iv) Object is placed in between $2F - F$
Image:
real ($2F - \infty$) inverted enlarged
 $|m| > 1 \text{ \& } m < 0$



- (v) Object is placed in between $F - 0$
Image: virtual (in front of lens) erected enlarge
($m > +1$)

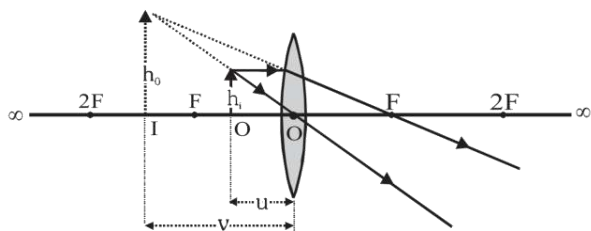


IMAGE FORMATION FOR CONCAVE LENS (DIVERGENT LENS)

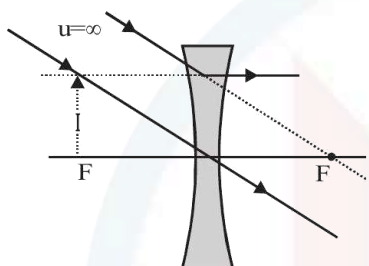
Image is virtual, diminished, erect, towards the object,

$m = +ve$

- (i) Object is placed at infinity

Image:

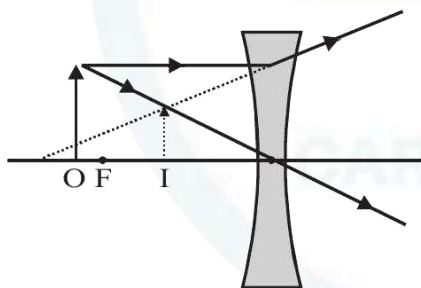
At F virtual erected
diminished ($m \ll +1$)



- (ii) Object is placed in front of lens

Image:

between F and optical centre virtual erected
diminished ($m < +1$)



Sign convention for object/image for lens

Real object	$u - ve$
Real image	$v + ve$
Virtual object	$u + ve$
Virtual image	$v - ve$

POWER OF LENS

Reciprocal of focal length in meter is known as power of lens.

SI UNIT: dioptre (D)

Power of lens: $P = \frac{1}{f(m)} = \frac{100}{f(cm)}$ dioptre [in air]

COMBINATION OF LENSES

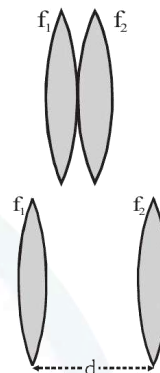
Two thin lens are placed in contact to each other power of combination. $P = P_1 + P_2 \Rightarrow \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$

Use sign convention when solve numericals

Two thin lens are placed in at a small distance d (provided incident rays are parallel to principal axis).

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \Rightarrow P = P_1 + P_2 - dP_1 P_2$$

Use sign convention when solving numericals

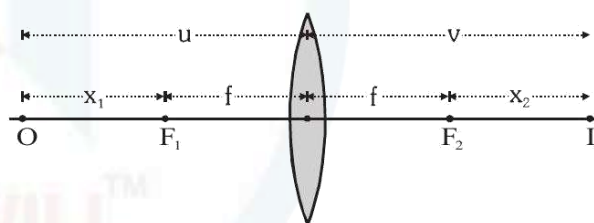


Newton's Formula

$$f = \sqrt{x_1 x_2}$$

x_1 = distance of object from focus.

x_2 = distance of image from focus.



SOME SPECIAL CASES

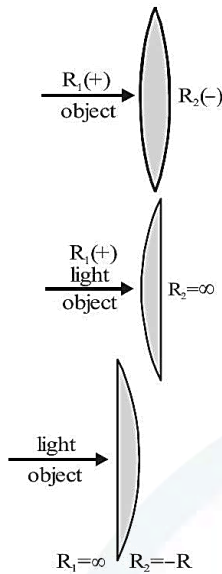
- (i) The focal length of equiconvex lens placed in air refractive index of lens

$\mu_L = \mu$ refractive index of medium $\mu_M = 1$

$R_1 = +R, R_2 = -R$

- (ii) Focal length of planoconvex lens placed in air

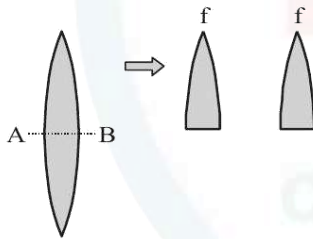
$$\frac{1}{f} = (\mu - 1) \left[\frac{1}{R} - \frac{1}{\infty} \right] \Rightarrow \text{Focal length } f = \frac{R}{(\mu - 1)}$$



If object is placed towards plane surface

$$\frac{1}{f} = (\mu - 1) \left[\frac{1}{\infty} - \left(-\frac{1}{R} \right) \right] \Rightarrow \text{Focal length } f = \frac{R}{(\mu - 1)}$$

- (iii) If an equiconvex lens of focal length f is cut into equal parts by a horizontal plane AB then the focal length of each part will be equal to that of initial lens. Because μ , R_1 and R_2 will remain unchanged. Only intensity will be reduced.



$\therefore$ intensity $I \propto (\text{apertures})^2$

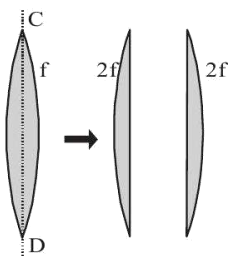
$\therefore$ intensity through a single part will be reduced

- (iv) If the same lens is cut into equal parts by a vertical plane CD the focal length of each part will be double of initial value but intensity will remain unchanged.

For equiconvex lens $\frac{1}{f} = \frac{(\mu-1)2}{R}$ For plano convex lens

$$\frac{1}{f_1} = \frac{\mu-1}{R}$$

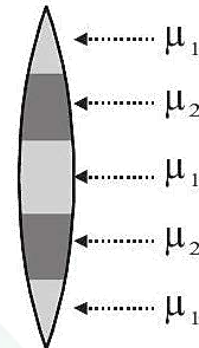
$$\text{So } \frac{1}{f} = \frac{2}{f_1} \Rightarrow f_1 = 2f \Rightarrow \text{Focal length of each part} = 2f$$



focal length of each part become $2f$
(focal length of original lens)

- (v) If a lens is made of number of layers of different refractive index for a given wavelength then no. of images is equal to number of refractive index, as $\frac{1}{f} \propto (\mu - 1)$

In figure number of images = 2

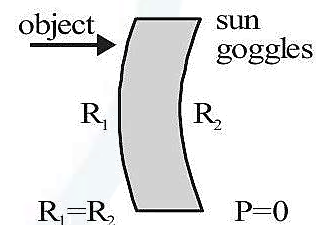


- (vi) Focal length of lens depends on wavelength. $\therefore \frac{1}{f} \propto (\mu - 1) \propto \frac{1}{\lambda} [f \propto \lambda] f_R > f_V$ will be reduced but complete image will be formed.

- (viii) Sun-goggles:
radius of curvature of two surfaces is equal with centre on the same side

$$R_1 = R_2 = +R \text{ so } \frac{1}{f} = (\mu - 1) \left[\frac{1}{R} - \frac{1}{R} \right]$$

$$\Rightarrow \frac{1}{f} = 0 \Rightarrow f = \infty \text{ and } P = 0 \Rightarrow \text{sun goggles have no power}$$



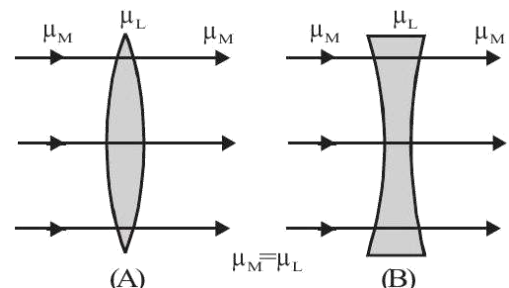
- (ix) If refractive index of medium < Refractive index of lens If $\mu_M < \mu_L$ then $f = +ve$
Convex lens behave as convex lens.
While concave lens behave as concave lens.

- (x) Refractive index of medium = Refractive index of lens

$$(\mu_M = \mu_L) \frac{1}{f} = \left(\frac{\mu_L}{\mu_M} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right); \frac{1}{f} = 0$$

$$\Rightarrow f = \infty \text{ \& } P = 0$$

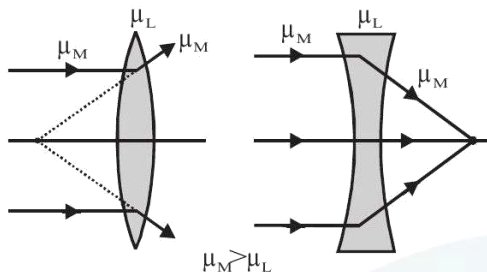
Lens will behave as plane glass plate



- (xi) Refractive index of medium > Refractive index of lens

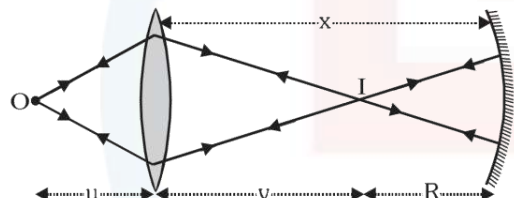
$$\mu_M > \mu_L \Rightarrow \frac{\mu_L}{\mu_M} < 1 [f \text{ will be negative}]$$

convex lens will behave as concave lens and concave lens will behave as convex lens. If a air bubble is formed in water it behaves as concave lens.

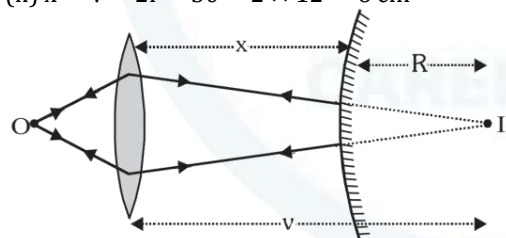


Example: A point source S is placed at distance of 15 cm from a converging lens of focal length 10 cm. Where should a (i) concave mirror (ii) convex mirror of focal length 12 cm be placed so that real image is formed on object itself.

Solution: $u = -15 \text{ cm}, f = +10 \text{ cm}; \frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
 $\Rightarrow \frac{1}{v} - \frac{1}{(-15)} = \frac{1}{10} \Rightarrow v = 30 \text{ cm}$
 (i) $x = v + 2f \Rightarrow 30 + 2 \times 12 = 54 \text{ cm}$



(ii) $x = v - 2f = 30 - 2 \times 12 = 6 \text{ cm}$



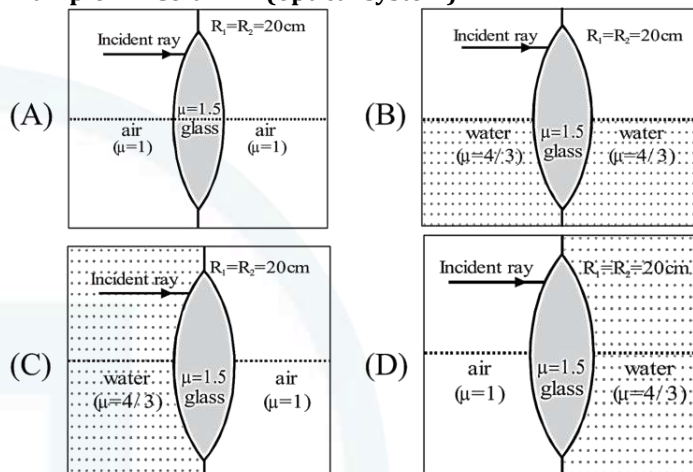
Example: A convex lens of focal length f is producing real image which is $\frac{1}{n}$ times of the size of the object. Find out position of the object.

Solution: Image is real so
 From lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = \frac{1}{u}$
 $\Rightarrow \frac{1}{-\frac{u}{n}} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{-(n+1)}{u} = \frac{1}{f} \Rightarrow u = -f(1+n)$

Example: (a) If $f = +0.5 \text{ m}$, what is the power of the lens?
 (b) The radii of curvature of the faces of a double convex lens are 10 cm and 15 cm. Its focal length is 12 cm. What is the refractive index of glass?
 (c) A convex lens has 20 cm focal length in air. What is the focal length in water? (Refractive index of air-water = 1.33, refractive index for air glass is 1.5)

Solution: (a) $P = \frac{1}{f(m)} = \frac{1}{0.5} = +2D$
 (b) $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \Rightarrow \frac{1}{12} = (\mu - 1)$
 $\left(\frac{1}{10} - \frac{1}{-15} \right) \Rightarrow \mu = 1.5$
 (c) $\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$ so $\frac{1}{f} \propto (\mu - 1); \frac{f_w}{f_a} = \frac{(\mu_a - 1)}{(\mu_w - 1)} \Rightarrow f_w = \frac{(1.5 - 1)}{(1.33 - 1)} \times 20 = 78.2 \text{ cm}$

Example: Column I (optical system)



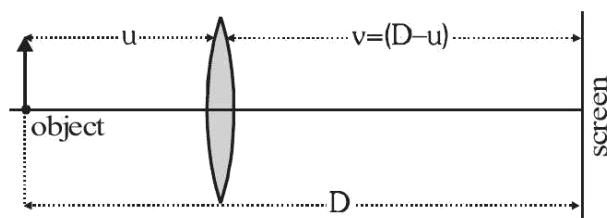
Column II (focal length)

- (P) 80 cm
 (Q) 40 cm
 (R) 30 cm
 (S) 20 cm

Solution: Ans. (A) -S (B) -P (C) -R
 For (A): $\frac{1}{f} = (\mu - 1) \left(\frac{1}{p_1} + \frac{1}{p_2} \right)$
 $= (1.5 - 1) \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{20} \Rightarrow f = 20 \text{ cm}$
 For (B): $\frac{1}{f} = \left(\frac{1.5}{4/3} - 1 \right) \left(\frac{1}{20} + \frac{1}{20} \right) = \frac{1}{80}$
 $\Rightarrow f = 80 \text{ cm}$
 For (C): $\frac{1.5}{v_1} - \frac{4}{\infty} = \left(1.5 - \frac{4}{3} \right) \left(\frac{1}{20} \right) \& \frac{1}{f} - \frac{1.5}{v_1}$
 $= (1 - 1.5) \left(\frac{1}{20} \right) \Rightarrow f = 30 \text{ cm}$
 For (D): $\frac{1.5}{v_1} - \frac{1}{\infty} = (1.5 - 1) \left(\frac{1}{20} \right) \& \frac{4}{3} - \frac{1.5}{v_1}$
 $= \left(\frac{4}{3} - 1.5 \right) \left(-\frac{1}{20} \right) \Rightarrow f = 40 \text{ cm}$

DISPLACEMENT METHOD

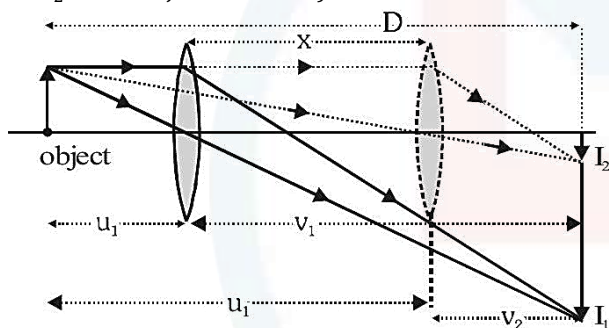
It is used for determination of focal length of convex lens in laboratory. A thin convex lens of focal length f is placed between an object and a screen fixed at a distance D apart. If $D > 4f$ there are two position of lens at which a sharp image of the object is formed on the screen



By lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{D-u} - \frac{1}{-u} = \frac{1}{f}$
 $\Rightarrow u^2 - Du + Df = 0 \Rightarrow u = \frac{D \pm \sqrt{D(D-4f)}}{2}$ there are three possibilities

- for $D < 4f$ u will be imaginary hence physically no position of lens is possible
- for $D = 4f$ $u = \frac{D}{2} = 2f$ so only one position of lens is possible and since $v = D - u = 4f - 2f = u = 2f$
- for $D > 4f$ $u_1 = \frac{D - \sqrt{D(D-4f)}}{2}$ and $u_2 = \frac{D + \sqrt{D(D-4f)}}{2}$

So there are two positions of lens for which real image will be formed on the screen. (for two distances u_1 and u_2 of the object from lens)



If the distance between two positions of lens is x then
 $x = u_2 - u_1 = \frac{D + \sqrt{D(D-4f)}}{2} - \frac{D - \sqrt{D(D-4f)}}{2}$
 $= \sqrt{D(D-4f)} \Rightarrow x^2 = D^2 - 4Df$
 $\Rightarrow f = \frac{D^2 - x^2}{4D}$

Distance of image corresponds to two positions of the lens:

$$v_1 = D - u_1 = D - \frac{1}{2} [D - \sqrt{D(D-4f)}]$$

$$= \frac{1}{2} [D + \sqrt{D(D-4f)}] = u_2 \Rightarrow v_1 = u_2$$

$$v_2 = D - u_2 = D - \frac{1}{2} [D + \sqrt{D(D-4f)}]$$

$$= \frac{1}{2} [D - \sqrt{D(D-4f)}] = u_1 \Rightarrow v_2 = u_1$$

for two positions of the lens distances of object and image are interchangeable.

Now $x_2 = u_2 - u_1$ and $D = v_1 + u_1 = u_2 + u_1$
 $[\because v_1 = u_2]$

so $u_1 (= v_2) = \frac{D-x}{2}$ and $v_1 = \frac{D+x}{2} = u_2$;

$$m_1 = \frac{I_1}{O} = \frac{v_1}{u_1} = \frac{\frac{D+x}{2}}{\frac{D-x}{2}} \text{ and } m_2 = \frac{I_2}{O} = \frac{v_2}{u_2} = \frac{\frac{D-x}{2}}{\frac{D+x}{2}}$$

$$\text{Now } m_1 \times m_2 = \frac{\frac{D+x}{2}}{\frac{D-x}{2}} \times \frac{\frac{D-x}{2}}{\frac{D+x}{2}} \Rightarrow \frac{I_1 I_2}{O^2} = 1 \Rightarrow O = \sqrt{I_1 I_2}$$

Example: A convex lens is placed between an object and a screen which are at a fixed distance apart for one position of the lens. The magnification of the image obtained on the screen is m_1 . When the lens is moved by a distance d the magnification of the image obtained on the same screen is m_2 , Find the focal length of the lens.

Solution: If D is the distance between the object and the screen, d the separation of the two position of lens throwing two images on the screen then
 $m_1 = \frac{(D+d)}{(D-d)}$ and $m_2 = \frac{(D-d)}{(D+d)} \therefore m_1 - m_2 = \frac{4Dd}{D^2 - d^2}$
 but $\frac{D^2 - d^2}{4D} = f$ so $m_1 - m_2 = \frac{d}{f} \Rightarrow f = \frac{d}{m_1 - m_2}$

Example: In a displacement method using lens, we obtain two images for separation of the lens d . One image is magnified as much as the other is diminished. If m is the magnifications of one image, find the focal length of the lens.

Solution: From above question $f = \frac{d}{m_1 - m_2}$
 here if m_1 is taken as m , $m_2 = \frac{1}{m}$,

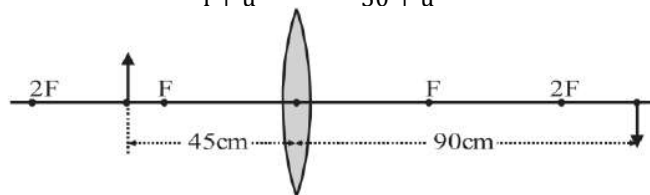
so f becomes $\frac{md}{(m^2 - 1)}$

Example: In the displacement method the distance between the object and the screen is 70 cm and the focal length of the lens is 16 cm, find the separations of the magnified and diminished image position of the lens.

Solution: $d = \sqrt{D^2 - 4fd} = \sqrt{(70)^2 - 4 \times 16 \times 70}$
 $= \sqrt{420} = 20.5 \text{ cm}$

Example: An object 25 cm high is placed in front of a convex lens of focal length 30 cm. If the height of image formed is 50 cm, find the distance between the object and the image (real and virtual)?

Solution: As object is in front of the lens, it is real and as $h_1 = 25 \text{ cm}$, $f = 30 \text{ cm}$, $h_2 = -50 \text{ cm}$;
 $m = \frac{h_2}{h_1} = \frac{-50}{25} = -2$
 $m = \frac{f}{f + u} \Rightarrow -2 = \frac{30}{30 + u} \Rightarrow u = -45 \text{ cm}$



$$\Rightarrow m = \frac{v}{u} \Rightarrow -2 = \frac{v}{-45} \Rightarrow v = 90 \text{ cm}$$

As in this situation object and image are on opposite sides of lens, the distance between object and image

$d_1 = u + v = 45 + 90 = 135 \text{ cm}$. If the image is erect (i.e., virtual)

$$m = \frac{f}{f+u} \Rightarrow 2 = \frac{30}{30+u} \Rightarrow u = -15 \text{ cm} \Rightarrow m = -\frac{v}{u} \Rightarrow 2 = \frac{-v}{-15} \Rightarrow v = 30 \text{ cm}$$

As in the situation both image and object are in front of the lens, the distance between object and image $d_2 = v - u = 30 - 15 = 15 \text{ cm}$

COMBINATION OF LENSES AND MIRRORS

When several lenses or mirrors are used, the image formation is considered one after another in steps. The image formed by the lens facing the object serves as an object for the next lens or mirror, the image formed by the second lens acts as an object for the third, and so on. The total magnification in such situations will be given by

$$m = \frac{I}{O} = \frac{I_1}{O} \times \frac{I_2}{I_1} \times \dots \Rightarrow m = m_1 \times m_2 \times \dots$$

Power of Lens [in air] $P_L = \frac{1}{f_L}$

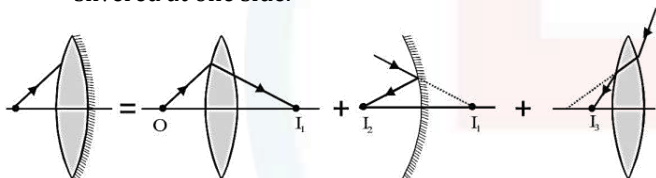
Converging lens $P_L = +ve$ Diverging lens $P_L = -ve$

Power For mirror $P_M = -\frac{1}{f_m}$

Convex mirror $P_M = -ve$ Concave mirror $P_M = +ve$

SILVERING OF LENS

Calculate equivalent focal length of a equiconvex lens silvered at one side.



$$P = P_L + P_M + P_L = 2P_L + P_M$$

$$\frac{1}{F} = \frac{1}{f_L} + \frac{1}{f_m} + \frac{1}{f_L} = \frac{2}{f_L} + \frac{1}{f_m} = \frac{2(\mu-1) \times 2}{R} + \frac{2}{R}$$

$$= \frac{4\mu-4+2}{R} \Rightarrow F = \frac{R}{4\mu-2}$$

Example: Calculate equivalent focal length of plano convex lens for following case:-

- When curved surface is silvered.
- When plane surface is silvered.

Solution:

(i)

$$P = 2P_L + P_M$$

$$\Rightarrow \frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_m} \Rightarrow \frac{1}{F} = \frac{2(\mu-1)}{R} + \frac{2}{R}$$

$$\Rightarrow \frac{1}{F} = \frac{2\mu}{R} \Rightarrow F = \frac{R}{2\mu}$$

(ii)

$$P = 2P_L + P_M$$

$$\Rightarrow \frac{1}{F} = \frac{2}{f_L} + \frac{1}{f_m} \Rightarrow \frac{1}{F} = \frac{2(\mu-1)}{R} + \frac{1}{\infty}$$

$$\Rightarrow F = \frac{R}{2(\mu-1)}$$

Example:

The radius of curvature of the convex face of a plano-convex lens is 12 cm and its refractive index is 1.5.

- Find the focal length of this lens. The plane surface of the lens is now silvered.
- At what distance from the lens will parallel rays incident on the convex face converge?
- Sketch the ray diagram to locate the image, when a point object is placed on the axis 20 cm from the lens.
- Calculate the image distance when the object is placed as in (c).

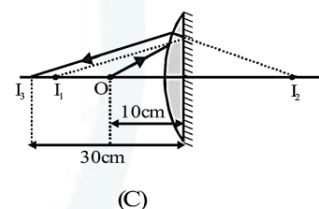
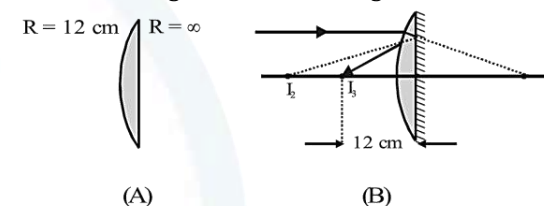
Solution:

(a) As for a lens, by lens-maker's formula

$$\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

Here $\mu = 1.5$; $R_1 = 12 \text{ cm}$ and $R_2 = \infty$

So $\frac{1}{f} = (1.5 - 1) \left[\frac{1}{12} - \frac{1}{\infty} \right]$ i.e. $f = 24 \text{ cm}$ i.e., the lens as convergent with focal length 24 cm.



(b) As light after passing through the lens will be incident on the mirror which will reflect it back through the lens again, so $P = P_L + P_M + P_L = 2P_L + P_M$ But $P_L = \frac{1}{f_L} = \frac{1}{0.24}$ and $P_M = -\frac{1}{\infty} = 0$ [as $f_m = \frac{R}{2} = \infty$]

So $P = 2 \times \frac{1}{0.24} + 0 = \frac{1}{0.12}$ D. The system is equivalent to a concave mirror of focal length

$$F, P = -\frac{1}{F}$$

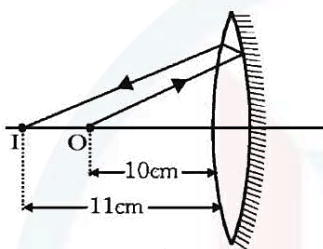
i.e., $F = -\frac{1}{P} = -0.12 \text{ m} = -12 \text{ cm}$ i.e., the rays will behave as a concave mirror of focal length 12 cm.

So as for parallel incident rays $u = -\infty$, from mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ we have $\frac{1}{v} + \frac{1}{-\infty} = \frac{1}{-12}$
 $\Rightarrow v = -12 \text{ cm}$ i.e., parallel incident rays will focus will at a distance of 12 cm in front of the lens as shown in Figure (c) and (d) When object is at 20 cm in front of the given silvered lens which behaves as a concave mirror of focal length 12 cm, from mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ we have $\frac{1}{v} + \frac{1}{-20} = \frac{1}{-12}$

$\Rightarrow v = -30$ cm i.e., the silvered lens will form image at a distance of 30 cm in front of it as shown in fig. (C)

Example: A pin is placed 10 cm in front of a convex lens of focal length 20 cm, made of material having refractive index 1.5. The surface of the lens farther away from the pin is silvered and has a radius of curvature 22 cm. Determine the position of the final image. Is the image real or virtual?

Solution: As radius of curvature of silvered surface is 22 cm, so $f_M = \frac{R}{2} = \frac{-22}{2} = -11$ cm = -0.11 m and hence, $M = -\frac{1}{f_M} = \frac{1}{-0.11} = \frac{1}{0.11}$ D Further as the focal length of lens is 20 cm, i.e., 0.20 m its power will be given by:



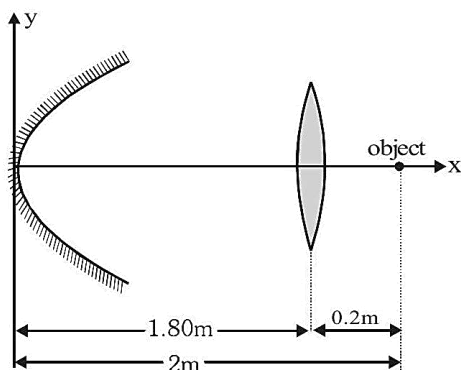
$$P_L = \frac{1}{f_L} = \frac{1}{0.20} \text{ D.}$$

Now as in image formation, light after passing through the lens will be reflected back by the curved mirror through the lens again $P = P_L + P_M + P_L = 2P_L + P_M$ i.e. $P = \frac{2}{0.20} + \frac{1}{0.11} = \frac{210}{11}$ D.

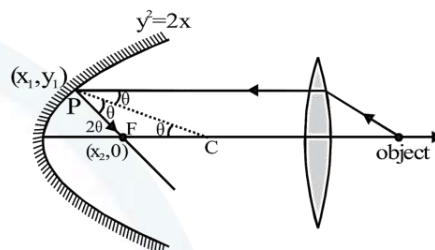
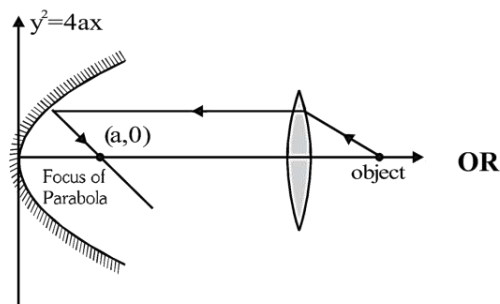
So the focal length of equivalent mirror $F = -\frac{1}{P} = -\frac{11}{210}$ m = $-\frac{110}{21}$ cm i.e., the silvered lens behave as a concave mirror of focal length (110/21)cm. So for object at a distance 10 cm in front of it, $\frac{1}{v} + \frac{1}{-10} = -\frac{21}{110}$

i.e., $v = -11$ cm i.e., image will be 11 cm in front of the silvered lens and will be real as shown in Figure.

Example: A point object is kept at a distance of 2 m from a parabolic reflecting surface $y^2 = 2x$. An equiconvex lens is kept at a distance of 1.80 m from the parabolic surface. The focal length of the lens is 20 cm. Find the position from origin of the image in cm, after reflection from the surface.



Solution:



Comparing with $y^2 = 4ax$

$$\Rightarrow a = 0.5$$

PC is a normal so

$$\tan(\pi - \theta) = \frac{-1}{(dy/dx)_{x_1, y_1}} = -y_1$$

final position of image = 0.5 m = 50 cm

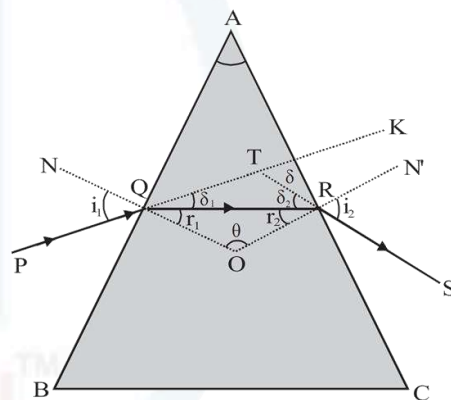
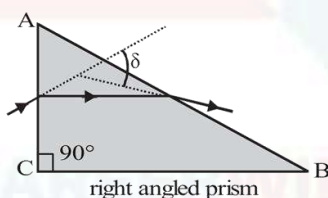
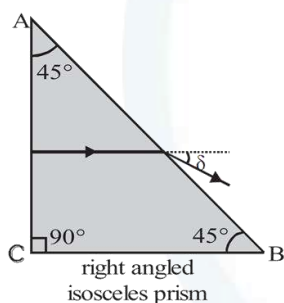
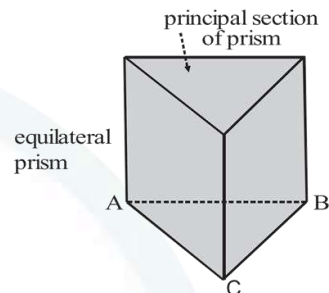
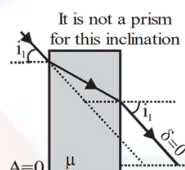
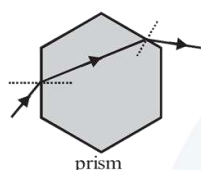
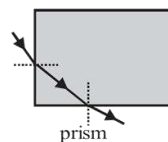
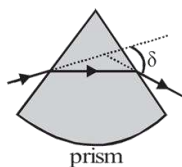
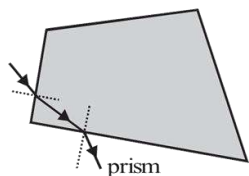
But $\tan 2\theta = \frac{y_1 - 0}{x_2 - x_1}$ &

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\Rightarrow \frac{y_1}{x_2 - x_1} = \frac{2(y_1)}{1 - y_1^2} x_2 = \frac{1}{2} \text{ m}$$

PRISM

A prism is a homogeneous, transparent medium (such as glass) enclosed by two plane surfaces inclined at an angle. These surfaces are called the 'refracting surfaces' and the angle between them is called the 'refracting angle' or the 'angle of prism'. The section cut by a plane perpendicular to the refracting surfaces is called the 'principal section' of the prism.



DEVIATION

PQ = incident ray
 QR = Refracted ray
 RS = emergent ray
 A = Prism angle
 i_1 = incident angle on face AB
 i_2 = emergent angle on face AC
 r_1 = refracted angle on face AB
 r_2 = incident angle on face AC
 Angle of deviation on face AB. $\delta_1 = i_1 - r_1$
 Angle of deviation on face AC $\delta_2 = i_2 - r_2$
 Total angle of deviation
 $\delta = \delta_1 + \delta_2 \Rightarrow \delta = (i_1 - r_1) + (i_2 - r_2)$
 $\quad \quad \quad = i_1 + i_2 - (r_1 + r_2)$
 In $\triangle QOR$ $r_1 + r_2 + \theta = 180^\circ$
 In $\triangle AQR$ $A + \theta = 180^\circ$
 from (ii) and (iii) $r_1 + r_2 = A$
 from (i) and (iv) Total angle of deviation
 $\delta = i_1 + i_2 - A$
 from Snell's law at surface AB $\mu_1 \sin i_1 = \mu_2 \sin r_1$
 and at surface AC $\mu_2 \sin r_2 = \mu_1 \sin i_2$

CONDITION OF MINIMUM DEVIATION

For minimum deviation

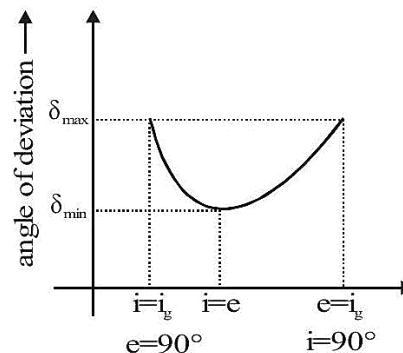
In this condition $i_1 = i_2 = i \Rightarrow r_1 = r_2 = r$ and since

$$r_1 + r_2 = A \therefore r + r = A \Rightarrow 2r = A \Rightarrow r = \frac{A}{2}$$

$$\text{Minimum deviation } \delta_{\min} = 2i - A; i = \frac{A + \delta_{\min}}{2}, r = \frac{A}{2}$$

if prism is placed in air $\mu_1 = 1; 1 \times \sin i = \mu \sin r$

$$\sin \left[\frac{A + \delta_{\min}}{2} \right] = \mu \sin \frac{A}{2} \Rightarrow \mu = \frac{\sin \left[\frac{A + \delta_{\min}}{2} \right]}{\sin \frac{A}{2}}$$



if angle of prism is small $A < 10^\circ$ then $\sin \theta \approx \theta$

$$\mu = \frac{\frac{A+\delta_{\min}}{2}}{\frac{A}{2}} = \frac{A+\delta_{\min}}{A} \Rightarrow A + \delta_{\min} = \mu A$$

$$\Rightarrow \delta_{\min} = (\mu - 1)A$$

CONDITION FOR MAXIMUM DEVIATION/GRAZING EMERGENCE

Angle of incidence (i_g) for grazing emergence

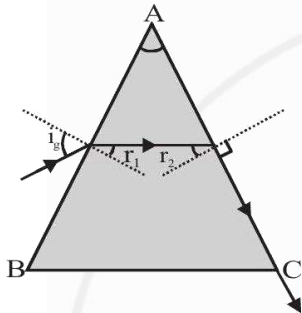
For i_g , $e = 90^\circ$

Applying Snell's law at face AC

$$\mu \sin r_2 = 1 \times 1 \Rightarrow \sin r_2 = \frac{1}{\mu}; r_2 = \sin^{-1} \left(\frac{1}{\mu} \right) = \theta_c$$

$$r_1 + r_2 = A \Rightarrow r_1 = A - \theta_c$$

Again, Applying Snell's law at face AB



$$1 \times \sin i_g = \mu \sin r_1; 1 \times \sin i_g = \mu \sin (A - \theta_c)$$

$$\sin i_g = \mu [\sin A \cos \theta_c - \cos A \sin \theta_c]$$

$$i_g = \sin^{-1} \left[\sqrt{\mu^2 - 1} \sin A - \cos A \right]$$

$$\left[\left(\sin \theta_c = \frac{1}{\mu}, \cos \theta_c = \frac{\sqrt{\mu^2 - 1}}{\mu} \right) \right]$$

If i increases beyond i_g , r_1 increases thus r_2 decreases and becomes less than θ_c and ray emerges.

Thus $i \geq i_g \Rightarrow$ ray emerges, otherwise TIR.

$$\Delta_{\max} = i_g + 90^\circ - A$$

NO EMERGENCE CONDITION

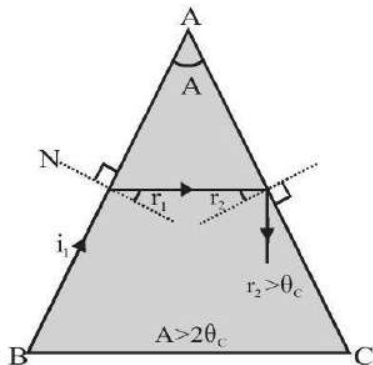
Let maximum incident angle on the face AB $i_{\max} = 90^\circ$

$$1 \times \sin 90^\circ = \mu \sin r_1; \sin r_1 = \frac{1}{\mu} = \sin \theta_c; r_1 = \theta_c$$

if TIR occur at face AC then $r_2 > \theta_c$

$$r_1 + r_2 = A$$

$$\text{from (i) and (ii)} \quad r_1 + r_2 > \theta_c + \theta_c \Rightarrow r_1 + r_2 > 2\theta_c$$



$$\text{from (iii) and (iv)} \quad A > 2\theta_c \Rightarrow \frac{A}{2} > \theta_c \Rightarrow \sin \frac{A}{2} >$$

$$\sin \theta_c \Rightarrow \sin \frac{A}{2} > \frac{1}{\mu} \Rightarrow \frac{1}{\sin \frac{A}{2}} < \mu$$

Example: A ray of light passes through an equilateral prism such that angle of incidence is equal of emergence and the later is equal to $3/4^{\text{th}}$ of the angle of prism. Calculate the angle of deviation. Refractive index of prism is 1.5.

Solution: $A = 60^\circ, \mu = 1.5; i_1 = i_2 = \frac{3}{4} A = 45^\circ, \delta = ?$
 $\therefore A + \delta = i_1 + i_2 \therefore 60^\circ + \delta = 45^\circ + 45^\circ$
 $\Rightarrow \delta = 90^\circ - 60^\circ = 30^\circ$

Example: A prism of refractive index 1.53 is placed in water of refractive index 1.33. If the angle of prism is 60° , calculate the angle of minimum deviation in water. ($\sin 35.1^\circ = 0.575$)

Solution: Here, ${}^a\mu_g = 1.33, {}^a\mu_w = 1.53, A = 60^\circ$,
 $\delta_m = ?$ ${}^w\mu_g = \frac{{}^a\mu_g}{{}^a\mu_w} = \frac{1.33}{1.53} = 1.15$
 $\therefore {}^w\mu_g = \frac{\sin \frac{A+\delta_m}{2}}{\sin \frac{A}{2}}$
 $\therefore \frac{\sin(A+\delta_m)}{2} = {}^w\mu_g \times \sin \frac{A}{2} = 1.15 \sin \frac{60^\circ}{2}$
 $= 0.575 \Rightarrow \frac{A+\delta_m}{2} = \sin^{-1}(0.575) = 35.1^\circ$
 $\therefore \delta_m = 35.1 \times 2 - 60 = 10.2^\circ$

DISPERSION OF LIGHT

When white light is incident on a prism then it is splitted into seven colours. This phenomenon is known as dispersion. Prism introduces different refractive index with different wavelength

$$\text{As } \delta_{\min} = (\mu - 1)A \therefore \lambda_R > \lambda_V$$

$$\text{So } \mu_V > \mu_R \Rightarrow \delta_m(\text{violet}) > \delta_m(\text{red})$$

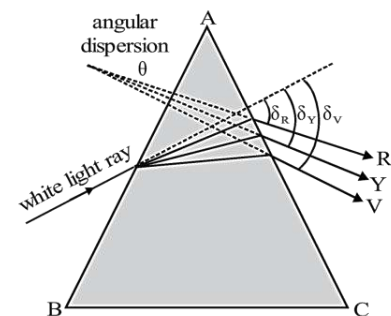
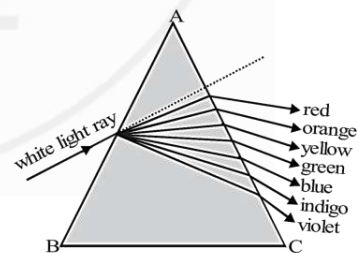
ANGULAR DISPERSION

It is the difference of angle of deviation for violet colour and red colour

Angular dispersion

$$\theta = \delta_V - \delta_R = (\mu_V - 1)A - (\mu_R - 1)A = (\mu_V - \mu_R)A$$

It depends on prism material and on the angle of prism
 $\theta = (\mu_V - \mu_R)A$



DISPERSIVE POWER (ω)

It is ratio of angular dispersion (θ) to mean colour deviation (δ_y)

$$\text{Dispersive power } \omega = \frac{\theta}{\delta_y} \Rightarrow \omega = \frac{(\mu_V - \mu_R)A}{(\mu_y - 1)A} = \frac{\mu_V - \mu_R}{\mu_y - 1}$$

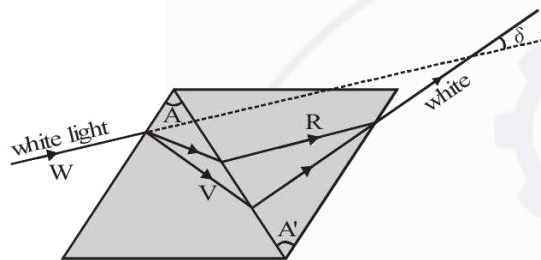
$$\Rightarrow \omega = \frac{\mu_V - \mu_R}{\mu_y - 1}$$

Refractive index of mean colour $\mu_y = \frac{\mu_V + \mu_R}{2}$. Dispersive power depends only on the material of the prism.

COMBINATION OF PRISM

Deviation without dispersion ($\theta = 0^\circ$)

Two or more than two thin prism are combined in such a way that deviation occurs i.e. emergent light ray makes angle with incident light ray but dispersion does not occur i.e., light is not splitted into seven colours.



$$\text{Total dispersion } = \theta = \theta_1 + \theta_2 = (\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A'$$

$$\text{For no dispersion } \theta = 0; (\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A' = 0$$

$$\text{Therefore, } A' = -\frac{(\mu_V - \mu_R)A}{\mu'_V - \mu'_R}$$

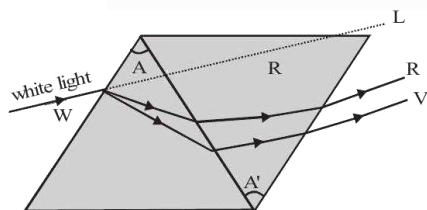
-ve sign indicates that prism angles are in opposite direction.

Dispersion without deviation ($\delta = 0^\circ$)

Two or more than two prisms combine in such a way that dispersion occurs i.e., light is splitted into seven colours but deviation do not occur i.e., emergent light ray becomes parallel to incident light ray.

$$\text{Total deviation } \delta = \delta_1 + \delta_2$$

$$\Rightarrow \delta = 0; (\mu - 1)A + (\mu' - 1)A' = 0 \Rightarrow A' = -\frac{(\mu - 1)A}{\mu' - 1}$$



-ve sign indicates that prism angles are in opposite direction.

Example: White light is passed through a prism of angle 5° . If the refractive indices for red and blue colours are 1.641 and 1.659 respectively, calculate the angle of dispersion between them.

Solution: As for small angle of prism $\delta = (\mu - 1)A$,

$$\delta_B = (1.659 - 1) \times 5^\circ = 3.295^\circ \text{ and}$$

$$\delta_R = (1.641 - 1) \times 5^\circ = 3.205^\circ$$

$$\text{so } \theta = \delta_B - \delta_R = 3.295^\circ - 3.205^\circ = 0.090^\circ$$

Example: Prism angle of a prism is 10° . Their refractive index for red and violet color is 1.51 and 1.52 respectively. Then find the dispersive power.

Solution: Dispersive power of prism

$$\omega = \left(\frac{\mu_V - \mu_R}{\mu_y - 1} \right) \text{ but } \mu_y = \frac{\mu_V + \mu_R}{2} = \frac{1.52 + 1.51}{2} = 1.515$$

$$\text{Therefore } \omega = \frac{1.52 - 1.51}{1.515 - 1} = \frac{0.01}{1.515} = 0.019$$

Example: The refractive indices of flint glass for red and violet colours are 1.644 and 1.664. Calculate its dispersive power.

Solution: Here, $\mu_r = 1.644$, $\mu_v = 1.664$, $\omega = ?$

$$\text{Now } \mu_y = \frac{\mu_V + \mu_R}{2} = \frac{1.664 + 1.644}{2} = 1.654$$

$$\therefore \omega = \frac{\mu_V - \mu_R}{\mu_y - 1} = \frac{1.664 - 1.644}{1.654 - 1} = 0.0305$$

Example: In a certain spectrum produced by a glass prism of dispersive power 0.031, it was found that $\mu_r = 1.645$ and

$\mu_v = 1.665$. What is the refractive index for yellow colour?

Solution: Here, $\omega = 0.031$, $\mu_r = 1.645$, $\mu_v = 1.665$, $\mu_y = ?$

$$\therefore \omega = \frac{\mu_V - \mu_R}{\mu_y - 1} \therefore \mu_y - 1 = \frac{\mu_V - \mu_R}{\omega} = \frac{1.665 - 1.645}{0.031}$$

$$= \frac{0.020}{0.031} = 0.645 \therefore \mu_y = 0.645 + 1 = 1.645$$

Example: A combination of two prisms, one of flint and other of crown glass produces dispersion without deviation. The angle of flint glass prism is 15° . Calculate the angle of crown glass prism and angular dispersion of red and violet. (μ for crown glass = 1.52, μ for flint glass = 1.65, ω for crown glass 0.20, ω for flint glass = 0.03).

Solution: Here, $A = 15^\circ$, $A' = ?$, $\omega = 0.03$, $\omega' = 0.02$,

$$\mu = 1.65, \mu' = 1.52, \text{ For no deviation, } \delta + \delta' = 0$$

$$(\mu - 1)A + (\mu' - 1)A' = 0$$

$$\Rightarrow (1.65 - 1)15^\circ + (1.52 - 1)A' = 0$$

$$\Rightarrow A' = \frac{-0.65 \times 15}{0.52} = -18.75^\circ$$

Negative sign indicates that two prisms must be joined in opposition. Net angular dispersion

$$(\mu_V - \mu_R)A + (\mu'_V - \mu'_R)A'$$

$$= \omega(\mu - 1)A + \omega'(\mu' - 1)A'$$

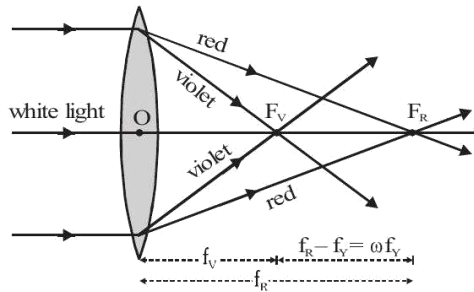
$$= 0.03(1.65 - 1)15^\circ + 0.02$$

$$(1.52 - 1)(-18.75^\circ)$$

$$= 0.2925 - 0.195 = 0.0975^\circ$$

CHROMATIC ABERRATION

The image of an object in white light formed by a lens is usually colored and blurred. This defect of image is called chromatic aberration and arises due to the fact that focal length of a lens is different for different colors. For a single lens $\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$ and as μ of lens is maximum for violet while minimum for red, violet is focused nearest to the lens while red farthest from it. It is defect of lens.



Longitudinal or Axial Chromatic Aberration

When a white object O is situated on the axis of a lens, then images of different colors are formed at different points along the axis. The formation of images of different colors at different positions is called 'axial' or longitudinal chromatic aberration. The axial distance between the red and the violet images $I_R - I_V$ is known as longitudinal aberration. When white light is incident on lens, image is obtained at different point on the axis because focal length of lens depend on wavelength.

$$f \propto \lambda \Rightarrow f_R > f_V$$

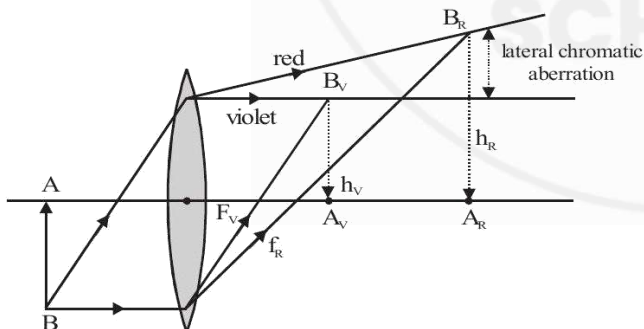
$$f_R - f_V = \omega f_V$$

$\Rightarrow$ Axial or longitudinal chromatic aberration

If the object is at infinity, then the longitudinal chromatic aberration is equal to the difference in focal-lengths ($f_R f_V$) for the red and the violet rays.

LATERAL CHROMATIC ABERRATION

As the focal-length of the lens varies from color to color, the magnification $m = \left[\frac{f}{u+f} \right]$ produced by the lens also varies from color to color. Therefore, for a finite-size white object AB, the images of different colors formed by the lens are of different sizes.



The formation of images of different colors in different sizes is called lateral chromatic aberration. The difference in the height of the red image $B_R A_R$ and the violet image $B_V A_V$ is known as lateral chromatic aberration.

$$LCA = h_R - h_V$$

ACHROMATISM

If two or more lens combined together in such a way that this combination produce image at a same point then this combination is known as achromatic combination of lenses.

$$\frac{\omega}{f_y} + \frac{\omega'}{f'_y} = 0 \Rightarrow \frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0 \Rightarrow \frac{\omega_1}{\omega_2} = -\frac{f_1}{f_2}$$

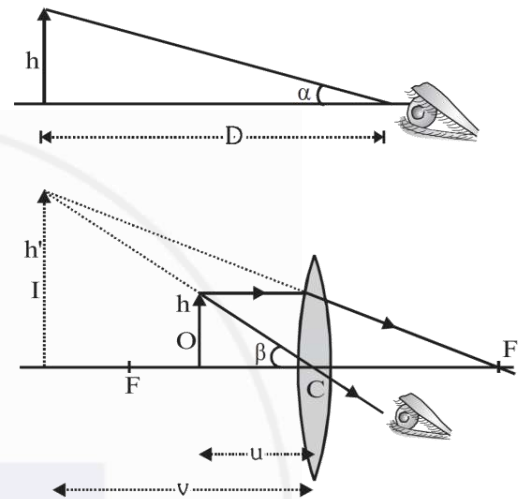
$$\text{For combination of lens. } \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2}$$

(Apply sign convention in numerical)

OPTICAL INSTRUMENTS

Simple microscope

When object is placed between focus and optical centre a virtual, magnified and erect image is formed



Magnifying power (MP)

$$= \frac{\text{visual angle with instrument } (\beta)}{\text{maximum visual angle for unaided eye } (\alpha)}$$

$$\Rightarrow MP = \frac{\frac{h}{-u}}{\frac{h}{-D}} = \frac{D}{u}$$

(i) When the image is formed at infinity:

$$\text{by lens equation } \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{-\infty} - \frac{1}{-u} = \frac{1}{f}$$

$$\Rightarrow u = f \text{ So } MP = \frac{D}{u} = \frac{D}{f}$$

(ii) If the image is at minimum distance of clear vision D:

$$\frac{1}{-D} - \frac{1}{-u} = \frac{1}{f} \Rightarrow \frac{1}{u} = \frac{1}{D} + \frac{1}{f} \quad [v = -D \text{ and } u = -ve]$$

$$\text{Multiplying by } D \text{ both the sides } \frac{D}{u} = 1 + \frac{D}{f}$$

$$\Rightarrow MP = \frac{D}{u} = 1 + \frac{D}{f}$$

Example: A man with normal near point 25 cm reads a book with small print using a magnifying glass, a thin convex lens of focal length 5 cm.

(a) What is the closest and farthest distance at which he can read the book when viewing through the magnifying glass?

(b) What is the maximum and minimum MP possible using the above simple microscope?

Solution: (a) As for normal eye far and near point are ∞ and 25 cm respectively, so for magnifier $v_{\max} = -\infty$ and $v_{\min} = -25$ cm. However, for a lens as $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{f} \Rightarrow u = \frac{f}{(f/v)-1}$$

So u will be minimum when v = minimum = -25 cm i.e.

$$(u)_{\min} = \frac{5}{-(5/25)-1} = -\frac{25}{6} = -4.17 \text{ cm}$$

Ans u will be maximum when $v = \text{maximum} = \infty$

$$\text{i.e., } u_{\max} = \frac{5}{(\frac{5}{\infty}-1)} = -5 \text{ cm}$$

So the closest and farthest distance of the book from the magnifier (or eye) for clear viewing are 4.17 cm and 5 cm respectively.

(b) As in case of simple magnifier $MP = (D/u)$.

So MP will be minimum when $u = \text{max} = 5 \text{ cm}$

$$\Rightarrow (MP)_{\min} = \frac{-25}{-5} = 5 \left[= \frac{D}{f} \right] \text{ and}$$

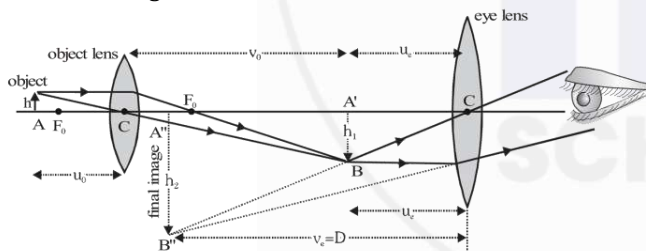
MP will be maximum when " $u = \text{min} = (25/6) \text{ cm}$ "

$$\Rightarrow (MP)_{\max} = \frac{-25}{-(\frac{25}{6})}$$

$$= 6 \left[= 1 + \frac{D}{f} \right]$$

COMPOUND MICROSCOPE

Compound microscope is used to get more magnified image. Object is placed in front of objective lens and image is seen through eye piece. The aperture of objective lens is less as compared to eye piece because object is very near so collection of more light is not required. Generally object is placed between $F - 2F$ due to this a real inverted and magnified image is formed between $2F - \infty$. It is known as intermediate image $A'B'$. The intermediate image acts as an object for eye piece. Now the distance between both the lenses is adjusted in such a way that intermediate image falls between the optical centre of eye piece and its focus. In this condition, the final image is virtual, inverted and magnified.



Total magnifying power = Linear magnification $\times$ angular magnification $MP = m_o m_e = \frac{v_o}{u_o} \frac{D}{u_e}$

- (i) When final image is formed at minimum, distance of distinct vision.

$$MP = \frac{v_o}{u_o} \left[1 + \frac{D}{f_e} \right] = \frac{f_o}{f_o + u_o} \left[1 + \frac{D}{f_e} \right] = \frac{f_o - v_o}{f_o} \left[1 + \frac{D}{f_e} \right]$$

$$= \frac{h_2}{h_1} \left(1 + \frac{D}{f_e} \right)$$

Length of the tube = $v_o + |u_e|$

- (ii) When final image is formed at infinity $\frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e}$

$$\Rightarrow \frac{1}{\infty} + \frac{1}{u_e} = \frac{1}{f_e} \Rightarrow u_e = f_e$$

$$MP = \frac{v_o}{u_o} \left[\frac{D}{f_e} \right] = \frac{f_o}{f_o + u_o} \left[\frac{D}{f_e} \right] = \frac{f_o - v_o}{f_o} \left[\frac{D}{f_e} \right] = \frac{h_2}{h_1} \left[\frac{D}{f_e} \right]$$

Length of the tube $L = v_o + f_e$

Sign convention for solving numerical $u_o = -ve, v_o = +ve, f_o = +ve$
 $u_e = -ve, v_e = -ve, f_e = +ve, m_o = -ve, m_e = +ve, M = -ve$

Example: A thin convex lens of focal length 5 cm is used as a simple microscope by a person with normal near point (25 cm). What is the magnifying power of the microscope?

Solution: Here, $f = 5 \text{ cm}; D = 25 \text{ cm}$,
 $M = MP = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$

Example: A compound microscope consists of an objective lens of focal length 2.0 cm and an eye piece of focal length 6.25 cm, separated by a distance of 15 cm. How far from the objective should an object be placed in order to obtain the final image at (a) the least distance of distinct vision (25 cm) (b) infinity?

Solution: Here, $f_o = 2.0 \text{ cm}; f_e = 6.25 \text{ cm}, u_o = ?$

$$(a) v_e = -25 \text{ cm} \because \frac{1}{v_e} - \frac{1}{u_e} = \frac{1}{f_e} \therefore \frac{1}{u_e} = \frac{1}{v_e} - \frac{1}{f_e}$$

$$= \frac{1}{-25} - \frac{1}{6.25} = \frac{-1-4}{25} = \frac{-5}{25} \Rightarrow u_e = -5 \text{ cm}$$

As distance between objective and eye piece = 15 cm; $v_o = 15 - 5 = 10 \text{ cm}$

$$\therefore \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \therefore \frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{10} - \frac{1}{2} = \frac{1-5}{10}$$

$$\Rightarrow u_o = \frac{-10}{4} = -2.5 \text{ cm}$$

$$\text{Magnifying power} = \frac{v_o}{|u_o|} \left[1 + \frac{D}{f_e} \right]$$

$$= \frac{10}{2.5} \left[1 + \frac{25}{6.25} \right] = 20$$

(b) $\because v_e = \infty, u_e = f_e = 6.25 \text{ cm}$

$$\therefore v_o = 15 - 6.25 = 8.75 \text{ cm}$$

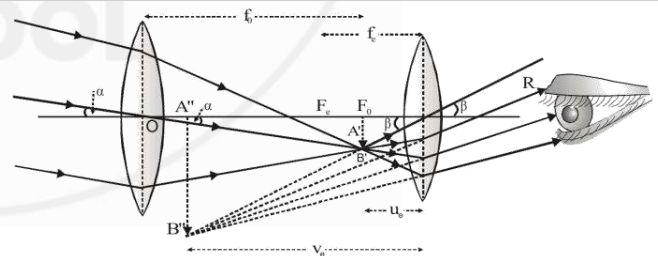
$$\therefore \frac{1}{v_o} - \frac{1}{u_o} = \frac{1}{f_o} \Rightarrow \frac{1}{u_o} = \frac{1}{v_o} - \frac{1}{f_o} = \frac{1}{8.75} - \frac{1}{2}$$

$$= \frac{2-8.75}{17.5} \Rightarrow u_o = \frac{-17.5}{6.75} = -2.59 \text{ cm}$$

$$\text{Magnifying power} = \frac{v_o}{|u_o|} \times \left[1 + \frac{D}{f_e} \right] = \frac{v_o}{|u_o|} \times \frac{D}{|u_e|}$$

$$= \frac{8.75}{2.59} \times \frac{25}{6.25} = 13.51$$

ASTRONOMICAL TELESCOPE



A telescope is used to see distant object, objective lens forms the image $A'B'$ at its focus. This image $A'B'$ acts as an object for eyepiece and it forms final image $A''B''$.

$$MP = \frac{\text{visual angle with instrument } (\beta)}{\text{visual angle for unaided eye } (\alpha)}$$

$$\Rightarrow MP = \frac{\frac{h'}{h}}{\frac{h'}{h'}} = -\frac{f_o}{u_e} \left[\frac{D}{f_e} \right] = -\frac{f_o}{u_e} \left[\frac{D}{f_e} \right]$$

- (i) If the final image is at infinity $v_e = -\infty, u_e = -ve$

$$\frac{1}{-\infty} - \frac{1}{-u_e} = \frac{1}{f_e} \Rightarrow u_e = f_e. \text{ So } MP = -\frac{f_o}{f_e} \text{ and}$$

length of the tube $L = f_o + f_e$

- (ii) If the final image is at D: $v_e = -D, u_e = -ve$

$$\frac{1}{-D} - \frac{1}{-u_e} = \frac{1}{f_e} \Rightarrow \frac{1}{u_e} = \frac{1}{f_e} + \frac{1}{D} = \frac{1}{f_e} \left[1 + \frac{f_e}{D} \right]$$

$$\text{So MP} = -\frac{f_o}{u_e} = -\frac{f_o}{f_e} \left[1 + \frac{f_e}{D} \right]$$

$$\text{Length of the tube is } L = f_o + u_e$$

S.No.	Compound Microscope	S.No.	Astronomical Telescope
1.	It is used to increase visual angle of near tiny object.	1.	It is used to increase visual angle of distant large objects.
2.	In it field and eye lens both are convergent, of short focal length and aperture.	2.	In it objective lens is of large focal length and aperture while eye lens of short focal length and aperture and both are convergent.
3.	Final image is inverted, virtual and enlarged and at a distance D to ∞ from the eye.	3.	Final image is inverted, virtual and enlarged at a distance D to ∞ from the eye.
4.	MP does not change appreciably if objective and eye lens are interchanged as $[MP \sim (LD/f_o f_e)]$	4.	MP becomes $(1/m^2)$ times of its initial value if objective and eye-lenses are interchanged as $MP \sim [f_o/f_e]$
5.	MP is increased by decreasing the focal length of both the lenses.	5.	MP is increased by increasing the focal length of objective lens and by decreasing the focal length of eyepiece
6.	RP is increased by decreasing the wavelength of light used. $(\because RP = \frac{2\mu \sin \theta}{\lambda})$	6.	RP is increased by increasing the aperture of objective. $(\because RP = \frac{D}{1.22\lambda})$

Example: Diameter of the moon is 3.5×10^3 km and its distance from earth is 3.8×10^5 km. It is seen by a telescope whose objective and eyepiece have focal lengths 4 m and 10 cm respectively. What will the angular diameter of the image of the moon.

Solution: $MP = -\frac{f_o}{f_e} = -\frac{400}{10}$
 $= -40$.
 Angle subtended by the moon at the objective radian.
 $= \frac{3.5 \times 10^3}{3.8 \times 10^5} = 0.009$
 Thus angular diameter of the image
 $= MP \times \text{visual angle} = 40 \times 0.009 = 0.36$
 radian $= \frac{0.36 \times 180}{3.14} \approx 21^\circ$

Example: A telescope consisting of an objective of focal length 60 cm and a single-lens eyepiece of focal length 5 cm is focussed at a distant object in such a way that parallel rays emerge from the eye piece. If the object subtends an angle of 2° at the objective, then find the angular width of the image.

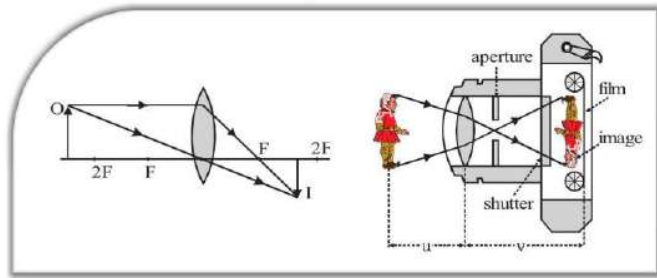
Solution: $MP = \frac{f_o}{f_e} = \frac{\beta}{\alpha}$
 $\Rightarrow \beta = \alpha \frac{f_o}{f_e} = 2^\circ \times \frac{60}{5} = 24^\circ$

Example: The focal lengths of the objective and the eye piece of an astronomical telescope are 60 cm and 5 cm respectively. Calculate the magnifying power and the length of the telescope when the final image is formed at (i) infinity, (ii) least distance of distinct vision (25 cm)

Solution: (i) When the final image is at infinity, then:
 $MP = -\frac{f_o}{f_e} = -\frac{60}{5}$
 $= -12$ and length of the telescope is
 $L = f_o + f_e = 60 + 5 = 65$ cm
 (ii) For least distance of distinct vision, the magnifying power is:
 $MP = -\frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right) = -\frac{60}{5} \left(1 + \frac{5}{25} \right)$
 $= -\frac{12 \times 6}{5} = -14.4$
 Now $\frac{1}{f_e} = \frac{1}{v_e} - \frac{1}{u_e}$
 $\Rightarrow \frac{1}{5} = -\frac{1}{25} - \frac{1}{u_e}$
 $\Rightarrow \frac{-1}{u_e} = \frac{1}{25} + \frac{1}{5}$
 $\Rightarrow u_e = -4.17$ cm $\Rightarrow |u_e| = 4.17$ cm
 The length of telescope in this position is
 $L = f_o + |u_e| = 60 + 4.17 = 64.17$ cm

LENS - CAMERA

There is a convex lens whose aperture and distance from the film can be adjusted. Object is real and placed between ∞ and $2F$, so the image is real, inverted diminished and between F and $2F$.



If I is the intensity of light, S is the light transmitting area of lens and t is the exposure time, then for proper exposure, $I \times S \times t = \text{constant}$ light transmitting area of a lens is proportional to the square of its aperture D ; $I \times D^2 \times t = \text{constant}$

If aperture is kept fixed, for proper exposure, $I \times t = \text{constant}$, i.e., $I_1 t_1 = I_2 t_2$ If intensity is kept fixed, for proper exposure, $D^2 \times t = \text{constant}$

$$\text{Time of exposure} \propto \frac{1}{(\text{aperture})^2} \quad (i)$$

The ratio of focal length to the aperture of lens is called f-number of the camera,

$$f\text{-number} = \frac{\text{focal length}}{\text{aperture}} \Rightarrow \text{Aperture} \propto \frac{1}{f\text{-number}} \quad (ii)$$

From equation (i) and (ii) $\Rightarrow$ Time of exposure $\propto (f\text{-number})^2$

Example: With diaphragm of the camera lens set at $\frac{f}{2}$, the correct exposure time is $\frac{1}{100}$, then with diaphragm set at $\frac{f}{4}$. Calculate the correct exposure time.

Solution: As exposure time $\propto \frac{1}{(\text{aperture})^2}$
 $\Rightarrow t_1 \propto \frac{1}{[f/2]^2}$ and $t_2 \propto \frac{1}{[f/4]^2}$
 here $t_1 = \frac{1}{100}$ s then $\frac{t_2}{t_1} = \frac{16}{4} = 4$
 $\Rightarrow t_2 = 4t_1 = \frac{4}{100}$ s

Example: A good photographic print is obtained by an exposure of two seconds at a distance of 20 cm from the lamp. Calculate the time of exposure required to get an equally good result at a distance of 40 cm.

Solution: We know that the intensity of light varies inversely as the $(\text{distance})^2$. When distance is doubled, the intensity becomes one-fourth. So, the time of exposure should be four times. Hence, time of exposure = $2 \times 4 = 8$ s

Example: Photograph of the ground are taken from an aircraft, flying at an altitude of 2000 m, by a camera with a lens of focal length 50 cm. The size of the film in the camera is 18 cm $\times$ 18 cm. What area of the ground can be photographed by this camera at any one time.

Solution: As here $u = -2000$ m, $f = 0.50$ m, so from lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$,
 we have $\frac{1}{v} - \frac{1}{(-2000)} = \frac{1}{0.5} \Rightarrow \frac{1}{v} = \frac{1}{0.5} - \frac{1}{2000} \approx \frac{1}{0.5}$

$$\left[\text{as } \frac{1}{0.5} \gg \frac{1}{2000} \right] \Rightarrow v = 0.5 \text{ m} = 50 \text{ cm} = f$$

$$\text{Now as in case of a lens, } m = \frac{v}{u} = \frac{0.5}{-2000} = -\frac{1}{4} \times 10^{-3}$$

$$\text{So } 1 = (ma)(mb) = m^2 A [\because A = ab]$$

$$A = \frac{I_1}{m^2} = \frac{18 \text{ cm} \times 18 \text{ cm}}{[(1/4) \times 10^{-3}]^2} = (720 \text{ m} \times 720 \text{ m})$$

Example: The proper exposure time for a photographic print is 20 s at a distance of 0.6 m from a 40 candle power lamp. How long will you expose the same print at a distance of 1.2 m from a 20 candle power lamp?

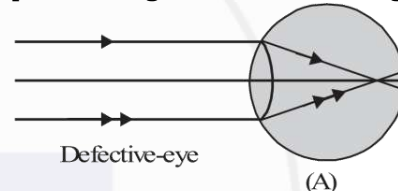
Solution: In case of camera, for proper exposure $I_1 D_1^2 t_1 = I_2 D_2^2 t_2$

As here D is constant and $I = \left(\frac{L}{r^2}\right)$;

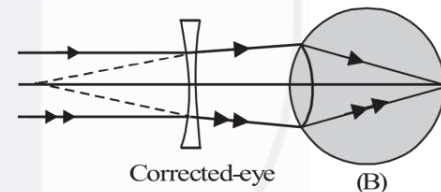
$$\frac{L_1}{r_1^2} \times t_1 = \frac{L_2}{r_2^2} \times t_2 \text{ So } \frac{40}{(0.6)^2} \times 20 = \frac{20}{(1.2)^2} t \Rightarrow t = 160 \text{ s}$$

DEFECTS OF EYES

MYOPIA [or Short-sightedness or Near-sightedness]

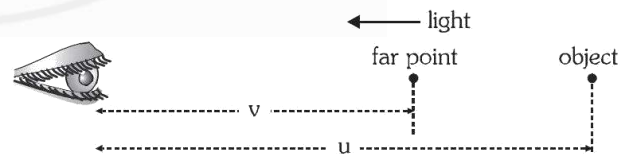


Defective-eye



Corrected-eye

- Distant objects are not clearly visible, but near objects are clearly visible because the image is formed before the retina.
- To remove the defect a concave lens is used. The maximum distance which a person can see without the help of spectacles is known as the far point.



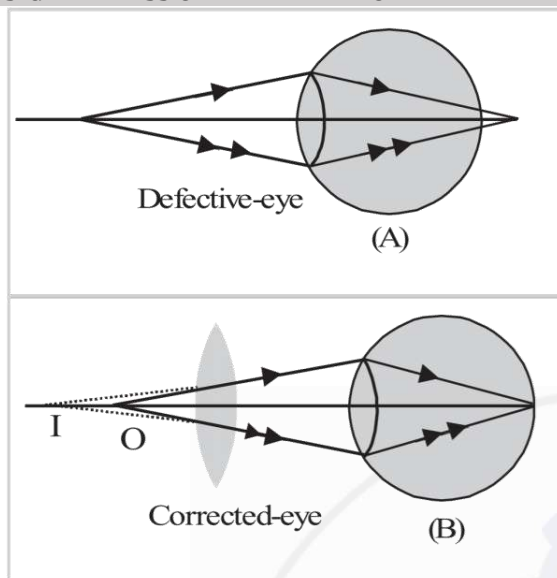
If the reference of object is not given then it is taken as infinity.

In this case the image of the object is formed at the far point of the person.

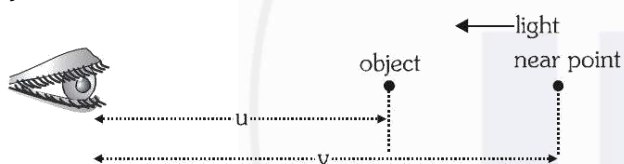
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = P \Rightarrow \frac{1}{\text{distance of far point (in m)}} - \frac{1}{\text{distance of object (in m)}} = \frac{1}{f} = P$$

$$\frac{1}{100} - \frac{1}{\infty} = \frac{1}{f} = P \Rightarrow \frac{1}{\text{distance of far point (in cm)}} - \frac{1}{\text{distance of object (in cm)}} = P$$

LONG-SIGHTEDNESS OR HYPERMETROPIA



- (i) Near objects are not clearly visible but far objects are clearly visible.
- (ii) The image of a near object is formed behind the retina.
- (iii) To remove this defect, a convex lens is used.



Near Point:-

The minimum distance which a person can see without the help of spectacles.

In this case, the image of the object is formed at the near point.

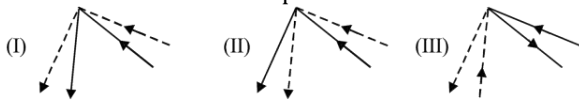
If the reference of the object is not given, it is taken as 25 cm.

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = P \Rightarrow \frac{1}{\text{distance of near point (in m)}} = P$$

$$- \frac{1}{\text{distance of object (in m)}} = \frac{1}{f} = P$$

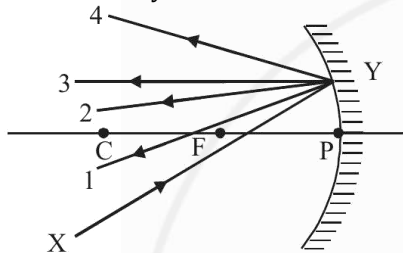
EXERCISE - I

1. Each of these diagrams is supposed to show two different rays being reflected from the same point on the same mirror. Which option is correct.



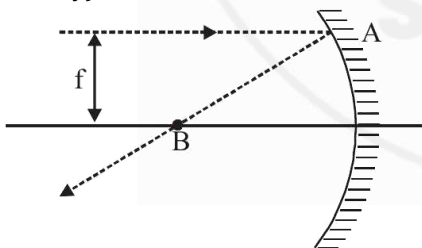
- (A) Only I
(B) Only II
(C) Only III
(D) All

2. Figure shows a small concave mirror with CP as its principal axis. A ray XY is incident on the mirror. Which of the four rays can be the reflected ray.



- (A) 1
(B) 2
(C) 3
(D) 4

3. A ray of light is incident on a concave mirror. It is parallel to the principal axis and its height from principal axis is equal to the focal length of the mirror. The ratio of the distance of point B to the distance of the focus from the centre of curvature is (AB is the reflected ray)

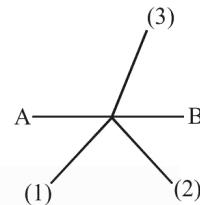


- (A) $\frac{2}{\sqrt{3}}$
(B) $\frac{\sqrt{3}}{2}$
(C) $\frac{2}{3}$
(D) $\frac{1}{2}$

4. The distance of an object from a spherical mirror is equal to the focal length of the mirror. Then the image:
(A) must be at infinity
(B) may be at infinity
(C) may be at the focus
(D) none

5. AB is a boundary separating two media of different refractive indices. A ray is incident on the boundary is partially reflected and partially transmitted. Choose the CORRECT statement.

- (A) 3 is incident ray and 1 is refracted ray
(B) 2 is incident ray and 1 is partially reflected ray
(C) 1 is incident ray and 3 is refracted ray
(D) 3 is incident ray and 2 is partially reflected ray



6. Statement: 1 You see a geostationary satellite above the horizon. You desire to communicate with the satellite by sending a beam of laser light. You should aim your laser slightly higher than the line of sight of the satellite.

Statement: 2 Light bends away from the normal while moving from denser to rarer medium.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

- (C) Statement-1 is true, statement-2 is false.

- (D) Statement-1 is false, statement-2 is true.

7. As light from the sun enters the atmosphere, it refracts due to the small difference between the speeds of light in air and in vacuum. The optical length of the day is defined as the time interval between the instant when the top of the sun is just visibly observed above the horizon to the instant at which the top of the sun just disappears below the horizon. The geometric length of the day is defined as the time interval between the instant when a geometric straight line drawn from the observer to the top of the sun just clears the horizon to the instant at which this line just dips below the horizon?

- (A) The optical length of a day is longer than geometrical length.

- (B) The geometric length of the day is longer than optical length

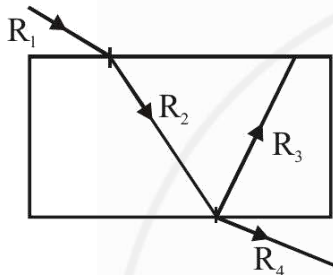
- (C) Optical length and geometrical length of day are equal.

- (D) Cannot be determined.

8. A ray of light travelling in a medium of refractive index μ is incident at an angle θ on a composite transparent plate consisting of 50 plates of R.I. $1.01\mu, 1.02\mu, 1.03\mu, \dots, 1.50\mu$. The ray emerges from the composite plate into a medium of refractive index 1.6μ at angle 'x'. Then:

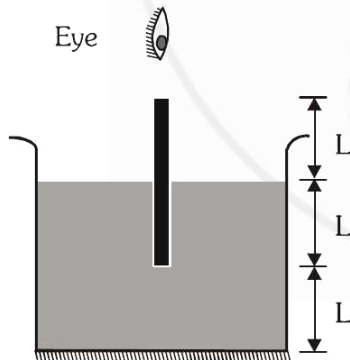
- (A) $\sin x = \left(\frac{1.01}{1.5}\right)^{50} \sin \theta$
 (B) $\sin x = \frac{5}{8} \sin \theta$
 (C) $\sin x = \frac{8}{5} \sin \theta$
 (D) $\sin x = \left(\frac{1.5}{1.01}\right)^{50} \sin \theta$

9. A ray R_1 is incident on the plane surface of the glass slab (kept in air) of refractive index $\sqrt{2}$ at an angle of incidence equal to the critical angle for this air glass system. The refracted ray R_2 undergoes partial reflection and refraction at the other surface. The angle between reflected ray R_3 and the refracted ray R_4 at that surface is:



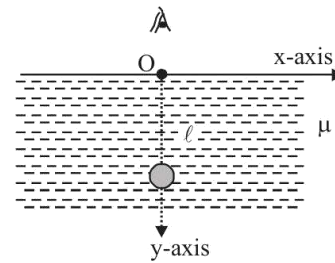
- (A) 45°
 (B) 135°
 (C) 105°
 (D) 75°

10. What is the length of the image of the rod in mirror, according to the observer in air? (refractive index of the liquid is μ)



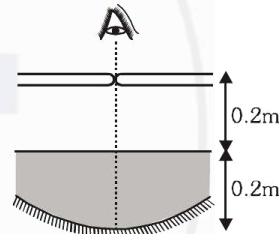
- (A) $\mu L + L$
 (B) $L + \frac{L}{\mu}$
 (C) $L\mu + \frac{L}{\mu}$
 (D) None of these

11. A pendulum of length is free to oscillate in vertical plane about point O in a medium of refractive index μ . An observer in air is viewing the bob of the pendulum directly from above. The pendulum is performing small oscillations about its equilibrium position. The equation of trajectory of bob as seen by observer is:



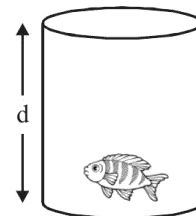
- (A) $x^2 + y^2 = \ell^2$
 (B) $\frac{x^2}{(\ell/\mu)^2} + \frac{y^2}{\ell^2} = 1$
 (C) $\frac{x^2}{\ell^2} + \left(\frac{y}{\ell/\mu}\right)^2 = 1$
 (D) $x^2 + y^2 = \left(\frac{\ell}{\mu}\right)^2$

12. When a pin is moved along the principal axis of a small concave mirror, the image position coincides with the object at a point 0.5 m from the mirror, refer figure. If the mirror is placed at a depth of 0.2 m in a transparent liquid, the same phenomenon occurs when the pin is placed 0.4 m from the mirror. The refractive index of the liquid is



- (A) $6/5$
 (B) $5/4$
 (C) $4/3$
 (D) $3/2$

13. A fish floats in liquid with its eye at the centre of an opaque walled full tank of liquid of circular cross section. When the fish look upwards, it can see a fish-eye view of the surrounding scene i.e. it is able to view the entire space above the liquid surface. The diameter of the tank is 30 cm, and the critical angle for liquid is 37° . At what maximum depth below the surface of the liquid, d, must the fish be floating? ($\sin 37^\circ = \frac{3}{5}$)



- (A) 16 cm
 (B) 20 cm
 (C) 11.25 cm
 (D) 25 cm

14. Statement-1: A point source of light is placed inside water. A light detector present out side, in air can detect light only in a conical region, with the apex at the source and circumscribing the circle of illuminance.

Statement-2: Ray incident from denser to rarer medium undergoes total internal reflection when the angle of incidence become more than the critical angle, this situation create a circular region from which light escapes and is called circle of illuminance.

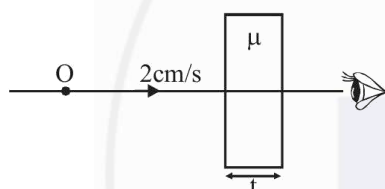
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

15. A glass slab of width 't', refractive index ' μ ' is placed as shown in the figure. If the point object, moves with a speed 2 cm/s towards the slab the speed observed will be



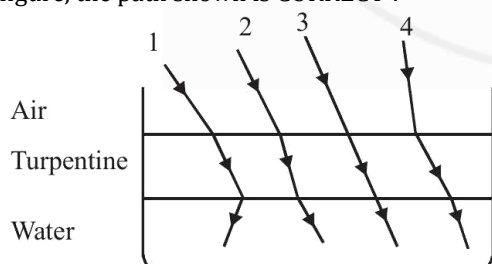
(A) 2 cm/s

(B) less than 2 cm/s

(C) greater than 2 cm/s

(D) dependent on the refractive index of surrounding medium

16. The optical density of turpentine is higher than that of water while its mass density is lower. Figure shows a layer of turpentine floating over water in a container. For which one of the four rays incident on turpentine in figure, the path shown is CORRECT?



17. An isosceles glass prism having refractive index μ has one of its faces coated with silver. A ray of light is incident normally on the other face (which is equal to the silvered face). The ray of light is reflected twice on the same sized faces and then emerges through the base of the prism perpendicularly. The angles of prism are

(A) $40^\circ, 70^\circ, 70^\circ$

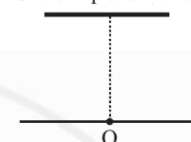
(B) $50^\circ, 65^\circ, 65^\circ$

(C) $36^\circ, 72^\circ, 72^\circ$

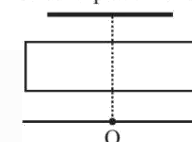
(D) data insufficient

18. In the diagram shown below, a point source O is placed vertically below the center of a circular plane mirror. The light rays starting from the source are reflected from the mirror such that a circular area A on the ground receives light. Now, a glass slab is placed between the mirror and the source O. What will the magnitude of the new area on the ground receiving light?

Circular plane mirror



Circular plane mirror



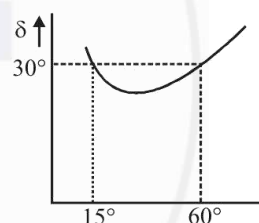
(A) A

(B) Greater than A

(C) Less than A

(D) Cannot tell, as the information given is insufficient

19. Figure shows graph of deviation δ versus angle of incidence for a light ray striking a prism. Angle of prism is



(A) 30°

(B) 45°

(C) 60°

(D) 75°

20. A given ray of light suffers minimum deviation in an equilateral prism P. If refractive index increases slightly then the ray will now suffer

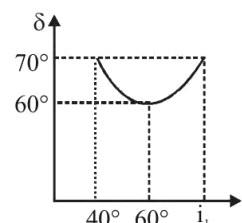
(A) greater deviation

(B) no deviation

(C) same deviation as before

(D) total internal reflection

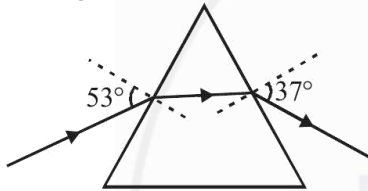
21. The curve of angle of incidence versus angle of deviation shown has been plotted for prism. The value of refractive index of the prism used is:



- (A) $\sqrt{3}$
 (B) $\sqrt{2}$
 (C) $\frac{\sqrt{3}}{\sqrt{2}}$
 (D) $\frac{2}{\sqrt{3}}$

22. A beam of monochromatic light is incident at $i = 50^\circ$ on one face of an equilateral prism, the angle of emergence is 40° , then the angle of minimum deviation is:
 (A) 30°
 (B) $< 30^\circ$
 (C) $\leq 30^\circ$
 (D) $\geq 30^\circ$

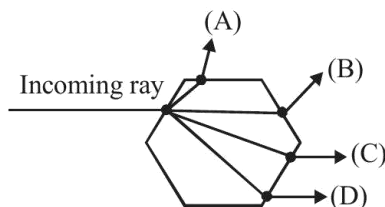
23. A ray incident at an angle 53° on a prism emerges at an angle 37° as shown. If the angle of incidence is made 50° , which of the following is a possible value of the angle of emergence?



- (A) 35°
 (B) 42°
 (C) 40°
 (D) 38°

24. Statement-1: When light falls on a sphere made of diamond total internal reflection takes place which makes it shine more than a similar sphere made of common glass.
 Statement-2: Refractive index for diamond is more than refractive index of cheap glass.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

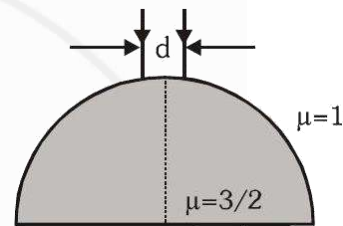
25. A light ray strikes a hexagonal ice crystal floating in the air as shown in the figure. The correct path of ray may be ?



26. A light ray is incident on a transparent sphere of index $= \sqrt{2}$, at an angle of incidence $= 45^\circ$. What is the deviation of a tiny fraction of the ray, which enters the sphere, undergoes two internal reflections and then refracts out into air?

- (A) 270°
 (B) 240°
 (C) 120°
 (D) 180°

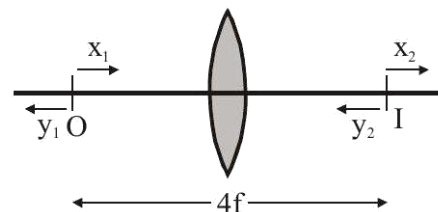
27. A beam of diameter 'd' is incident on a glass hemisphere as shown. If the radius of curvature of the hemisphere is very large in comparison to d, then the diameter of the beam at the base of the hemisphere will be



- (A) $\frac{3}{4}d$
 (B) d
 (C) $\frac{d}{3}$
 (D) $\frac{2}{3}d$

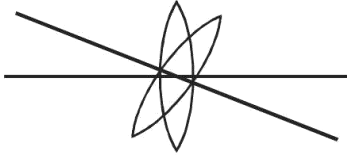
28. An air bubble is inside water. The refractive index of water is $4/3$. At what distance from the air bubble should a point object be placed so as to form a real image at the same distance from the bubble:
 (A) $2R$
 (B) $3R$
 (C) $4R$
 (D) The air bubble cannot form a real image

29. In a converging lens of focal length f , the distance between real object and its real image is $4f$. If the object moves x_1 distance towards lens its image moves x_2 distance away from the lens and when object moves y_1 distance away from the lens its image moves y_2 distance towards the lens, then choose the correct option:



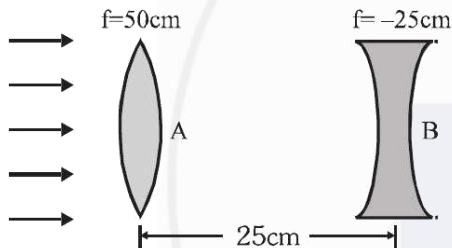
- (A) $x_1 > x_2$ and $y_1 > y_2$
 (B) $x_1 < x_2$ and $y_1 < y_2$
 (C) $x_1 < x_2$ and $y_1 > y_2$
 (D) $x_1 > x_2$ and $y_2 > y_1$

30. A point object is placed on the principal axis of a converging lens and its image (I_1) is formed on its principal axis. If the lens is rotated by a small angle θ about its optical centre such that its principal axis also rotates by the same amount then the image (I_2) of the same object is formed at point P. Choose the correct option.



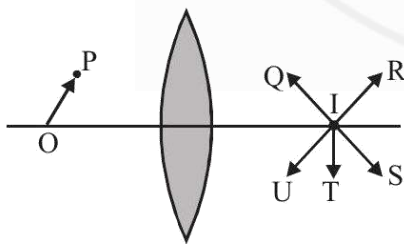
- (A) Point P lies on the new principal axis.
 (B) Point P lies on the old principal axis.
 (C) Point P is anywhere between the two principal axes
 (D) None of these

31. The two lenses shown are illuminated by a beam of parallel light from the left. Lens B is then moved slowly toward lens A.



The beam emerging from lens B is:

- (A) Initially parallel and then diverging
 (B) Always diverging
 (C) Initially converging and finally parallel
 (D) Always parallel
32. A point object O moves from the principal axis of a converging lens in a direction OP. The image of O, will move initially in the direction:

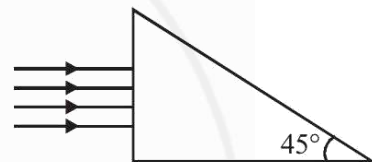


- (A) IQ
 (B) IR
 (C) IS
 (D) IU
33. When the object is at distances u_1 and u_2 the images formed by the same lens are real and virtual respectively and of the same size. Then focal length of the lens is:

- (A) $\frac{1}{2}\sqrt{u_1 u_2}$
 (B) $\frac{u_1 + u_2}{2}$
 (C) $\sqrt{u_1 u_2}$
 (D) $\sqrt{(u_1 + u_2)}$

34. A concave lens of glass, refractive index 1.5, has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a
 (A) convergent lens of focal length 3.5 R
 (B) convergent lens of focal length 3.0 R.
 (C) divergent lens of focal length 3.5 R
 (D) divergent lens of focal length 3.0 R

35. A beam of light consisting of red, green and blue and is incident on a right angled prism. The refractive index of the material of the prism for the above red, green and blue wavelengths are 1.39, 1.44 and 1.47 respectively. The prism will



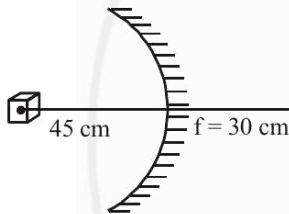
- (A) separate part of the red color from the green and blue colors
 (B) separate part of the blue color from the red and green colors
 (C) separate all the three colors from the other two colors
 (D) not separate even partially, any colors from the other two colors
36. A prism of angle 3° is made of glass having refractive index 1.64. Two thin prisms made of glass having refractive index 1.48 are intended to be coupled with the former prism to yield a combination without an average deviation. Which of the following angles cannot correspond to the two prism?
 (A) 2° and 2°
 (B) 1.5° and 2.5°
 (C) 6° and 2°
 (D) 5.5° and 3.5°

37. Two thin prisms of flint glass, with refracting angles of 6° and 8° respectively, possess dispersive powers in the ratio:
 (A) 4: 3
 (B) 3: 4
 (C) 1: 1
 (D) 9: 16

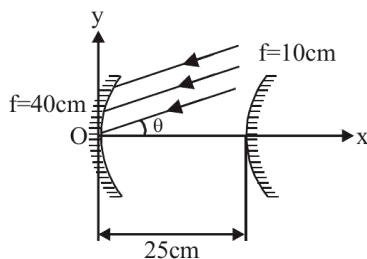
38. It is desired to make an achromatic combination of two lenses (L_1 & L_2) made of materials having dispersive powers ω_1 and $\omega_2 (< \omega_1)$. If the combination of lenses is converging then

- (A) L_1 is converging
 (B) L_2 is converging
 (C) Power of L_1 is greater than the power of L_2
 (D) None of these

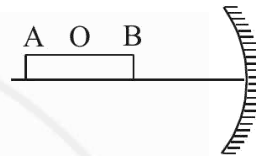
39. A concave mirror of focal length 20 cm is cut into two parts from the middle and the two parts are moved perpendicularly by a distance 1 mm from the previous principal axis AB. If an object is placed at a distance of 10 cm from the mirror on the line AB then, find the distance between the images formed by the two parts?
40. A thin rod of length $d/3$ is placed along the principal axis of a concave mirror of focal length = d such that its image, which is real and elongated, just touches the rod. Find the length of the image?
41. A cube of side length 1 mm is placed on the axis of a concave mirror at a distance of 45 cm from the pole as shown in the figure. One edge of the cube is parallel to the axis. The focal length of the mirror is 30 cm. Find approximate volume of the image.



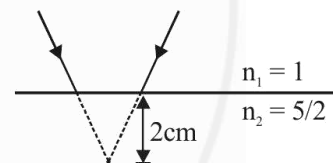
42. An experimentalist devises a method for finding the radius of curvature of a convex mirror. He uses a plane mirror strip between the object and the convex mirror and adjusts it till the two virtual images formed by reflection at both the mirrors coincide without parallax. In his observations, the object distance from the convex mirror is 0.5 m while it is 0.30 m in front of the plane mirror. Find the radius of curvature (in cm) of the convex mirror.
43. Consider a concave mirror kept at origin with focal length 40 cm. Parallel rays which subtend an angle $\theta = \frac{1}{40}$ radian are incident on it. A convex mirror is kept at a distance 25 cm from the first mirror as shown. Find the y-coordinate of the image (in cm) formed by the system of mirrors after two reflections.



44. A large temple has a depression in one wall. On the floor plan it appears as a indentation having spherical shape of radius 2.50 m. A worshiper stands on the center line of the depression, 2.00 m out from its deepest point, and whispers a prayer. Where is the sound concentrated after reflection from the back wall of the depression?
45. A rod of length ℓ is moving with constant speed 1 m/s towards the pole of a concave mirror of focal length ℓ . Find the rate of change in length of the image of the rod (mm/s) at the moment centre of the rod coincides with the centre of curvature of the mirror.



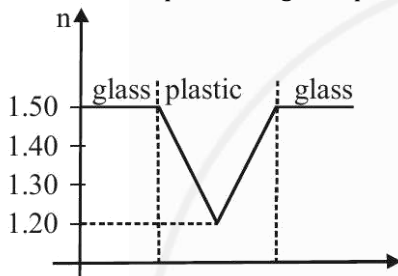
46. In the given figure rays incident on an interface would converge 2 cm below the interface if they continued to move in straight lines without bending. But due to refraction, the rays will bend and meet somewhere else. Find the distance of meeting point of refracted rays below the interface (in cm). (Assuming the rays to be making small angles with the normal to the interface)



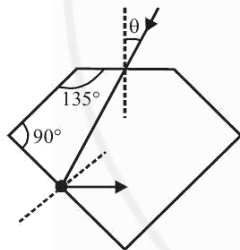
47. The $x-y$ plane is the boundary between two transparent media. Medium-1 with $z > 0$ has refractive index $\sqrt{2}$ and medium-2 with $z < 0$ has a refractive index $\sqrt{2}$. A ray of light in medium-1 given by the vector $A = 6\sqrt{3}\hat{i} + 8\sqrt{3}\hat{j} - 10\hat{k}$ is incident on the plane of separation. Find the unit vector in the direction of refracted ray in medium-2.
48. A beam of parallel rays of width b propagates in glass at an angle θ to its plane face. The beam width after it goes over to air through this face is _____ if the refractive index of glass is μ .
49. A convergent beam is incident on two slabs placed in contact as shown in figure. Where will the rays finally converge from the interface of B and air?
50. A point object is placed 33 cm from a convex mirror of curvature radius = 40 cm. A glass plate of thickness 6 cm and index 2.0 is placed between the object and mirror, close to the mirror. Find the distance of final image from the object?

51. A room contains air in which the speed of sound is 340 m/s. The walls of the room are made of concrete, in which the speed of sound is 1700 m/s.
 (a) Find the critical angle for total internal reflection of sound at the concrete-air boundary.
 (b) In which medium must the sound be traveling to undergo total internal reflection?

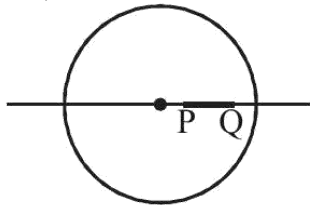
52. A thin plate of transparent plastic is embedded in a thick slab of glass. The index of refraction of the glass is $n = 1.50$; the index of refraction of the plate changes as shown in the diagram. A beam of light passes through glass and strikes the surface of the plastic plate. The maximum angle of incidence $\theta_{\max}$ which enables the beam to pass through the plate. Find $\theta_{\max}$.



53. A ray of light enters a diamond ($n = 2$) from air and is being internally reflected near the bottom as shown in the figure. Find maximum value of angle θ possible?

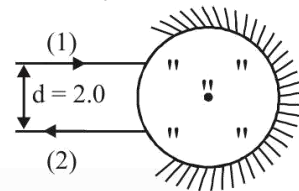


54. A small object of length 1 mm lies along the principal axis of a spherical glass of radius $R = 10$ cm and refractive index is $3/2$. The object is seen from air along the principal axis from left. The distance of object from the centre is 5 cm. Find the size of the image. Is real, inverted?



55. A narrow parallel beam of light is incident on a transparent sphere of refractive index 'n'. If the beam finally gets focussed at a point situated at a distance = $2 \times$ (radius of sphere) from the centre of the sphere, then find n?

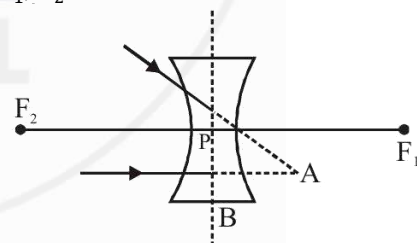
56. A transparent sphere of radius $R = 2.0$ m has a mirrored surface on its right half as shown in figure. A light ray travelling in air is incident on the left side of the sphere. The incident light ray
 (1) & exiting light ray
 (2) are parallel & separated by distance $d = 2.0$ m.
 Then find the refractive index of the material.
 (Take: $\sin 15^\circ = 0.25$)



57. A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens. If the lens is removed, the point where the rays meet will move 5 cm closer to the mounting that holds the lens. Find the focal length (in cm) of the lens without sign.

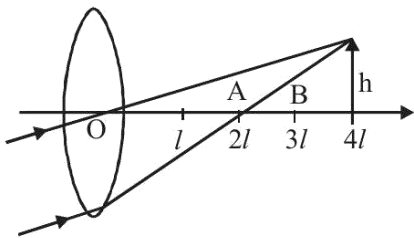
58. A lens is placed at origin, with x-axis as its principal axis. A ray of light is incident on it from the negative side of x-axis along the line $y = \frac{x}{400} + 0.1$, where x, y are in cm. Focal length of lens is 20 cm. Find the equation of the ray after passing the lens.

59. The rays of a converging beam meet at a point A. A diverging lens is placed in their path in the plane B. Plot the position of the point where the rays meet after passing through the lens. The position of the principal foci F_1, F_2 is known.

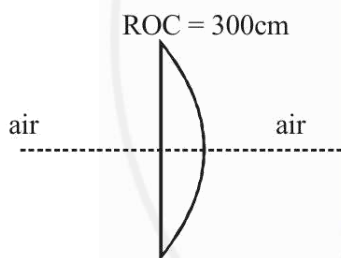


60. A converging bundle of light rays in the shape of a cone with the semi-vertex angle of 37° falls on a circular diaphragm of 14 cm diameter. A lens with a focal power of $25/3$ diopter is fixed in diaphragm. What will be the new semi vertex angle of cone (in degree)

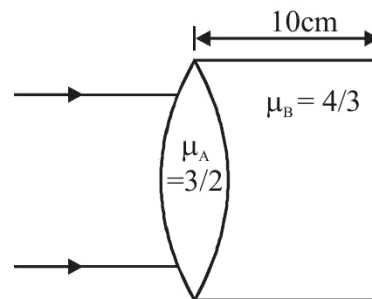
61. A thin converging lens L_1 forms a real image of an object located far away from the lens as shown in the figure. The image is located at a distance 4l and has height h. A diverging lens of focal length l is placed 2l from lens L_1 at A. Another converging lens of focal length 2l is placed 3l from lens L_1 at B. Find the height of final image thus formed?



62. There is a hole in middle of a thin converging lens whose radius is $\frac{1}{2} \times$ radius of the lens. The lens has a focal length of 4 cm. A point object is kept 9 cm from a screen and this lens is kept in between. It is seen that there is a single circular illuminated spot on the wall with a sharp edge (as seen in figure). What is the distance of the lens from the object (in cm)? Consider the case when the rays after refraction from the lens are converging in nature.
63. Plane surface of a thin planoconvex lens reflects 50% of light, while the curved surface is completely transparent, if final image of 'O' after refraction through thin lens coincides with the image formed due to partial reflection from plane surface. If distance between O and lens is x (in m) then find the value of x/4. (Given $\mu = 3/2$)



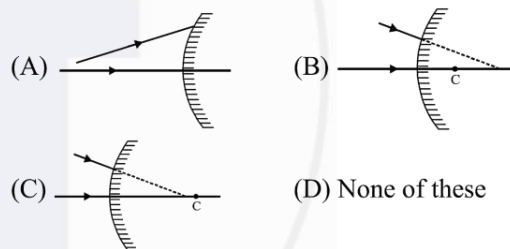
64. A plano convex lens ($\mu = 1.5$) has a maximum thickness of 1 mm. If diameter of its aperture is 4 cm. Find
(i) Radius of curvature of curved surface;
(ii) Its focal length in air.
65. A point source of light is kept at a distance of 15 cm from a converging lens, on its optical axis. The focal length of the lens is 10 cm and its diameter is 3 cm. A screen is placed on the other side of the lens, perpendicular to the axis of lens, at a distance 20 cm from it. Then find the area of the illuminated part of the screen?
66. A parallel paraxial beam of light is incident on the arrangement as shown $\mu_A = 3/2$, $\mu_B = 4/3$, the two spherical surfaces are very close and each has radius of curvature 10 cm. Find the point where the rays are focussed. (w.r.t. point of entry)



67. A flint glass convex lens of focal length 16 cm is placed in contact with crown glass lens. The refractive indices of crown glass for violet and red colour are $\mu_v = 1.525$, $\mu_r = 1.515$ and for flint glass $\mu_v' = 1.655$ and $\mu_r' = 1.645$.
(a) Find the nature and focal length of the crown glass lens which will form an achromatic combination with the flint glass lens.
(b) What is the focal length of the combination?

EXERCISE - II

1. In which of the following diagrams the image formed is virtual and inverted?

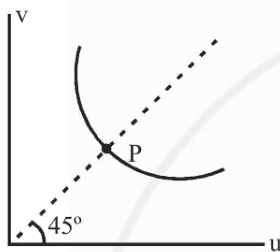


2. An infinitely long rod lies along the axis of a concave mirror of focal length f. The near end of the rod is at distance $u > f$ from the mirror. Its image will have a length
(A) $\frac{uf}{u-f}$
(B) $\frac{uf}{u+f}$
(C) $\frac{f^2}{u+f}$
(D) $\frac{f^2}{u-f}$
3. A short linear object of length b lies along the axis of a concave mirror of focal length f, at a distance u from the mirror. The size of the image is approximately
(A) $b \left(\frac{u-f}{f} \right)^{1/2}$
(B) $b \left(\frac{f}{u-f} \right)$
(C) $b \left(\frac{u-f}{f} \right)$
(D) $b \left(\frac{f}{u-f} \right)^2$

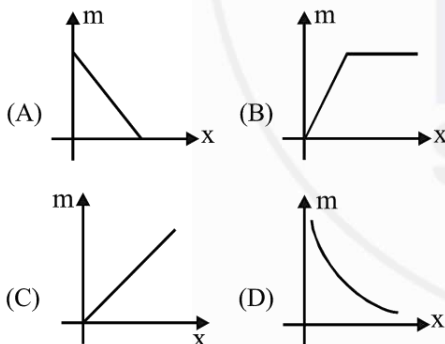
4. The table below lists object and image positions for four objects placed in front of mirrors, using Cartesian sign convention with pole of the mirror as origin. In the following cases, the case in which images is formed by a convex spherical mirror, is:

Object Position	Image Position
(A) -25.0 cm	-16.7 cm
(B) -5.0 cm	10.0 cm
(C) -20.0 cm	5.71 cm
(D) -40.0 cm	80.0 cm

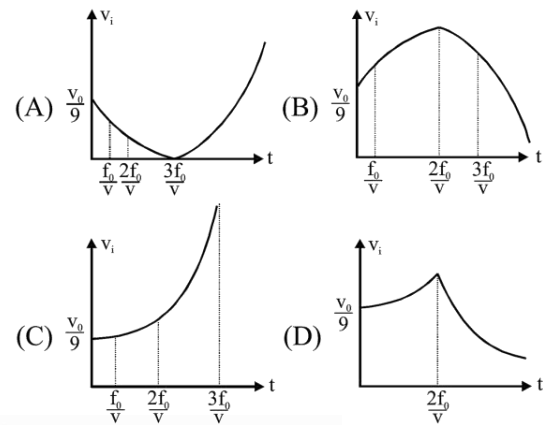
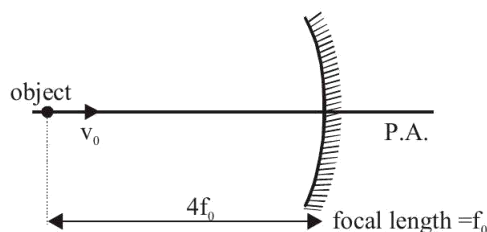
5. The graph shows the variation of v with change in u for a mirror. Points plotted above the point P on the curve are for values of v



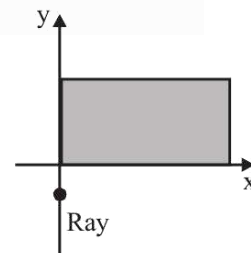
- (A) smaller than f
 (B) smaller than $2f$
 (C) larger than $2f$
 (D) larger than f
6. Which of the following graphs is the magnification of a real image against the distance of object from the focus of a concave mirror?



7. An object is moving with constant speed v_0 , towards a concave mirror along its principal axis, from a distance $4f_0$ as shown, where f_0 is the focal length of mirror. The correct graph of speed of image (v_i) with time (t) is

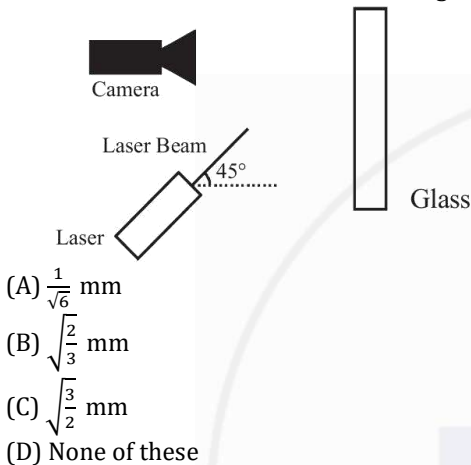


8. A point source of light is 60 cm from a screen and is kept at the focus of a concave mirror which reflects light on the screen. The focal length of the mirror is 20 cm. The ratio of average intensities of the illumination on the screen when the mirror is present and when the mirror is removed is:
- (A) 36:1
 (B) 37:1
 (C) 49:1
 (D) 10:1
9. A bird is flying 3 m above the surface of water. If the bird is diving vertically down with speed = 6 m/s, his apparent velocity as seen by a stationary fish underwater is:
- (A) 8 m/s
 (B) 6 m/s
 (C) 12 m/s
 (D) 4 m/s
10. A ray of light travelling in air is incident almost along y-axis on medium of variable refractive index at the origin. The refractive index of the medium changes according to the relation $\mu = 1 + x^2$. What is the light ray vector at the point where the x coordinate becomes equal to 1?

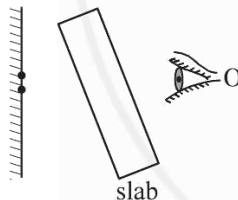


- (A) $\left(\frac{\sqrt{3}}{2}\right)\hat{i} + \left(\frac{1}{2}\right)\hat{j}$
 (B) $\left(\frac{1}{2}\right)\hat{i} + \left(\frac{\sqrt{3}}{2}\right)\hat{j}$
 (C) $\left(\frac{1}{2}\right)\hat{i} + \left(\frac{1}{\sqrt{2}}\right)\hat{j}$
 (D) None of these

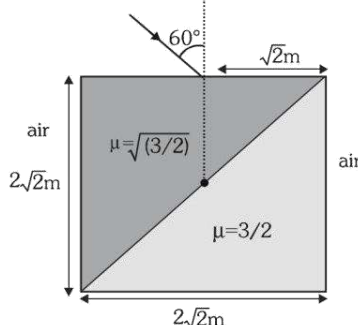
11. A quality control engineer working at the Davinci Glass Company has invented a new method to measure the thickness of glass panes. By directing a laser beam at an incident angle of 45° w.r.t. the glass, he notices that not one but two parallel beams are reflected. He measures the distance between the reflected beams with an automated camera system, as shown. (The camera is pointed directly at the reflected beams). Assuming that the index of refraction of glass is $\sqrt{2}$, and the distance between the reflected beams is measured to be 1 mm, how thick is the glass?



12. The observer at O views two closely spaced spots on a vertical wall through an angled glass slab as shown. As seen by observer, the spots appear.

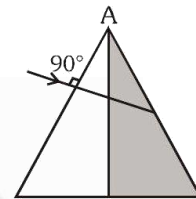


- (A) shifted upward
(B) shifted downward
(C) spaced farther apart
(D) spaced closer together
13. A ray of light strikes a cubical slab surrounded by air as shown in the figure. Then the geometrical path length traversed by the light in the slab will be:



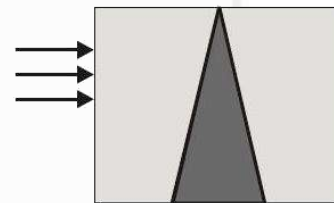
- (A) $2\sqrt{3}$ m
(B) 2m
(C) 6 m
(D) $\left(\frac{\sqrt{3}}{2} + \frac{3}{2}\right)$ m

14. A ray of light is incident normally on the first refracting face of the prism of refracting angle A. The ray of light comes out at grazing emergence. If one half of the prism (shaded position) is knocked off, the same ray will

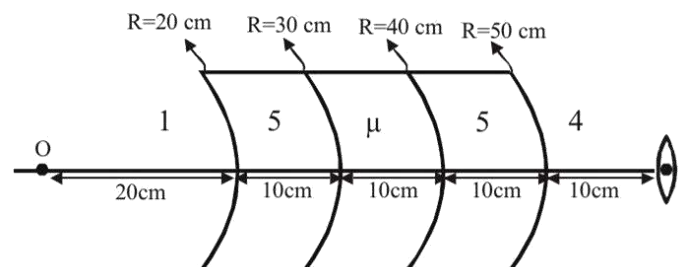


- (A) Emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec\frac{A}{2}\right)$
(B) Not emerge out of the prism
(C) Emerge at an angle of emergence $\sin^{-1}\left(\frac{1}{2}\sec\frac{A}{4}\right)$
(D) None of these

15. A thin isosceles prism with angle 4° and refractive index $5/4$ is placed inside a transparent tube with liquid (refractive index = 1.5) as shown. The deviation of light due to prism will be

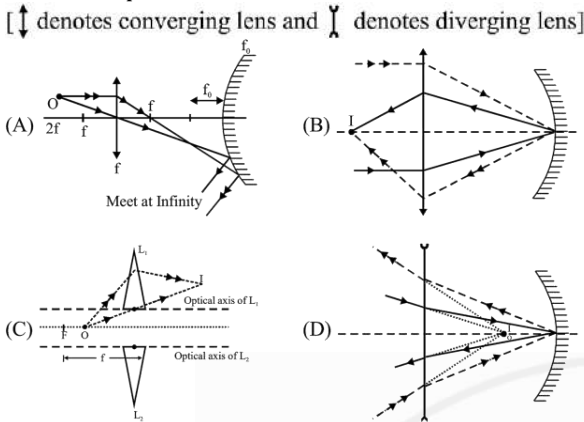


- (A) 0.8° upward
(B) 0.8° downward
(C) 0.67° upward
(D) 0.67° downward
16. In the arrangement shown, separation between observer and object as seen by observer is

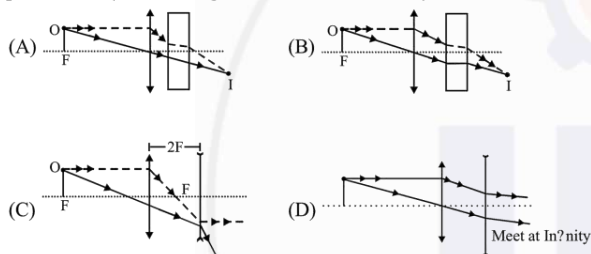


- (A) 60 cm
(B) more than 60 cm
(C) less than 60 cm
(D) depends on μ

17. Choose the incorrect ray diagram(s). All the rays shown are paraxial.



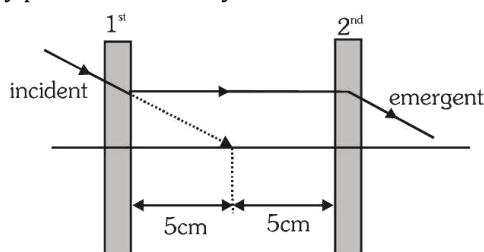
18. Choose incorrect ray diagram [q denotes converging lens and y denotes diverging lens] All symbols have their usual meaning and all the rays shown are paraxial. (focal length of each lens is F)



19. A point object is located 30 cm in front of a camera lens of aperture diameter 6 cm. The image is focussed on the film plate. If the film plate is moved back 10 cm the width of the image blurs to 1 cm wide. Find the focal length of the lens.

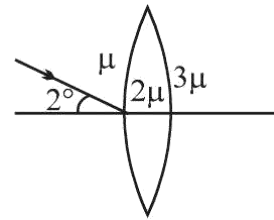
- (A) 20 cm
(B) 10 cm
(C) 30 cm
(D) 40 cm

20. Look at the ray diagram shown, what will be the focal length of the 1st and the 2nd lens, if the incident light ray passes without any deviation?



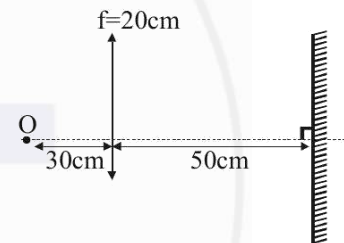
- (A) -5 cm and +10 cm
(B) +5 cm and +10 cm
(C) -5 cm and +5 cm
(D) +5 cm and -5 cm

21. A light ray hits the pole of a thin biconvex lens as shown in figure. The angle made by the emergent ray with the optic axis will be nearly



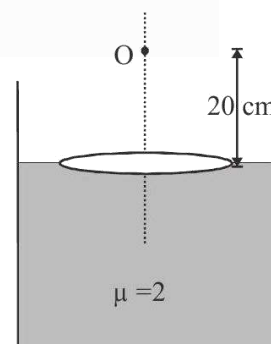
- (A) 0°
(B) (1/3)°
(C) (2/3)°
(D) 2°

22. In the given situation object and plane mirror are fixed while the lens is performing SHM perpendicular to principle axis with an amplitude 4 mm. At mean position of the lens the object lies on the principal axis. Distance between two extreme positions of image formed by mirror is (consider paraxial light rays)



- (A) 1.2 cm
(B) 2.4 cm
(C) 0.8 cm
(D) can not be determined

23. A point object O is placed at a distance of 20 cm in front of a equi-convex lens ($\mu_g = 1.5$) of focal length 10 cm. The lens is placed on a liquid of refractive index 2 as shown. Image will be formed at a distance h from lens. The value of h is



- (A) 5 cm
(B) 10 cm
(C) 20 cm
(D) 40 cm

24. Consider a common mirage formed by super-heated air just above a roadway. A truck driver is in a medium of $\mu = 1.0003$ looks forward. He perceives the illusion of a patch of water ahead on the road, where his line of sight makes an angle of 1.20° below the horizontal. Find the index of refraction of the air just above the road surface. Take $\cos(1.20^\circ) = 0.99978$. (Hint: Treat this as a problem in total internal reflection).

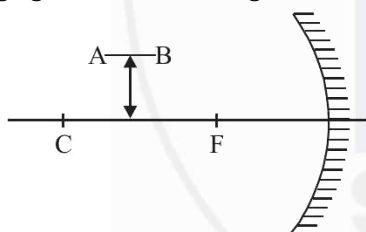
(A) 1.00006
(B) 1.0001
(C) 1.00008
(D) None of these

25. A thin lens with focal length f is to be used as a magnifying glass. Which of the following statements regarding the situation is true?

(A) A converging lens may be used, and the object be placed at a distance greater than $2f$ from the lens.
(B) A diverging lens may be used, and the object be placed between f and $2f$ from the lens.
(C) A converging lens may be used, and the object be placed at a distance less than f from the lens.
(D) A diverging lens may be used, and the object be placed at any point other than the focal point.

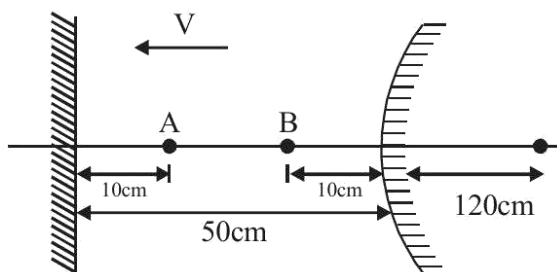
MULTIPLE CORRECT TYPE QUESTIONS

26. An object AB is placed parallel and close to the optical axis between focus F and centre of curvature C of a converging mirror of focal length f as shown in figure.



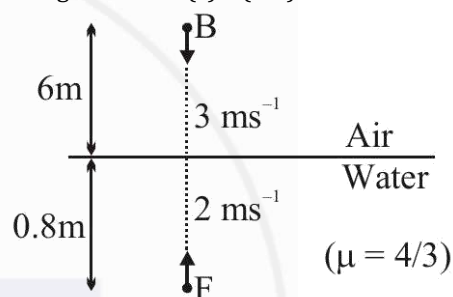
(A) Image of A will be closer than that of B from the mirror.
(B) Image of AB will be parallel to the optical axis.
(C) Length of image is equal to AB.
(D) Length of image is more than AB.

27. In the figure shown consider the first reflection at the plane mirror and second at the convex mirror. AB is object.



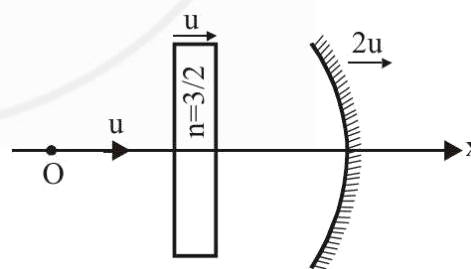
(A) the second image is real, inverted of $1/5^{\text{th}}$ magnification w.r.t AB
(B) the second image is virtual and erect with magnification $1/5$ w.r.t AB
(C) the second image moves towards the convex mirror
(D) the second image moves away from the convex mirror.

28. A fish, F in the pond, is at a depth of 0.8 m from water surface and is moving vertically upwards with velocity 2 ms^{-1} . At the same instant, a bird B is at a height of 6 m from water surface and is moving downwards with velocity 3 ms^{-1} . At this instant both are on the same vertical lines as shown in the figure. Which of the following statement(s) is(are) correct?



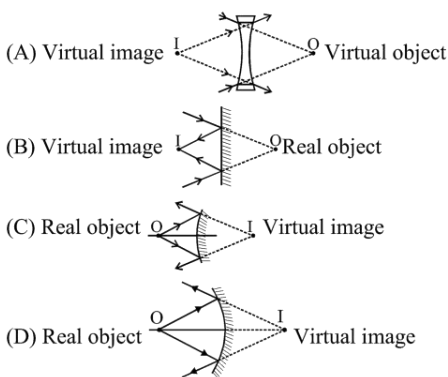
(A) Height of B, observed by F (from itself) is equal to 8.00 m.
(B) Depth of F, observed by B (from itself) is equal to 6.60 m.
(C) Velocity of B, observed by F (relative to itself) is equal to 5.00 ms^{-1} .
(D) Velocity of F, observed by B (relative to itself) if equal to 4.50 ms^{-1}

29. For the system as shown in the figure, the image formed by the concave mirror.

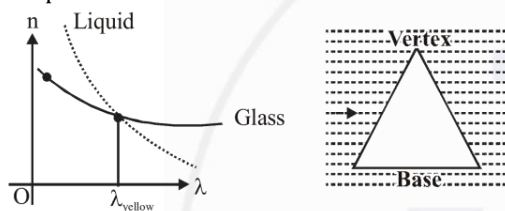


(A) will have speed greater than the speed of the object.
(B) will move in the direction of motion of the mirror.
(C) will have speed greater than the speed of the mirror.
(D) moves away from the mirror.

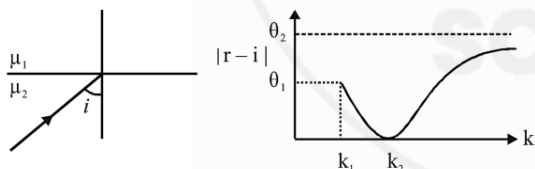
30. The nature of object and image given with each of the optical condition is shown. Choose the correct option(s)



31. A glass prism is immersed in a hypothetical liquid. The curves showing the refractive index n as a function of wavelength λ for glass and liquid are as shown in the figure. When a ray of white light is incident on the prism parallel to the base:

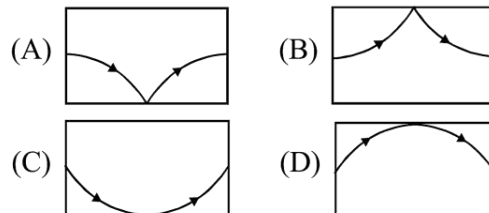


- (A) yellow ray travels without deviation
(B) blue ray is deviated towards the vertex
(C) red ray is deviated towards the base
(D) there is no dispersion
32. The figure shows a ray incident at an angle $i = \pi/3$. If the plot drawn shows the variation of $|r - i|$ versus $\frac{\mu_1}{\mu_2} = k$, (r = angle of refraction)

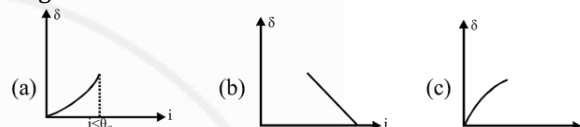


- (A) the value of k_1 is $\frac{2}{\sqrt{3}}$
(B) the value of $\theta_1 = \frac{\pi}{6}$
(C) the value of $\theta_2 = \frac{\pi}{3}$
(D) the value of k_2 is 1
33. For the refraction of light through a prism
(A) For every angle of deviation there are two angles of incidence.
(B) The light travelling inside an equilateral prism is necessarily parallel to the base when prism is set for minimum deviation.
(C) There are two angles of incidence for maximum deviation.
(D) Angle of minimum deviation will increase if refractive index of prism is increased keeping the outside medium unchanged if $\mu_p > \mu_s$

34. A cubic container is filled with a liquid whose refractive index increases linearly from top to bottom. Which of the following figures may represent the path of a ray of light inside the liquid?

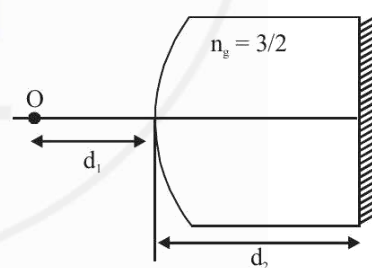


35. Following are graphs of angle of deviation versus angle of incidence



- Based on the above graphs mark the correct options
(A) Graph-a may be a part of the graph for ray of light that is incident on rarer medium.
(B) Graph-b may be for ray of light that is incident on rarer medium.
(C) Graph-c may be a part of the graph for ray of light that is incident on denser medium.
(D) Graph-b may be a part of the graph for ray of light that is reflected from a plane mirror

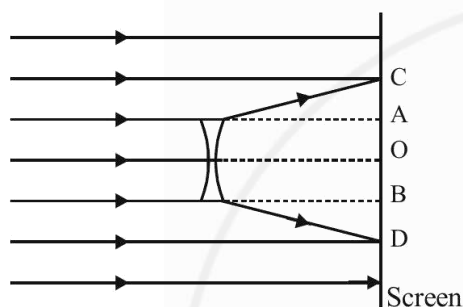
36. In the figure shown a point object O is placed in air on the principal axis. The radius of curvature of the spherical surface is 60 cm. I_f is the final image formed after all the refractions and reflections.



- (A) If $d_1 = 120$ cm, then the ' I_f ' is formed on ' O ' for any value of d_2 .
(B) If $d_1 = 240$ cm, then the ' I_f ' is formed on ' O ' only if $d_2 = 360$ cm.
(C) If $d_1 = 240$ cm, then the ' I_f ' is formed on ' O ' for all values of d_2 .
(D) If $d_1 = 240$ cm, then the ' I_f ' cannot be formed on ' O '.
37. Optical axis of a thin equi-convex lens is the X -axis. The co-ordinate of a point object and its image are $(-20 \text{ cm}, 1 \text{ cm})$ and $(25 \text{ cm}, -2 \text{ cm})$ respectively
(A) the lens is located at $x = 5 \text{ cm}$
(B) the lens is located at $x = -5 \text{ cm}$
(C) the focal length of the lens is 10 cm
(D) the focal length of the lens is 15 cm

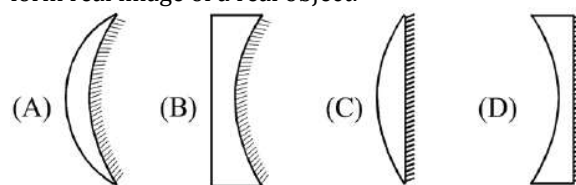
38. The principal axis of an optical device is along $y = -1$, image of a small body placed at $(-30, 3)$ is formed at a point $(60, -3)$. Then the optical device is
 (A) A convex lens of focal length 20 cm
 (B) A concave mirror of focal length 60 cm
 (C) A concave lens of focal length 20 cm
 (D) A convex mirror of focal length 60 cm

39. A concave lens is placed in the path of a uniform parallel beam of light falling on a screen as shown. Then



- (A) intensity of light will be the uniform everywhere on the screen.
 (B) intensity in the region AB will be lesser than what it would be in the absence of the lens.
 (C) in the region AC and BD, the intensity will be lesser than what it would be in the absence of the lens.
 (D) in the region AC and BD, the intensity will be more than what it would be in the absence of the lens.
40. In displacement method, the distance between object and screen is 96 cm. The ratio of length of two images formed by a convex lens placed between them is 4.84:
 (A) Ratio of the length of object to the length of shorter image is 11/5.
 (B) Distance between the two positions of the lens is 36 cm.
 (C) Focal length of the lens is 20.625 cm.
 (D) Distance of the lens from the shorter image is 30 cm.
41. Figure shows a convex lens cut symmetrically into two equal halves and separated laterally by a distance h . point object O placed symmetrically at a distance 30 cm, from the lens halves, within the focus, forms two images separated by a distance d . A plot of d versus h is shown in figure. The focal length of the lens is
 (A) 22.5 cm
 (B) 40 cm
 (C) 45 cm
 (D) 20 cm

42. Which of the following silvered lenses kept in air may form real image of a real object.

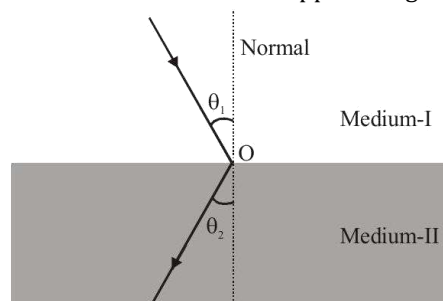


43. A man wanted to get a picture of a Zebra. He photographed a white donkey after fitting a glass with black streaks onto the objective of his camera.
 (A) the image will look like a white donkey on the photograph.
 (B) the image will look like a Zebra on the photograph.
 (C) the image will be more intense compared to the case in which no such glass is used.
 (D) the image will be less intense compared to the case in which no such glass is used.
44. A lens is formed by a material having refracting index $\frac{3}{2}$ & radii of curvature 20 cm & 10 cm. Then choose the correct option(s):
 (A) If lens is concavo-convex & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm
 (B) If lens is concavo-convex & light is falling on the surface having radius of curvature 10 cm, then focal length is -40 cm
 (C) If lens is convexo-concave & light is falling on the surface having radius of curvature 20 cm, then focal length is 40 cm
 (D) If lens is convexo-concave & light is falling on the surface having radius of curvature 10 cm, then focal length is -40 cm

COMPREHENSION TYPE QUESTIONS

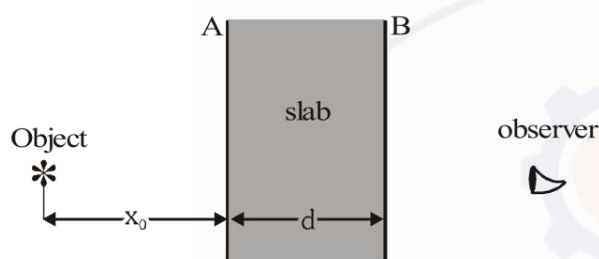
Paragraph for Question 45 and 46

If Snell's law is found valid for two media, whose refractive indexes have opposite signs, the incident and the refracted rays must lay on the same side of the normal at the point of incidence. In the figure is shown a ray crossing the interface between two media makes angle θ_1 and θ_2 with the normal in medium-I and medium-II of refractive indexes μ_1 and μ_2 . Refractive indexes of these media have opposite signs.



Snell's law is expressed for the above situation as usual by the equation $\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$. Therefore, to satisfy equation $\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$, the angles θ_1 and θ_2 must also have opposite signs.

45. A self-luminous point object placed at distance x_0 from a slab of transparent material of negative refractive index is viewed through it as shown in the figure. Thickness of the slab is d , refractive index of material of the slab with respect to the outside medium is $\mu = -1$. For different magnitudes of value of x_0 and d the observer may find real or virtual image of the object. Which of the following statement is true? Do not use any sign conventions.



- (A) If $x_0 > d$ image is virtual and for $x_0 < d$ image is real.
 (B) If $x_0 < d$ image is virtual and for $x_0 > d$ image is real.
 (C) If $x_0 \leq d$ image is virtual and for $x_0 > d$ image is real.
 (D) If $x_0 > d$ image is virtual and for $x_0 \leq d$ image is real.
46. Suppose a virtual object is placed instead of a real object at a distance x_0 from the surface A of the slab. Now which of the following statement is true? Do not use any sign conventions.
- (A) If $x_0 < d$ a real image is formed outside the slab and for $x_0 > d$ a virtual image is formed inside the slab.
 (B) If $x_0 < d$ a real image is formed and for $x_0 > d$ a virtual image is formed and in both the case the image is outside the slab.
 (C) For both the cases $x_0 < d$ and $x_0 > d$ a real and erect image is formed outside the slab.
 (D) For both the cases $x_0 < d$ and $x_0 > d$ a real and inverted image is formed outside the slab.

Paragraph for Question No. 47 to 49

When a wave reaches at boundary of any medium it get partially refracted and partially reflected. Direction of refracted as well as reflected ray can be given by Snell's law.

$$\frac{\sin i}{v_i} = \frac{\sin r}{v_r} \quad v_i = \text{velocity of incident wave}$$

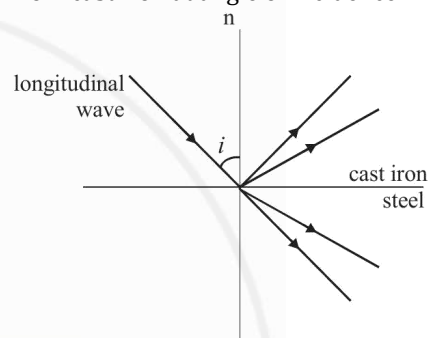
$$v_r = \text{velocity of reflected/refracted wave}$$

For the reflected wave of same type, $v_i = v_r$ as the wave returns to the same medium. Thus $\sin i = \sin r \Rightarrow i = r$.

It is observed that when a longitudinal wave reaches at a surface, it generates longitudinal and transverse wave. These two waves get reflected and transmitted in direction suggested by Snell's law.

Given:

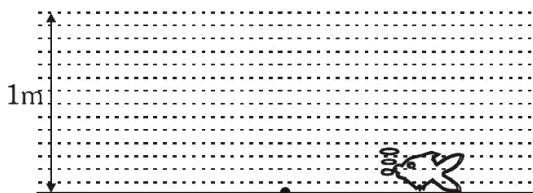
that velocity of longitudinal wave in steel = 4000 m/s
 velocity of longitudinal wave in cast iron = 3000 m/s
 velocity of transverse wave in steel = 3200 m/s
 velocity of transverse wave in cast iron = 1600 m/s
 Longitudinal waves come at boundary of cast iron and steel from cast iron at angle of incidence i .



47. If the longitudinal wave is incident at an angle of 30° , then the angle between the refracted longitudinal and reflected transverse wave will be
- (A) $\sin^{-1} \frac{2}{3} - \sin^{-1} \frac{4}{15}$
 (B) $\sin^{-1} \frac{2}{3} + \sin^{-1} \frac{4}{15}$
 (C) $\pi - \sin^{-1} \frac{2}{3} - \sin^{-1} \frac{4}{15}$
 (D) $\frac{5\pi}{6} - \sin^{-1} \frac{2}{3}$
48. If the angle of incidence of longitudinal wave is such that produced longitudinal wave just fails to enter steel, then the angle of reflection for transverse wave will be:
- (A) $\sin^{-1} \frac{3}{4}$
 (B) $\sin^{-1} \frac{2}{5}$
 (C) $\sin^{-1} \frac{4}{5}$
 (D) $\frac{\pi}{2}$
49. Choose the CORRECT statement:
- (A) For any angle of incidence, we can never get transverse wave in steel.
 (B) Reflected longitudinal wave will be closer to normal than reflected transverse wave (normal is taken towards cast iron)
 (C) Refracted transverse wave will bend towards the normal.
 (D) Refracted longitudinal wave will bend away from the normal.

Paragraph for Question 50 & 51

An object O is placed at the bottom of a pond of depth 1 m. A fish is lying at the bottom of the pond as shown in figure.

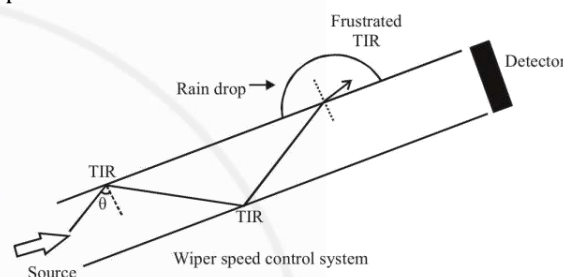


50. If refractive index of the liquid is $\sqrt{2}$, what will be minimum distance between fish and the object so that image of the object can be seen by the fish.
- (A) 3 m
(B) 2 m
(C) 1 m
(D) none of these
51. Suppose fish is observing the image and object starts moving vertically upwards. Pick the correct option
- (A) image will not move
(B) image will move vertically upwards
(C) image will move vertically downward
(D) none of these

Paragraph for Question 52 & 54

Frustrated TIR (F-TIR): In optics, when light rays traveling in a denser medium strike at medium boundary at an angle greater than critical angle, TIR occurs and a surface wave which is called Evanescent wave forms in rarer medium. An evanescent wave is a near-field standing wave with an intensity that exhibits exponential decay with distance (less than wavelength) from the boundary at which the wave is formed. Imagine that a beam of light traveling within a block of glass is internally reflected at a boundary. Presumably if you pressed another piece of glass against the first, the air-glass interface would be made to vanish and the beam would then propagate onward undisturbed. Further more, this transition from total to no reflection occurs gradually as the air film between them thinned out as explained above, when third medium with a higher refractive index (than the low-index second medium) is placed within less than several wavelengths distance from the interface between the first medium and the second medium, the evanescent wave will be different from the one under "ordinary conditions" and it will pass energy across the second into the third medium (evanescent wave coupling). This process is called "frustrated" total internal reflection (FTIR) and is very similar to quantum tunneling. An example of application of this principle is automatic Wiper speed control found in high end cars like Skoda, Audi, BMW etc.

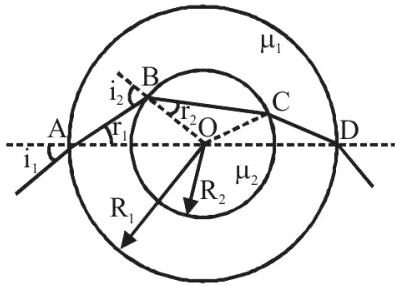
Figure shows an example of an optical system designed to detect the amount of water present on the windshield of a car to adjust the wiper speed. As shown in this figure, we can use the windshield as a waveguide to guide the light from a source located at one end (bottom of the windshield) to a detector located in the opposite end. The light suffers total-internal reflection (TIR) at the glass-air interface. However, when rain drops are present, some of the light will suffer frustrated TIR escaping outside the waveguide. Since we know the power of the light source, a given drop in power can be correlated to the amount of water present and used to adjust the wiper speed.



52. Choose the CORRECT statement.
- (A) The energy carried by evanescent wave is zero in both TIR and FTIR.
(B) The energy carried by evanescent wave is non-zero in both TIR and FTIR.
(C) The energy carried by evanescent wave is zero in TIR and non-zero in FTIR.
(D) The energy carried by evanescent wave is non-zero in TIR and zero in FTIR.
53. In the following systems which should not be based on FTIR?
- (A) Credit card swiping system
(B) Thumb attendance system
(C) Finger print scanner
(D) Multi touch screen
54. For the wiper speed control system to work, the angle of incidence on the glass air interface is θ , then
- (A) $\theta < \sin^{-1} \left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}} \right)$
(B) $\theta > \sin^{-1} \left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}} \right)$
(C) $\sin^{-1} \left(\frac{\mu_{\text{air}}}{\mu_{\text{glass}}} \right) < \theta < \sin^{-1} \left(\frac{\mu_{\text{water}}}{\mu_{\text{glass}}} \right)$
(D) $\theta > \sin^{-1} \left(\frac{\mu_{\text{air}}}{\mu_{\text{glass}}} \right)$

Paragraph for Questions 55 and 56

There is a spherical glass ball of refractive index μ_1 and another glass ball of refractive index μ_2 inside it as shown in figure. The radius of the outer ball is R_1 and that of inner ball is R_2 . A ray is incident on the outer surface of the ball at an angle i_1 .



55. Find the value of r_1

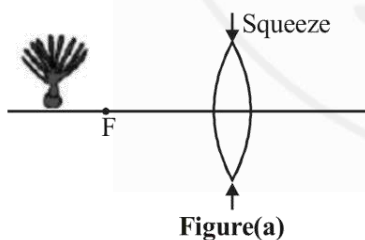
- (A) $\sin^{-1} \left(\frac{\sin i_1}{\mu_1} \right)$
 (B) $\sin^{-1} (\mu_1 \sin i_1)$
 (C) $\sin^{-1} \left(\frac{\mu_1}{\sin i_1} \right)$
 (D) $\sin^{-1} \left(\frac{1}{\mu_1 \sin i_1} \right)$

56. Find the value of r_2

- (A) $\sin^{-1} \left(\frac{R_1}{\mu_2 R_2} \sin i_1 \right)$
 (B) $\sin^{-1} \left(\frac{R_2}{\mu_2 R_1} \sin i_1 \right)$
 (C) $\sin^{-1} \left(\frac{R_1}{\mu_1 R_2} \frac{1}{\sin i_1} \right)$
 (D) $\sin^{-1} \left(\frac{R_2}{\mu_1 R_1} \sin i_1 \right)$

Paragraph for Questions 57 and 59

A turnip sits before a thin converging lens, outside the focal point of the lens. The lens is filled with a transparent gel so that it is flexible; by squeezing its ends toward its center [as indicated in figure(a)], you can change the curvature of its front and rear sides.



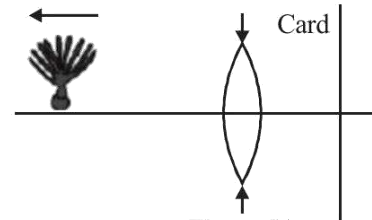
57. When you squeeze the lens, the image.

- (A) moves towards the lens
 (B) moves away from the lens
 (C) shifts up
 (D) remains as it is

58. When you squeeze the lens, the lateral height of image.

- (A) increases
 (B) decreases
 (C) remains same
 (D) data insufficient

59. Suppose that a sharp image must be formed on a card which is at a certain distance behind the lens [figure(b)], while you move the turnip away from the lens, then you should



Figure(b)

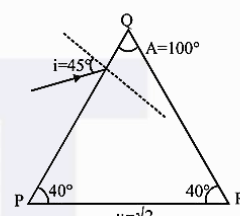
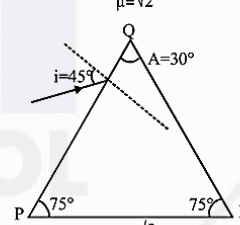
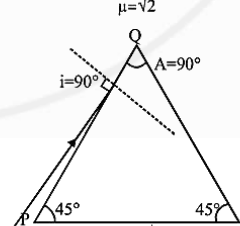
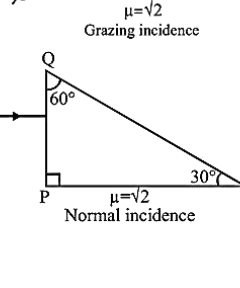
- (A) decrease the squeeze of the lens
 (B) increase the squeeze of the lens
 (C) keep the card and lens as it is.
 (D) move the card away from the lens

MATRIX MATCH TYPE QUESTIONS

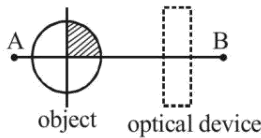
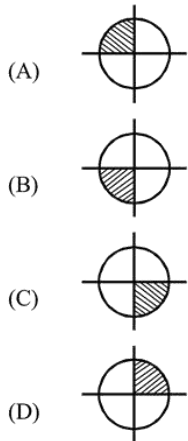
60. Light is incident at surface PQ of prism as shown in column I then match the column I with column II (surrounding medium is air in all cases)

Column I

Column II

- | | |
|---|---|
| <p>(A) </p> | <p>(P) Total internal reflection takes place at surface QR.</p> |
| <p>(B) </p> | <p>(Q) Light emerges normally from the surface QR</p> |
| <p>(C) </p> | <p>(R) Light emerges parallel to surface QR</p> |
| <p>(D) </p> | <p>(S) The light ray emerges from face PR perpendicularly
 (T) When light ray passes through the prism it is parallel to the base PR.</p> |

61. A very small circular object is kept in front of an optical device as shown in figure. The plane of object is parallel to the optical device. Match the images as seen by the observer (ignoring magnification):


Column-I (Probable image)
Column-II (Device)


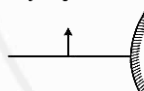
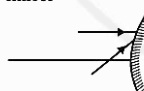
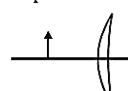
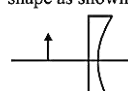
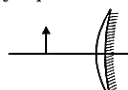
- (P) Plane mirror
(The observer is at A)
- (Q) Concave mirror
(The observer is at A)
- (R) Convex mirror
(The observer is at A)
- (S) Convex lens
(The observer is at A)
- (T) Concave lens
(The observer is at A)

62. Medium of lens in (R), (S), (T) is denser than surroundings

Column-I

- (A) Real erect image cannot be formed
- (B) Virtual erect image cannot be formed
- (C) Real inverted image cannot be formed
- (D) Virtual inverted image cannot be formed

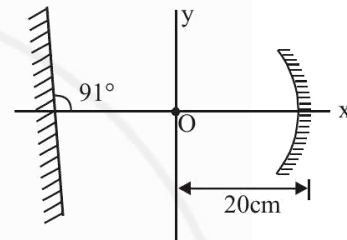
Column-II

- (P) Object placed in front of a convex mirror

- (Q) Converging beam incident on a convex mirror

- (R) Object placed in front of a lens having the shape as shown

- (S) Object placed in front of a lens having the shape as shown

- (T) Object placed in front of a silvered lens as shown


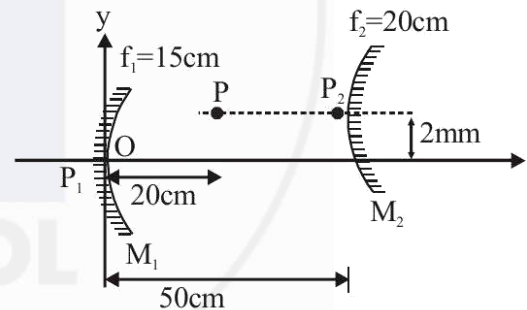
63. A light ray parallel to the x-axis strikes the outer reflecting surface of a sphere at a point (2,2,0). Its center is at the point (0,0,-1). The unit vector along the direction of the reflected rays is $x\hat{i} + y\hat{j} + z\hat{k}$. Find the value of $\frac{yz}{x^2}$

64. A point object is placed at the centre of curvature of a concave mirror (taken as origin). A plane mirror is also placed at a distance of 10 cm from the object as shown. Consider two reflection first at plane mirror and then at concave mirror (x_0, y_0).

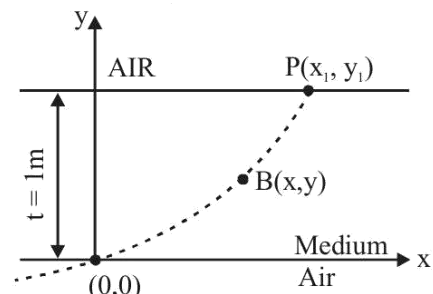
Find $\frac{\pi x_0}{y_0}$ the coordinates of the image thus formed are.



65. Find the co-ordinates of image of point object P formed after two successive reflection in situation as shown in figure considering first reflection at concave mirror and then at convex.

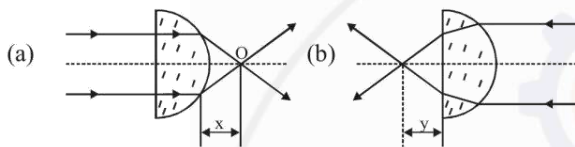


66. A ray of light travelling in air is incident at grazing angle (incident angle = 90°) on a long rectangular slab of a transparent medium of thickness $t = 1.0$ (see figure). The point of incidence is the origin A(0,0). The medium has a variable index of refraction $n(y)$ given by: $n(y) = [ky^{3/2} + 1]^{1/2}$, where $k = 1.0 \text{ m}^{-3/2}$. The refractive index of air is 1.0.

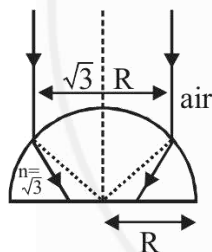


- (i) Obtain a relation between the slope (dy/dx) of the trajectory of the ray at a point $B(x, y)$ in the medium and the incident angle (i) at that point.
- (ii) Find the value of $n \sin i$.
- (iii) Obtain an equation for the trajectory $y(x)$ of the ray in the medium.
- (iv) Determine the coordinates (x_1, y_1) of the point P , where the ray intersects the upper surface of the slab-air boundary.
- (v) Indicate the path of the ray subsequently.

67. A narrow beam of light passing through the hemisphere of material with refractive index n , intersects at point O . Where does the beam converge (i.e. y in cm) if beam were to travel in the opposite direction as shown in figure (b)? The value of x is given to be 10 cm. Radius of the hemisphere is also 10 cm.

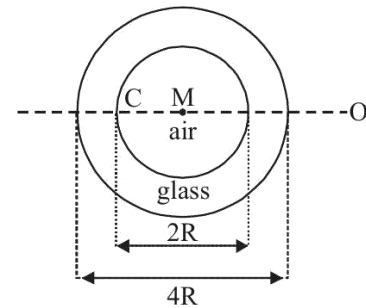


68. A light beam of diameter $\sqrt{3}R$ is incident symmetrically on a glass hemisphere of radius R and of refractive index $n = \sqrt{3}$. Find the radius of the beam at the base of hemisphere



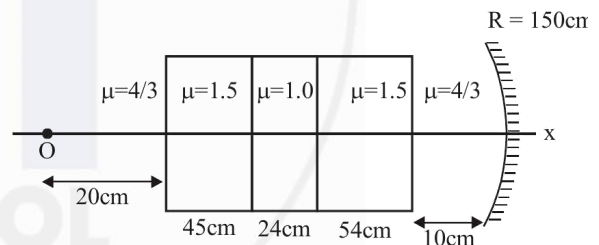
69. A surveyor on one bank of canal observes the images of the 4 inch mark and 17 ft mark on a vertical staff, which is partially immersed in the water and held against the bank directly opposite to him. He sees that reflected and refracted rays come from the same point which is the centre of the canal. If the 17 ft mark and the surveyor's eye are both 6 ft above the water level, estimate the width of the canal, assuming that the refractive index of the water is $4/3$. Zero mark is at the bottom of the canal.

70. A hollow sphere of glass of R.I. n has a small mark M on its interior surface which is observed by an observer O from a point outside the sphere. C is centre of the sphere. The inner cavity (air) is concentric with the external surface and thickness of the glass is everywhere equal to the radius of the inner surface. Find the distance by which the mark will appear nearer than it really is in terms of n and R assuming paraxial rays.

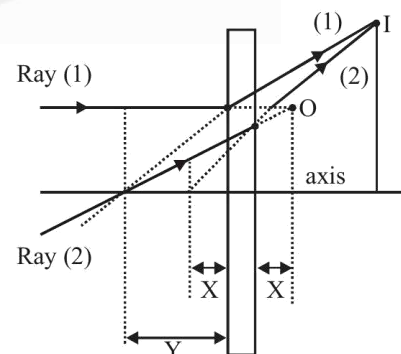


71. A parallel beam of light falls normally on the first face of a prism of small angle. At the second face it is partly transmitted and partly reflected, the reflected beam striking at the first face again, and emerging from it in a direction making an angle $6^\circ 30'$ with the reversed direction of the incident beam. The refracted beam is found to have undergone a deviation of $1^\circ 15'$ from the original direction. Find the refractive index of the glass and the angle of the prism.

72. A composite slab consisting of different media is placed in front of a concave mirror of radius of curvature 150 cm. The whole arrangement is placed in water. An object O is placed at a distance 20 cm from the slab. The R.I. of different media are given in the diagram. Find the position of the final image formed by the system.



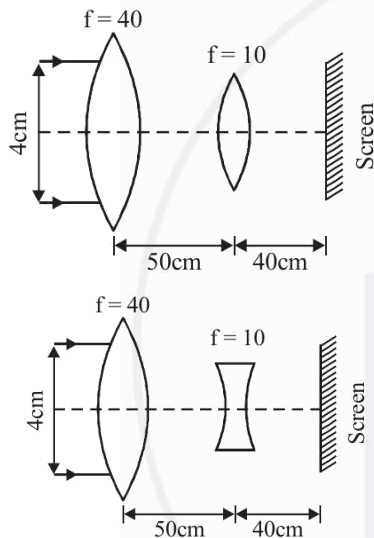
73. The rectangular box shown is the place of lens. By looking at the ray diagram, answer the following questions:



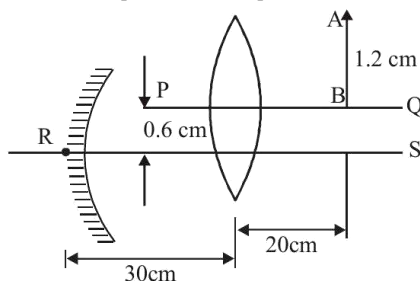
- (i) If X is 5 cm then what is the focal length of the lens?
- (ii) If the point O is 1 cm above the axis then what is the position of the image? Consider the optical center of the lens to be the origin.

74. A luminous point P is inside a circle. A ray enters from P and after two reflections by the circle, return to P. If θ be the angle of incidence, a the distance of P from the centre of the circle and b the distance of the centre from the point where the ray in its course crosses the diameter through P, prove that $\tan \theta = \sqrt{\frac{a-b}{a+b}}$

75. A parallel beam of light of diameter 4 cm falls on a thin convex lens of focal length 40 cm and then on a thin convex lens of focal length 10 cm as shown in figure (a). In figure (b) the second lens is an equiconcave lens of focal length 10 cm. In both the cases, the second lens has an aperture equal to 1 cm and $\mu = 1.5$. Compare the area illuminated by the beam of light on the screen, which passes through the second lens in the two cases.

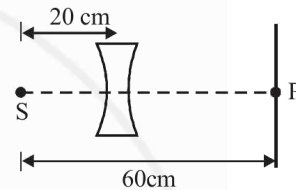


76. A convex lens of focal length 15 cm and a concave mirror of focal 30 cm are kept with their optic axes PQ and RS parallel but separated in vertical direction by 0.6 cm as shown. The distance between the lens and mirror is 30 cm. An upright object AB of height 1.2 cm is placed on the optic axis PQ of the lens at a distance of 20 cm from the lens. If A'B' is the image after refraction from the lens and reflection from the mirror, find the distance of A'B' from the pole of the mirror and obtain its magnification. Also locate positions of A' and B' with respect to the optic axis RS.

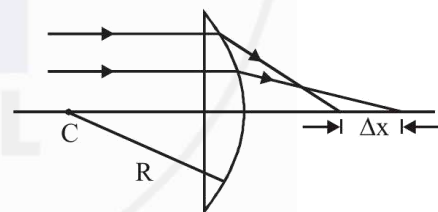


77. An observer holds in front of himself a thin symmetrical converging lens; a is the numerical value of the radius of curvature of each face. He sees two images of his eyes, one erect and the other inverted. Explain the formation of these images and assuming the refractive index of glass to be 1.50, prove that he will see two erect images if the distance of the lens is less than 0.25 a .

78. A point source of light is placed 60 cm away from screen. Intensity detected at point P is I . Now a diverging lens of focal length 20 cm is placed 20 cm away from S between S and P. The lens transmits 75% of light incident on it. Find the new value of intensity at P.

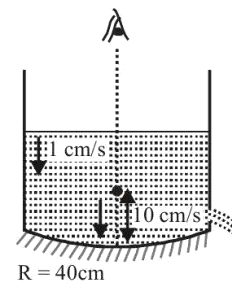


79. Two rays travelling parallel to the principal axis strike a large plano-convex lens having a refractive index of 1.60. If the convex face is spherical, a ray near the edge does not pass through the focal point (spherical aberration occurs). If this face has a radius of curvature of magnitude 20.0 cm and the two rays are $h_1 = 0.500$ cm and $h_2 = 12.0$ cm from the principal axis, find the difference in the positions where they cross the principal axis.

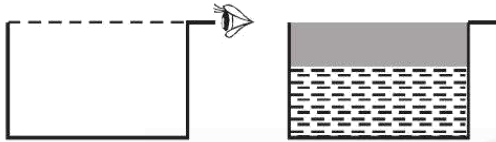


80. Water level in the tank is decreasing at a constant rate of 1 cm/s. A small metal sphere is moving downwards with a constant velocity 5 cm/s. Base of the tank is a concave mirror of radius 40 cm. Find the velocity of the image seen [Take $n_{\text{air}}/n_{\text{water}} = 4/3$].

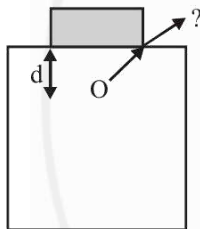
- (a) directly
(b) after reflection at the mirror



81. A man places his eye parallel to rim of a cylinder. When the cylinder is empty, the man cannot see anything inside the cylinder. Now, the cylinder is filled by two immiscible liquids to the top. Both the liquid layers have the same width. Now from the same position, the observer can just see the base of the cylinder. The upper layer has refractive index $\sqrt{2}$. If the height of the cylinder is $2H$ and the diameter of the base is $3H$ then find out the refractive index of the lower layer.

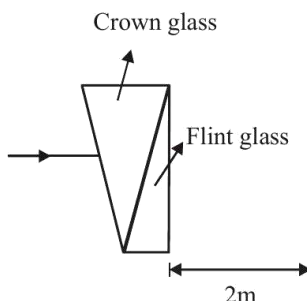


82. A large glass cube ($n = 1.5$) has a small air bubble below one surface. When a rupee coin (diameter 2 cm) is placed directly over the bubble on the outside of the cube, the bubble cannot be seen by looking down into the cube at any angle. However when a 50 paise coin (diameter 1.5 cm) is placed directly over it, the bubble can be seen by the looking down into the cube. What is the range of the possible depths d of the air bubble beneath the surface?

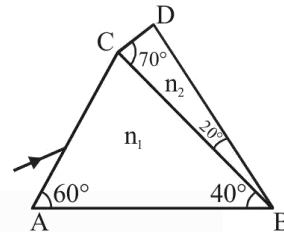


83. The refractive indices of the crown glass for violet and red lights are 1.51 and 1.49 respectively and those of the flint glass are 1.77 and 1.73 respectively. A prism of angle 6° is made of crown glass. A beam of white light is incident at a small angle on this prism. The other thin flint glass prism is combined with the crown glass prism such that the net mean deviation is 1.5° anticlockwise.

- (i) Determine the angle of the flint glass prism.
(ii) A screen is placed normal to the emerging beam at a distance of 2 m from the prism combination. Find the distance between red and violet spot on the screen. Which is the top most colour on screen.



84. A prism of refractive index n_1 & another prism of refractive index n_2 are stuck together without a gap as shown in the figure. The angles of the prisms are as shown. n_1 & n_2 depend on λ , the wavelength of light according to $n_1 = 1.20 + \frac{10.8 \times 10^4}{\lambda^2}$ & $n_2 = 1.45 + \frac{1.80 \times 10^4}{\lambda^2}$ where λ is in nm.

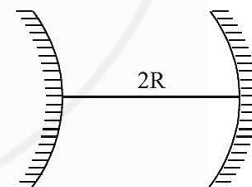


- (i) Calculate the wavelength λ_0 for which rays incident at any angle on the interface BC pass through without bending at that interface.

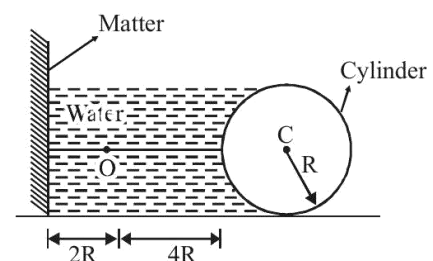
- (ii) For light of wavelength λ_0 , find the angle of incidence i on the face AC such that the deviation produced by the combination of prisms is minimum.

85. Two rays are incident on a spherical mirror of radius of $R = 5$ cm parallel to its optical axis at the distance $h_1 = 0.5$ cm and $h_2 = 3$ cm. Determine the distance Δx between the point at which these rays intersect the optical axis after being reflected at the mirror.

86. Two spherical mirrors, one convex and the other concave, each of same radius of curvature R are arranged coaxially at a distance of $2R$ from each other as shown in figure. A small circle of radius a is drawn on the convex mirror. What is the radii of first three images of the circle.

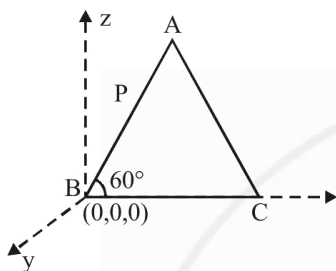


87. The figure shows a very long circular cylinder of radius R made of glass $\mu_g = 3/2$ fixed to the horizontal surface. The space between the cylinder and a vertical wall (plane mirror) is filled with water of refractive index $\mu_w = 4/3$, as shown in figure



Find the positions of the final image formed of an object O placed on the horizontal axis passing through the centre of mass C of the cylinder as shown in figure. Express your answer with respect to centre C of the cylinder. Consider only paraxial rays lying in the vertical plane containing the point O and C.

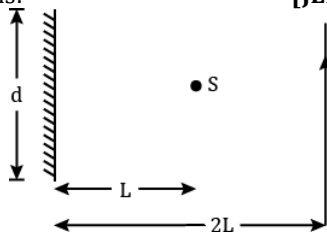
88. An equilateral prism ABC is placed in air with its base side C lying horizontally along X-axis as shown in the figure. A ray given by $\sqrt{3}z + x = 10$ is incident at a point P on face AB of prism



- Find the value of μ for which the ray grazes the faces AC.
- Find direction of the finally refracted ray if $\mu = 3/2$.
- Find the equation of ray coming out of prism if bottom BC is silvered?

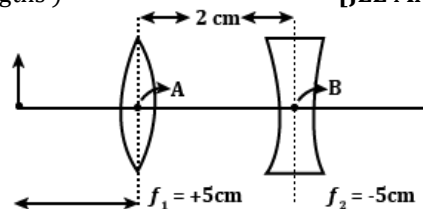
EXERCISE – JEE MAIN

- A convex lens is put 10 cm from a light source and it makes a sharp image on a screen, kept 10 cm from the lens. Now a glass block (refractive index 1.5) of 1.5 cm thickness is placed in contact with the light source. To get the sharp image again, the screen is shifted by a distance d. Then d is: [JEE MAIN 2019]
 - 1.1 cm away from the lens
 - 0
 - 0.55 cm towards the lens
 - 0.55 cm away from the lens
- A point source of light, S is placed at a distance L in front of the centre of plane mirror of width d which is hanging vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror, at a distance 2L as shown below. The distance over which the man can see the image of the light source in the mirror is: [JEE MAIN 2019]



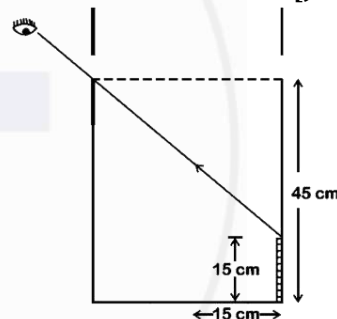
- d
- 2d
- 3d
- $\frac{d}{2}$

- What is the position and nature of image formed by lens combination shown in figure? (f_1, f_2 are focal lengths) [JEE MAIN 2019]



- 70 cm from point B at left; virtual
- 40 cm from point B at right; real
- $\frac{20}{3}$ cm from point B at right; real
- 70 cm from point B at right; real

- An observer can see through a small hole on the side of a jar (radius 15 cm) at a point at height of 15 cm from the bottom (see figure). The hole is at a height of 45 cm. When the jar is filled with a liquid up to a height of 30 cm the same observer can see the edge at the bottom of the jar. If the refractive index of the liquid is $N/100$, where N is an integer, the value of N is [JEE MAIN 2020]



- When an object is kept at a distance of 30 cm from a concave mirror, the image is formed at a distance of 10 cm from the mirror. If the object is moved with a speed of 9 cm s^{-1} , the speed (in cm s^{-1}) with which image moves at that instant is [JEE MAIN 2020]
- A point like object is placed at a distance of 1 m in front of a convex lens of focal length 0.5 m. A plane mirror is placed at a distance of 2 m behind the lens. The position and nature of the final image formed by the system is [JEE MAIN 2020]
 - 2.6 m from the mirror; real
 - 1 m from the mirror; real
 - 2.6 m from the mirror; virtual
 - 1 m from the mirror; virtual
- Two immiscible liquids of refractive index $\sqrt{2}$ and $2\sqrt{2}$ are filled with equal height h in a vessel. Then apparent depth of bottom surface of the container given that outside medium is air: [JEE MAIN 2020]

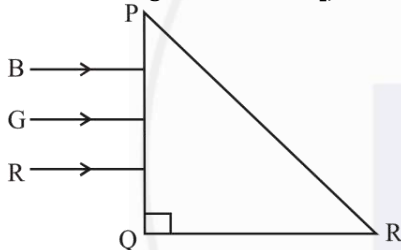
- $\frac{3\sqrt{2}h}{4}$
- $\frac{3h}{4}$
- $\frac{3h}{2}$
- $\frac{3h}{4\sqrt{2}}$

8. A short straight object of height 100 cm lies before the central axis of a spherical mirror whose focal length has absolute value $|f| = 40$ cm. The image of object produced by the mirror is of height 25 cm and has the same orientation of the object. One may conclude from the information: [JEE MAIN 2021]

(A) Image is real, same side of concave mirror.
(B) Image is virtual, opposite side of convex mirror.
(C) Image is virtual, opposite side of concave mirror.
(D) Image is real, same side of convex mirror.

9. The image of an object placed in air formed by a convex refracting surface is at a distance of 10 m behind the surface. The image is real and is at $\frac{2^{\text{rd}}}{3}$ of the distance of the object from the surface. The wavelength of light inside the surface is $\frac{2}{3}$ times the wavelength in air. The radius of the curved surface is $\frac{x}{13}$ m. the value of 'x' is [JEE MAIN 2021]

10. Three rays of light, namely red (R), green (G) and blue (B) are incident on the face PQ of a right-angled prism PQR as shown in figure. [JEE MAIN 2021]



The refractive indices of the material of the prism for red, green and blue wavelength are 1.27, 1.42 and 1.49 respectively. The colour of the ray(s) emerging out of the face PR is :

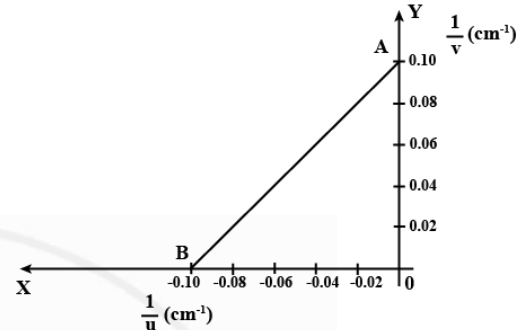
(A) green (B) red
(C) blue and green (D) blue

11. An object viewed from a near point distance of 25 cm, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 m, if the focal length of the eyepiece is equal to cm. [JEE MAIN 2021]

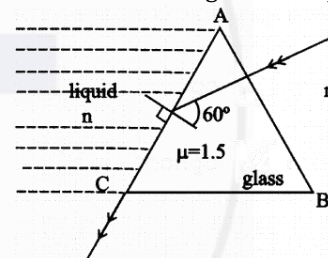
12. A prism of refractive index μ and angle of prism A is placed in the position of minimum angle of deviation. If minimum angle of deviation is also A, then in terms of refractive index [JEE MAIN 2021]

(A) $2\cos^{-1}\left(\frac{\mu}{2}\right)$
(B) $\sin^{-1}\left(\frac{\mu}{2}\right)$
(C) $\sin^{-1}\left(\sqrt{\frac{\mu-1}{2}}\right)$
(D) $\cos^{-1}\left(\frac{\mu}{2}\right)$

13. The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of lens is 1.5 and its both the surfaces have same radius of curvatures R. The value of R will be cm. [JEE MAIN 2022]
(Where u = object distance, v = image distance)



14. In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle 60° at the side AC, the refracted light beam just grazes along face AC. The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$. The value of x is [JEE MAIN 2022]
(Given refractive index of glass = 1.5)



15. In normal adjustment, for a refracting telescope, the distance between objective and eye piece is 30 cm. The focal length of the objective, when the angular magnification of the telescope is 2, will be: [JEE MAIN 2022]

(A) 20 cm
(B) 30 cm
(C) 10 cm
(D) 15 cm

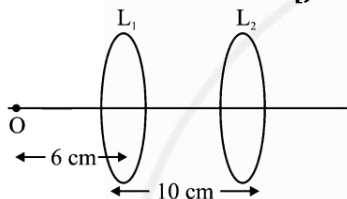
16. A thin prism P_1 with an angle 6° and made of glass of refractive index 1.54 is combined with another prism P_2 made from glass of refractive index 1.72 to produce dispersion without average deviation. The angle of prism P_2 is : [JEE MAIN 2023]

(A) 6° (B) 1.3°
(C) 7.8° (D) 4.5°

17. A pole is vertically submerged in swimming pool, such that it gives a length of shadow 2.15 m within water when sunlight is incident at an angle of 30° with the surface of water. If swimming pool is filled to a height of 1.5 m, then the height of the pole above the water surface in centimeters is $\left(n_w = \frac{4}{3}\right)$ [JEE MAIN 2023]

18. A 2 meter long scale with least count of 0.2 cm is used to measure the locations of objects on an optical bench. While measuring the focal length of a convex lens, the object pin and the convex lens are placed at 80 cm mark and 1 m mark, respectively. The image of the object pin on the other side of lens coincides with image pin that is kept at 180 cm mark. The % error in the estimation of focal length is: [JEE MAIN 2023]
 (A) 0.85 (B) 1.70 (C) 1.02 (D) 0.51

19. A point object O is placed in front of two thin symmetrical coaxial convex lenses L_1 and L_2 with focal length 24 cm and 9 cm respectively. The distance between two lenses is 10 cm and the object is placed 6 cm away from lens L_1 as shown in the figure. The distance between the object and the image formed by the system of two lenses is cm [JEE MAIN 2023]



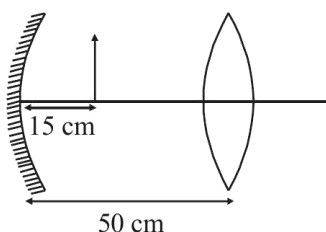
20. An ice cube has a bubble inside. When viewed from one side the apparent distance of the bubble is 12 cm. When viewed from the opposite side, the apparent distance of the bubble is observed as 4 cm. If the side of the ice cube is 24 cm, the refractive index of the ice cube is [JEE MAIN 2023]

(A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{6}{5}$ (D) $\frac{4}{3}$

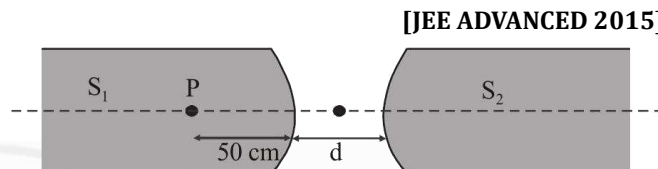
21. A bi convex lens of focal length 10 cm is cut in two identical parts along a plane perpendicular to the principal axis. The power of each lens after cut is [JEE MAIN 2023]

EXERCISE - JEE ADVANCED

1. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1) as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image formed by this combination has magnification M_1 . When the set-up is kept in a medium of refractive index $7/6$, the magnification becomes M_2 . The magnitude $\left| \frac{M_2}{M_1} \right|$ is [JEE ADVANCED 2015]

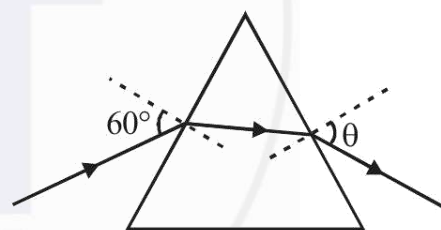


2. Two identical glass rods S_1 and S_2 (refractive index = 1.5) have one convex end of radius of curvature 10 cm. They are placed with the curved surfaces at a distance d as shown in the figure, with their axes (shown by the dashed line) aligned. When a point source of light P is placed inside rod S_1 on its axis at a distance of 50 cm from the curved face, the light rays emanating from it are found to be parallel to the axis inside S_2 . The distance d is [JEE ADVANCED 2015]



(A) 60 cm
 (B) 70 cm
 (C) 80 cm
 (D) 90 cm

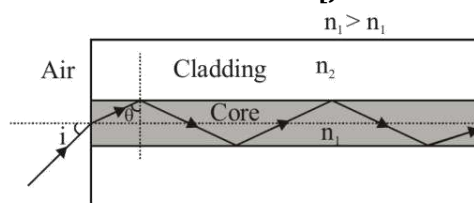
3. A monochromatic beam of light is incident at 60° on one face of an equilateral prism of refractive index n and emerges from the opposite face making an angle $\theta(n)$ with the normal (see the figure). For $n = \sqrt{3}$ the value of θ is 60° and $\frac{d\theta}{dn} = m$. The value of m is [JEE ADVANCED 2015]



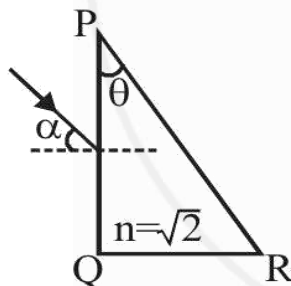
Paragraph (For questions No. 4 & 5)

Light guidance in an optical fiber can be understood by considering a structure comprising of thin solid glass cylinder of refractive index n_1 surrounded by a medium of lower refractive index n_2 . The light guidance in the structure takes place due to successive total internal reflections at the interface of the media n_1 and n_2 as shown in the figure. All rays with the angle of incidence i less than a particular value i_m are confined in the medium of refractive index n_1 . The numerical aperture (NA) of the structure is defined as $\sin i_m$.

[JEE ADVANCED 2015]

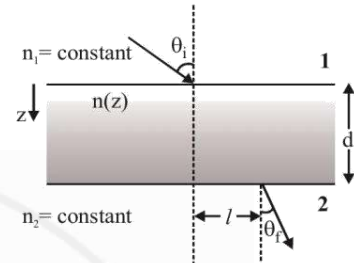


4. For two structures namely S_1 with $n_1 = \sqrt{45}/4$ and $n_2 = 3/2$ and S_2 with $n_1 = 8/5$ and $n_2 = 7/5$ and taking the refractive index of water to be $4/3$ and that of air to be 1, the correct option(s) is(are)
- (A) NA of S_1 immersed in water is the same as that of S_2 immersed in a liquid of refractive index $\frac{16}{3\sqrt{15}}$
- (B) NA of S_1 immersed in liquid of refractive index $\frac{6}{\sqrt{15}}$ is the same as that of S_2 immersed in water
- (C) NA of S_1 placed in air is the same as that of S_2 immersed in liquid of refractive index $\frac{4}{\sqrt{15}}$
- (D) NA of S_1 placed in air is the same as that of S_2 placed in water
5. If two structures of same cross-sectional area, but different numerical apertures NA_1 and NA_2 ($NA_2 < NA_1$) are joined longitudinally, the numerical aperture of the combined structure is
- (A) $\frac{NA_1 NA_2}{NA_1 + NA_2}$
- (B) $NA_1 + NA_2$
- (C) NA_1
- (D) NA_2
6. A parallel beam of light is incident from air to an angle α on the side PQ of a right angled triangular prism of refractive index $n = \sqrt{2}$. Light undergoes total internal reflection in the prism at the face PR when α has a minimum value of 45° . The angle θ of the prism is
- [JEE ADVANCED 2016]



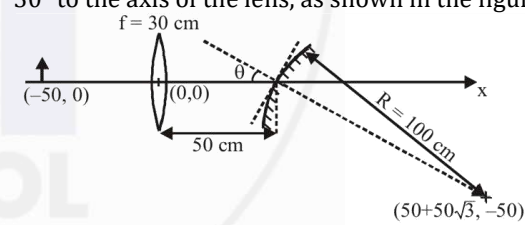
- (A) 15°
- (B) 22.5°
- (C) 30°
- (D) 45°
7. A plano-convex lens is made of a material of refractive index n . When a small object is placed 30 cm away in front of the curved surface of the lens, an image of double the size of the object is produced. Due to reflection from the convex surface of the lens, another faint image is observed at a distance of 10 cm away from the lens. Which of the following statement(s) is(are) true?
- [JEE ADVANCED 2016]
- (A) The refractive index of the lens is 2.5
- (B) The radius of curvature of the convex surface is 45 cm
- (C) The faint image is erect and real
- (D) The focal length of the lens is 20 cm

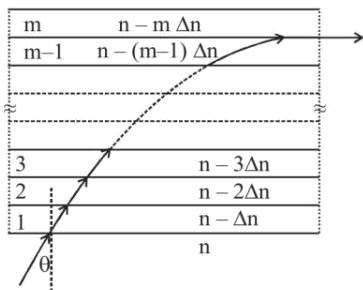
8. A transparent slab of thickness d has a refractive index $n(z)$ that increases with z . Here z is the vertical distance inside the slab, measured from the top. The slab is placed between two media with uniform refractive indices n_1 and $n_2 (> n_1)$, as shown in the figure. A ray of light is incident with angle θ_i from medium 1 and emerges in medium 2 with refraction angle θ_f with a lateral displacement l .



Which of the following statement (s) is (are) true?

[JEE ADVANCED 2016]

- (A) $n_1 \sin \theta_i = n_2 \sin \theta_f$
- (B) $n_1 \sin \theta_i = (n_2 - n_1) \sin \theta_f$
- (C) l is independent of n_2
- (D) l is dependent on $n(z)$
9. A small object is placed 50 cm to the left of a thin convex lens of focal length 30 cm. A convex spherical mirror of radius of curvature 100 cm is placed to the right of the lens at a distance of 50 cm. The mirror is tilted such that the axis of the mirror is at an angle $\theta = 30^\circ$ to the axis of the lens, as shown in the figure.
- 
- (A) (0,0)
- (B) $(50 - 25\sqrt{3}, 25)$
- (C) $(25, 25\sqrt{3})$
- (D) $(\frac{125}{3}, \frac{25}{\sqrt{3}})$
10. If the origin of the coordinate system is taken to be at the centre of the lens, the coordinates (in cm) of the point (x,y) at which the image is formed are
- [JEE ADVANCED 2016]
- (A) (0,0)
- (B) $(50 - 25\sqrt{3}, 25)$
- (C) $(25, 25\sqrt{3})$
- (D) $(\frac{125}{3}, \frac{25}{\sqrt{3}})$
11. A monochromatic light is travelling in a medium of refractive index $n = 1.6$. It enters a stack of glass layers from the bottom side at an angle $\theta = 30^\circ$. The interfaces of the glass layers are parallel to each other. The refractive indices of different glass layers are monotonically decreasing as $n_m = n - m\Delta n$, where n_m is the refractive index of the m^{th} slab and $\Delta n = 0.1$ (see the figure). The ray is refracted out parallel to the interface between the $(m-1)^{\text{th}}$ and m^{th} slabs from the right side of the stack. What is the value of m ?
- [JEE ADVANCED 2017]



12. For an isosceles prism of angle A and refractive index μ , it is found that the angle of minimum deviation $\delta_m = A$. Which of the following options is/are correct?

[JEE ADVANCED 2017]

(A) At minimum deviation, the incident angle i_1 and the refracting angle r_1 at the first refracting surface are related by $r_1 = (i_1/2)$

(B) For this prism, the refractive index μ and the angle of prism A are related as $A = \frac{1}{2} \cos^{-1} \left(\frac{\mu}{2} \right)$

(C) For this prism the emergent ray at the second surface will be tangential to the surface when the angle of incidence at the first surface is $i_1 = \sin^{-1} \left[\sin A \sqrt{4 \cos^2 \frac{A}{2} - 1} - \cos A \right]$

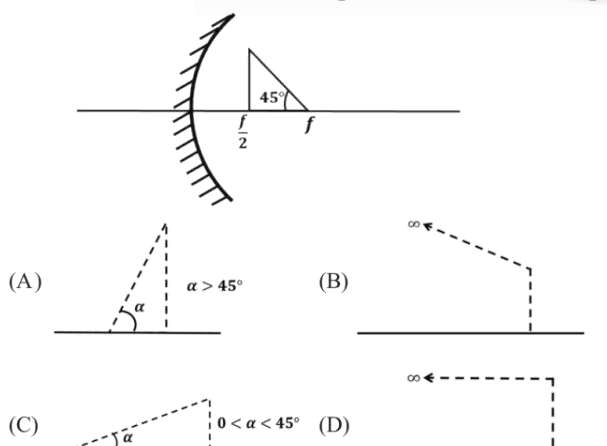
(D) For the angle of incidence $i_1 = A$, the ray inside the prism is parallel to the base of the prism

13. Sunlight of intensity 1.3 kW m^{-2} is incident normally on a thin convex lens of focal length 20 cm. Ignore the energy loss of light due to the lens and assume that the lens aperture size is much smaller than its focal length. The average intensity of light, in kW m^{-2} , at a distance 22 cm from the lens on the other side is _____.

[JEE ADVANCED 2018]

14. A wire is bent in the shape of a right angled triangle and is placed in front of a concave mirror of focal length f , as shown in the figure. Which of the figures shown in the four options qualitatively represent(s) the shape of the image of the bent wire? (These figures are not to scale.)

[JEE ADVANCED 2018]



15. A thin convex lens is made of two materials with refractive indices n_1 and n_2 , as shown in figure. The radius of curvature of the left and right spherical surfaces are equal, f is the focal length of the lens when $n_1 = n_2 = n$. The focal length is $f + \Delta f$ when $n_1 = n$ and $n_2 = n + \Delta n$. Assuming $\Delta n \ll (n - 1)$ and $1 < n < 2$, the correct statement(s) is/are,

[JEE ADVANCED 2019]



(1) If $\frac{\Delta n}{n} < 0$ then $\frac{\Delta f}{f} > 0$

(2) For $n = 1.5$, $\Delta n = 10^{-3}$ and $f = 20 \text{ cm}$, the value of $|\Delta f|$ will be 0.02 cm (round off to 2nd decimal place).

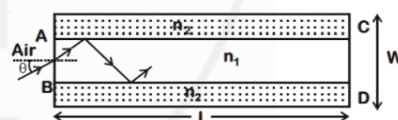
(3) The relation between $\frac{\Delta f}{f}$ and $\frac{\Delta n}{n}$ remains unchanged if both the convex surfaces are replaced by concave surfaces of the same radius of curvature.

(4) $\left| \frac{\Delta f}{f} \right| < \left| \frac{\Delta n}{n} \right|$

16. A planar structure of length L and width W is made of two different optical media of refractive indices $n_1 = 1.5$ and $n_2 = 1.44$ as shown in figure. If $L \gg W$, a ray entering from end AB will emerge from end CD only if the total internal reflection condition is met inside the structure. For $L = 9.6 \text{ m}$, if the incident angle θ is varied, the maximum time taken by a ray to exit the plane CD is $t \times 10^{-9} \text{ s}$, where t is _____.

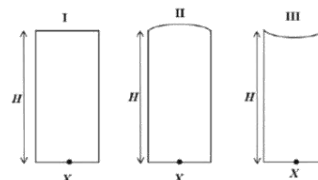
[Speed of light $c = 3 \times 10^8 \text{ m/s}$]

[JEE ADVANCED 2019]



17. Three glass cylinders of equal height $H = 30 \text{ cm}$ and same refractive index $n = 1.5$ are placed on a horizontal surface as shown in figure. Cylinder I has a flat top, cylinder II has a convex top and cylinder III has a concave top. The radii of curvature of the two curved tops are same ($R = 3 \text{ m}$). If H_1, H_2 , and H_3 are the apparent depths of a point X on the bottom of the three cylinders, respectively, the correct statement(s) is/are:

[JEE ADVANCED 2019]



(1) $H_2 > H_1$

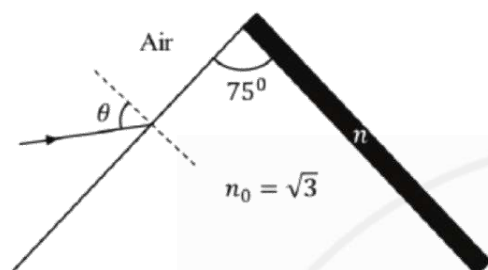
(2) $H_2 > H_3$

(3) $H_3 > H_1$

(4) $0.8 \text{ cm} < (H_2 - H_1) < 0.9 \text{ cm}$

18. A monochromatic light is incident from air on a refracting surface of a prism of angle 75° and refractive index $n_0 = \sqrt{3}$. The other refracting surface of the prism is coated by a thin film of material of refractive index n as shown in figure. The light suffers total internal reflection at the coated prism surface for an incidence angle of $\theta \leq 60^\circ$. The value of n^2 is ____.

[JEE ADVANCED 2019]



19. An optical bench has 1.5 m long scale having four equal divisions in each cm. While measuring the focal length of a convex lens, the lens is kept at 75 cm mark of the scale and the object pin is kept at 45 cm mark. The image of the object pin on the other side of the lens overlaps with image pin that is kept at 135 cm mark. In this experiment, the percentage error in the measurement of the focal length of the lens is ____.

[JEE ADVANCED 2019]

ANSWER KEY

EXERCISE-I

1. A 2. D 3. A 4. B 5. C 6. D 7. A 8. B 9. C 10. B 11. C 12. D
 13. B 14. D 15. A 16. B 17. C 18. A 19. B 20. A 21. A 22. B 23. D 24. D
 25. C 26. A 27. D 28. D 29. C 30. B 31. A 32. C 33. B 34. A 35. A 36. D
 37. C 38. B 39. 2 mm 40. $d/2$ 41. 16 mm^3 42. 25 43. 2 44. $\left(\frac{10}{3}\right) \text{ m}$ 45. $\frac{32}{9}$, Expanding
 46. 5 47. $\vec{r} = \frac{3}{5\sqrt{2}}\hat{i} + \frac{2\sqrt{2}}{5}\hat{j} - \frac{1}{\sqrt{2}}\hat{k}$ (angle of incidence = 60° ; $r = 45^\circ$) 48. $\frac{b(1-\mu^2\cos^2\theta)^{1/2}}{\sin\theta}$
 49. 8 50. 42 cm 51. (a) $\sin^{-1}\left(\frac{1}{5}\right)$ (b) air 52. $\theta = 53^\circ$ 53. $\theta < \sin^{-1}(2\sin 15^\circ)$
 54. $\frac{8}{3} \text{ mm}$ 55. $\frac{4}{3}$ 56. 2 57. 30 58. $y = -\frac{x}{400} + 0.1$
 59. Real, below principal axis, anywhere b/w P & F_1 60. 53 61. $2h$ 62. $d = 6 \text{ cm}$
 63. 3 64. (i) 0.2 m; (ii) 0.4 m 65. $\left(\frac{\pi}{4}\right) \text{ cm}^2$ 66. 17.5 cm
 67. (a) $f = -20 \text{ cm}$, (b) $+80 \text{ cm}$, convergent achromatic lens

EXERCISE-II

1. B,C 2. D 3. D 4. C 5. C 6. D 7. C 8. D 9. A 10. A 11. C 12. A
 13. C 14. A 15. C 16. A 17. ABC 18. ACD 19. A 20. C 21. C 22. B 23. D 24. C
 25. C 26. AD 27. BC 28. BD 29. ABC 30. ACD 31. ABC 32. BCD 33. BCD 34. AD 35. ABD 36. AB
 37. BC 38. AB 39. BD 40. ABCD 41. BD 42. AC 43. AD 44. AD 45. D 46. C 47. C 48. B
 49. D 50. B 51. C 52. C 53. A 54. C 55. A 56. A 57. A 58. B 59. A
 60. (A) $\rightarrow P$; (B) $\rightarrow Q$; (C) $\rightarrow RT$; (D) $\rightarrow PT$;
 61. (A) $\rightarrow PQR$; (B) $\rightarrow Q$; (C) $\rightarrow S$; (D) $\rightarrow ST$;
 62. (A) $\rightarrow PRST$; (B) $\rightarrow Q$; (C) $\rightarrow PQST$; (D) $\rightarrow PRST$;
 63. 32 64. 0.180 65. 30 cm, -14 mm
 66. (i) $\tan\theta = \frac{dy}{dx} = \cot i$; (ii) 1; (iii) $y = k^2\left(\frac{x}{4}\right)^4$ (iv) 4.0, 1; (v) It will become parallel to x-axis
 67. $v_1 = 20 \text{ cm}$; $d_0 = 10 \text{ cm}$; $\frac{10}{2} = \frac{d_1}{1} \Rightarrow d_1 = 5 \text{ cm}$
 68. $\frac{R}{\sqrt{3}}$ 69. 16 feet 70. $(n-1)R/(3n-1)$ 71. $\mu = \frac{13}{8}$, $A = 2^\circ$ 72. On the object itself
 73. (i) 10 cm, (ii) (10, 2) 75. 81 76. 15 cm left of the mirror, magnification
 = 1.5 (-15 cm , -1.5 cm ; -15 cm , 0.3 cm)
 77. 0.25 a 78. 0.27 l 79. $\frac{52}{3} \text{ cm}$
 80. (a) $v_i = 4 \text{ cm/s}$ downwards; (b) $\frac{59}{4} \text{ cm/s}$ upwards
 81. $\frac{\sqrt{5}}{2}$ 82. $\left(\frac{3\sqrt{5}}{8} \text{ cm} \leq d \leq \frac{\sqrt{5}}{2} \text{ cm}\right)$ 83. (i) 2° , (ii) $\frac{4\pi}{9} \text{ mm}$
 84. (i) $\lambda_0 = 600 \text{ nm}$, $n = 1.5$ (ii) $i = \sin^{-1}(0.75) = 48.59^\circ$
 85. $\frac{5}{8} \text{ cm}$ 86. $\frac{a}{3}$; $\frac{a}{11}$; $\frac{a}{41}$ 87. $\frac{15R}{4}$, $3R$ 88. (a) $\frac{2}{\sqrt{3}}$ (b) Normal to surface (c) Retrace the path, $\sqrt{3}z + x = 10$

EXERCISE-JEE MAIN

1. D 2. C 3. D 4. 158 5. 1 6. A 7. A 8. B 9. 30 10. B 11. 25 12. A
 13. 10 14. 27 15. A 16. D 17. 50 18. B 19. 34 20. A 21. 5

EXERCISE-JEE ADVANCED

1. B 2. 7 3. B 4. 2 5. AC 6. D 7. A 8. AD 9. ACD 10. C 11. 8 12. ACD
 13. 130.00 14. D 15. 1, 2, 3 16. 50 17. 1, 2 18. 1.50 19. 0.69